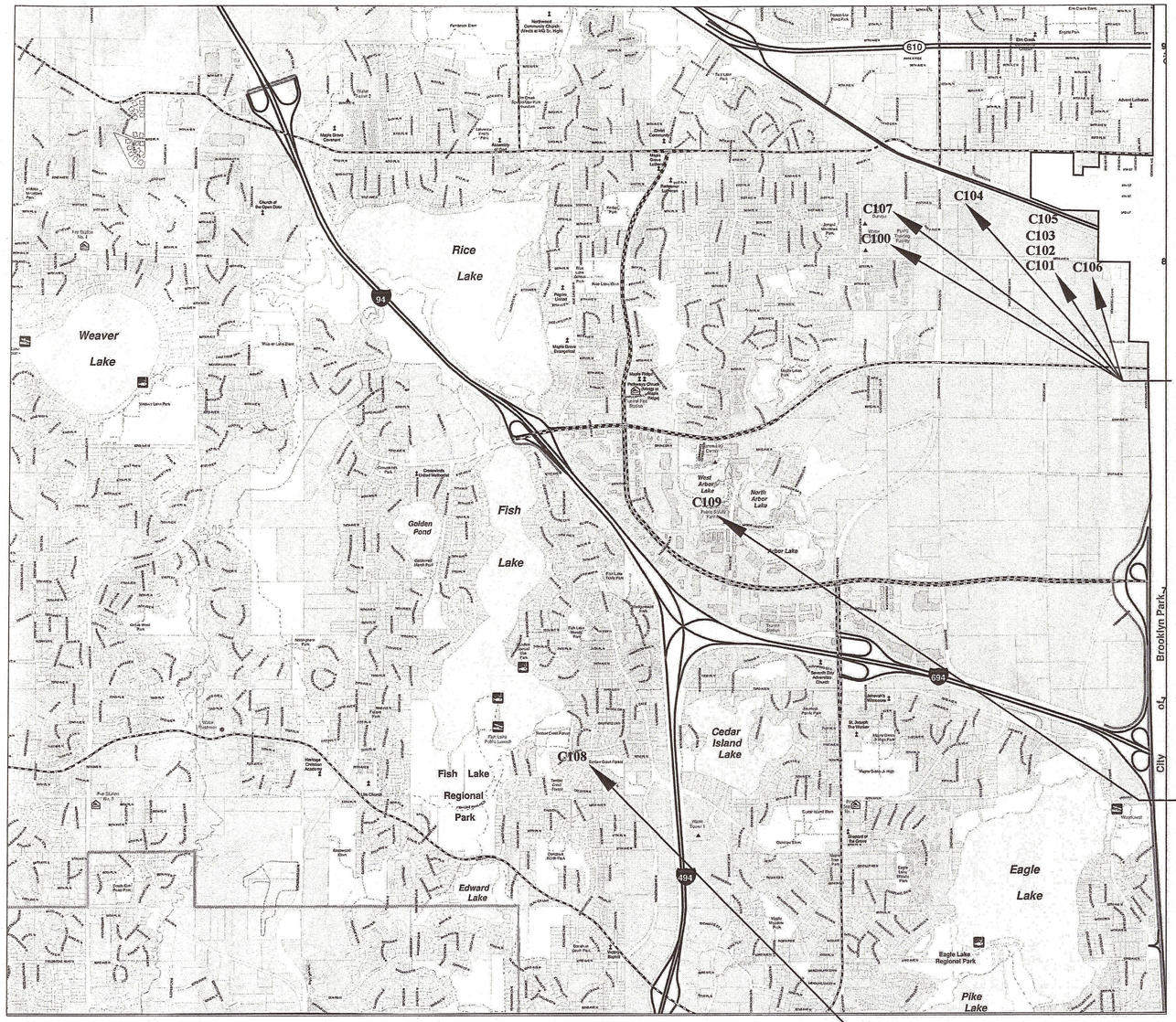
CITY OF MAPLE GROVE

STANDBY GENERATOR PROJECT

City of mg Terry → Gene
Gov Center
612-9799 0736

CITY PROJECT 12-08
TKDA PROJ. NO. 15088.000



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CITY OF MAPLE GROVE, MN

N

I HEREBY CERTIFY THAT THIS PLAN WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MINNESOTA

VERNE E. JACOBSEN
DATE 11-14-2012 LIC. NO. 06988

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TKDA

EXISTING UTILITY LOCATIONS

THE CONTRACTOR SHALL VERIFY ALL EXISTING UNDERGROUND UTILITY LOCATIONS AND ELEVATIONS PRIOR TO CONSTRUCTION. ALL INPLACE UTILITIES MAY NOT BE SHOWN ON THIS PLAN & THOSE THAT ARE SHOWN, MAY NOT BE SHOWN IN THE EXACT LOCATIONS.

THE SUBSURFACE UTILITY INFORMATION IN THIS PLANSET IS UTILITY QUALITY LEVELS "C" & "D" AS DEFINED IN ASCE "STANDARD GUIDELINE FOR THE DEPICTION OF EXISTING SUBSURFACE UTILITY DATA".

GOPHER STATE ONE CALL: 1-800-252-1166

SPECIFICATION REFERENCE

THE 2000 EDITION OF THE MINNESOTA DEPARTMENT OF TRANSPORTATION "STANDARD SPECIFICATIONS FOR HIGHWAY CONSTRUCTION", DIVISIONS II & III, AND ALL AMENDMENTS THERETO, SUPPLEMENTAL SPECIFICATIONS, AND THE "STANDARD UTILITIES SPECIFICATIONS FOR SANITARY SEWER, STORM DRAIN AND WATERMAIN INSTALLATIONS" AS PROPOSED BY THE CITY ENGINEERS ASSOCIATION OF MINNESOTA, 1999 EDITION.

FIELD BOOK:
Plot Date: 11/13/2012
Drawing name: K:\gm\Maple Grove\15088000\mun\dwgs\PlanSet\C001.dwg
Xrefs:

SYMBOLS

COMMUNICATION

- J–BOX, 4–1/16"x4–1/16"x2–1/2" DEEP WITH 2 GANG RING
- JACK, TWO–WAY RADIO
- JACK, DATA/VOICE
- JACK, TELEPHONE (VOICE)
- JACK, TELEVISION
- MICROPHONE
- PROJECTOR
- SPEAKER, CEILING MOUNTED
- SPEAKER, HORN TYPE
- SPEAKER, WALL MOUNTED

FIRE ALARM

- DETECTOR, SMOKE / HEAT (S– SMOKE, B– BEAM, D– DUCT, F– FIXED, RR– RATE/RISE)
- STROBE – WITH CANDELA RATING
- HORN
- HORN/STROBE – WITH CANDELA RATING
- INDIVIDUAL ADDRESSABLE MODULE
- MAGNETIC DOOR HOLDER
- MANUAL PULL STATION
- SWITCH, FLOW
- SWITCH, TAMPER
- FIRE/SMOKE DAMPER W/ POWER CONNECTION FROM FACP TO ACTUATOR AND DUCT DETECTOR.

SECURITY/ACCESS CONTROL

- AUTOMATIC DOOR OPERATOR
- BUZZER
- CARD READER
- CHIME
- CLOSED CIRCUIT CAMERA
- DOOR CONTACTS
- ELECTRIC STRIKE
- HANDICAP DOOR PUSH–BUTTON
- KEY PAD
- REQUEST TO EXIT – MOTION SENSOR
- ELECTRONIC MORTISE LOCK
- MOTION DETECTOR
- SECURITY SIREN
- INTERCOM

MISCELLANEOUS

- FLOOR BOX
- HAND HOLE
- JUNCTION BOX
- NOTE, REFERRAL
- NOTE, REVISION
- POWER POLE
- PULL BOX
- CONDUIT UNDERGROUND OR UNDERFLOOR
- CONDUIT EXPOSED
- CONDUIT UP
- CONDUIT DOWN

LIGHTING

- DAYLIGHT SENSOR
- EXIT LIGHT, CEILING
- EXIT LIGHT, WALL MNT w/EM
- LUMINAIRE TYPE
- CIRCUIT NUMBER
- SWITCH IDENTIFICATION
- LUMINAIRE, CEILING
- LUMINAIRE, NIGHTLIGHT (½ SHADING)
- LUMINAIRE, POLE MOUNTED
- LUMINAIRE, FLUORESCENT
- LUMINAIRE, WALL MOUNT
- LUMINAIRE, WALL WASH
- LUMINAIRE TYPE AND CIRCUIT #
- OCCUPANCY SENSOR OS1 – EQUAL TO WATTSTOPPER WT SERIES
- SWITCH, DIMMER
- SWITCH, SINGLE GANG SP (b–SWITCH IDENTIFICATION)
- SWITCH, SINGLE GANG SP (DUAL RELAY) (OS–OCCUPANCY SENSOR EQUAL TO WATTSTOPPER WA–300, b–SWITCH IDENTIFICATION)
- PHOTOSENSOR
- SWITCH WITH PILOT

DEVICES

- CONTROL DEVICE (PL– PILOT LIGHT, PC PHOTOCELL, PB– PUSHBUTTON,MS–MOTION SENSOR,)
- RECEPTACLE, CEILING DUPLEX
- RECEPTACLE, DOUBLE DUPLEX
- RECEPTACLE, DUPLEX
- RECEPTACLE, SINGLE
- RECEPTACLE, SPECIAL, NEMA# AS NOTED
- RECEPTACLE, SPLIT WIRED
- J–BOX, 4"X4"X2–1/2" DEEP WITH 2 GANG RING

EQUIPMENT

- CONTROL MODULE
- DISCONNECT SWITCH
- EQUIPMENT
- KEYED SWITCH
- MOTOR
- MOTOR STARTER, COMBINATION
- MOTOR STARTER, MAGNETIC
- MOTOR STARTER, MANUAL
- PANEL, ANNUNCIATOR/ALARM
- PANEL, AS DESIGNATED
- PANEL, POWER PANEL
- PANEL, POWER (FLUSH)
- PANEL, TERMINAL CABINET
- TIME SWITCH
- TRANSFORMER

NOTE:

ITEMS SHOWN ON THIS SHEET ARE STANDARD SYMBOLS AND ABBREVIATIONS AND MAY NOT ALL APPEAR ON THE PROJECT DRAWINGS.

ABBREVIATIONS

AFF	ABOVE FINISHED FLOOR
AGF	ABOVE FINISHED GRADE
BFG	BELOW FINISHED GRADE
CUH	CABINET UNIT HEATER
CB	CIRCUIT BREAKER
CCT	DENOTES CIRCUIT(S)
CP	CONTROL PANEL
DP	DISTRIBUTION PANEL
EF	EXHAUST FAN
EM	EMERGENCY
ELEV	ELEVATOR
EPO	EMERGENCY POWER OFF
EWC	ELECTRIC WATER COOLER
GFI	GROUND FAULT INTERRUPTING
GND	GROUND
MC	MOMENTARY CONTACT
MT	EMPTY
NC	NORMALLY CLOSED
NO	NORMALLY OPEN
PB	PULLBOX
SAS	SAFETY SWITCH
SS	SELECTOR SWITCH
TT	TELEPHONE BOARD
TC	TERMINAL CABINET
VFD	VARIABLE FREQUENCY DRIVE
WH	WATER HEATER
WP	WEATHERPROOF
X	INDICATES EXISTING DEVICES, FIXTURE OR WIRING TO REMAIN, UNLESS NOTED OTHERWISE
R	REMOVE EXISTING EQUIPMENT OR DEVICE
RL	EXISTING EQUIPMENT OR DEVICE TO BE RELOCATED
NL	NEW LOCATION OF RELOCATED EQUIPMENT OR DEVICE
MD	MOTORIZED DAMPER
CU	CONDENSING UNIT
F	FURNACE
FSD	FIRE/SMOKE DAMPER

GENERAL NOTES

1. UNLESS OTHERWISE NOTED, REMOVE ALL EQUIPMENT DESIGNATED WITH A 'R' AND/OR WITH A SOLID HEAVY DASHED LINE. REMOVE ALL EQUIPMENT, CONDUCTORS, AND RACEWAY BACK TO LAST ACTIVE SOURCE.
2. REPLACE EQUIPMENT IN SAME LOCATION WHERE DESIGNATED WITH A "RP" NOTATION.
3. RELOCATE EQUIPMENT IN NEW LOCATION WHERE DESIGNATED WITH A "RL" AND CORRESPONDING "NL" NOTATION.
4. EXISTING EQUIPMENT SHOWN IN SOLID LIGHT LINES AND/OR DESIGNATED WITH A 'X' SHALL REMAIN.
5. CONDUIT SHOWN IN DASHED LINES OR BY NOTE REPRESENT HIDDEN, UNDERFLOOR, OR UNDERGROUND RUNS. ROUTING IS SCHEMATIC.
6. EXISTING EQUIPMENT AND CONDUIT LOCATIONS ARE SCHEMATIC, APPROXIMATE, AND SHOWN FOR REFERENCE ONLY; FIELD VERIFY ALL LOCATIONS.
7. NOT ALL CONDUIT SHOWN. ALL EXISTING CONDUIT AND WIRE SIZES ARE APPROXIMATIONS AND SUBJECT TO FIELD VERIFICATION. ALL SCHEDULED CONDUIT AND WIRE SIZES ARE PRELIMINARY AND REQUIRE CONFIRMATION AT TIME OF INSTALLATION.
8. IF POWER OR CONTROL SOURCES OF EXISTING DEVICES ARE REMOVED, THEN REMOVE WIRING BACK TO LAST ACTIVE SOURCE. IF NECESSARY, EXTEND CIRCUITS TO REFEED DEVICES/EQUIPMENT THAT ARE TO REMAIN.
9. NUMBERS AT RECEPTACLES AND LUMINARIES CORRESPOND WITH ASSOCIATED PANELBOARDS. VERIFY EXISTING PANELBOARD CIRCUITRY. BRANCH CIRCUIT CONDUIT RUNS ARE NOT SHOWN TO SIMPLIFY DRAWING.
10. REFER TO SPECIFICATIONS FOR MATERIALS AND INSTALLATION REQUIREMENTS OR AS DESCRIBED ON DRAWINGS.
11. CIRCUITS SUPPLYING LINE TO NEUTRAL LOADS, BOTH 120V AND 277V, SHALL HAVE DEDICATED NEUTRALS. NEUTRAL CONDUCTORS SHALL HAVE TRACER STRIPE ALONG ENTIRE LENGTH INDICATING ASSOCIATED PHASE.
12. ALL RACEWAYS SHALL CONTAIN A GROUNDING CONDUCTOR.
13. ALL CIRCUIT CONDUCTORS SIZING SHALL BE INCREASED WHERE REQUIRED FOR VOLTAGE DROP AND REDUCED TO ACCOUNT FOR DERATING DUE TO AMBIENT TEMPERATURE. ALL BRANCH CIRCUIT AND FEEDER WIRING SHALL LIMIT THE VOLTAGE DROP TO 3% FOR EITHER SEGMENT AND 5% OVERALL.

WELL BUILDINGS	100F
LIFT STATION	80F
GOVERNMENT CENTER	80F
PUBLIC WORKS	80F
WATER TREATMENT PLANT	80F

14. SEE RELATED DRAWINGS FOR LOCATION OF EQUIPMENT REQUIRING ELECTRICAL CONNECTIONS. COORDINATE INTERCONNECTION REQUIREMENTS WITH CIVIL AND FROM EQUIPMENT SUBMITTAL DOCUMENTS.
15. UPDATE PANELBOARD SCHEDULES (TYPEWRITTEN AND DATED) TO REFLECT CHANGES. UNLESS OTHERWISE NOTED, UNUSED PANELBOARD CIRCUIT BREAKERS REMAINING FROM REMOVALS SHALL REMAIN FOR FUTURE USE. LABEL SUCH CIRCUITS AS "SPARE". PROVIDE EQUIPMENT LABELS AS NEEDED. PANELBOARD SCHEDULE CIRCUIT DESCRIPTION TO BE UNIQUE PER NEC.
16. LABEL ALL EXISTING AND NEW PROJECT AREA JUNCTION BOXES, CONDULETS, AND RECEPTACLE COVERS WITH SOURCE PANELBOARD NAME(S) AND BRANCH CIRCUIT NUMBER(S).
17. LABEL ALL PROJECT AREA SIGNAL OUTLET COVERS WITH SOURCE EQUIPMENT NAME AND CABLE DESIGNATOR AS APPROPRIATE.
18. CONDUIT TYPES SHALL MATCH EXISTING OR BE AS FOLLOWS FOR PROJECT WIRING:

UNDERGROUND POWER	CONCRETE COVERED PVC
BUILDING POWER	EMT

19. FLEXIBLE METAL CONDUIT MAY BE USED FOR FINAL CONNECTIONS ABOVE DROP CEILINGS AND CONTINUOUSLY IN WALLS IN FINISHED AREAS; OTHERWISE, LIQUID TIGHT FMC SHALL BE USED IN PROCESS AREAS FOR FINAL CONNECTIONS (NO CONTINUOUS RUNS).
20. PROVIDE EMT FOR COMMUNICATION WIRING FOR SIGNAL SYSTEMS (FIRE ALARM, SECURITY, OFFICE PHONE/DATA, AND NETWORK).
21. POWER, CONTROL, SIGNAL, AND COMMUNICATION WIRING SHALL NOT BE RUN IN THE SAME CONDUIT. WIRING WITHIN A SINGLE GROUP MAY SHARE A CONDUIT (UNLESS PROHIBITED BY THE NEC), BUT DIFFERENT WIRING GROUPS CANNOT SHARE A CONDUIT UNLESS EXPLICITLY APPROVED BY THE ENGINEER. WIRING SHALL BE CATEGORIZED ACCORDING TO THE FOLLOWING GROUPS:

GROUP I POWER
120VAC–480VAC

GROUP IIA CONTROL PROCESS
24VAC/VDC, 2–5A; 120VAC, 0–5A

GROUP IIB SIGNAL FA (NEC SEPARATION)
24VDC, 2–5A; 120VAC, 0–5A

GROUP IIC SIGNAL SECURITY
24VDC, 0–2A; VIDEO

GROUP IIIA SIGNAL INSTRUMENTATION (LOW CURRENT)
0–30mA (uA), 24VDC (mV)

GROUP IV COMMUNICATION
ETHERNET, MODBUS, RS485, DH+

22. ALL PROPOSED OUTAGES MUST BE PRE–APPROVED BY THE OWNER AND SCHEDULED SO AS TO MINIMIZE THE DURATION. IF A PROPOSED OUTAGE IS DEEMED UNACCEPTABLE OR TOO LONG BY OWNER, THEN THE CONTRACTOR MUST PROVIDE TEMPORARY WIRING AND/OR GENERATOR(S) TO MAINTAIN POWER TO THE AFFECTED EQUIPMENT AT NO ADDITIONAL COST TO THE OWNER.

MAXIMUM ALLOWED OUTAGE	
LOCATION	HOURS
WELL BUILDINGS	N.A.
PUBLIC WORKS	4
WATER TREATMENT	8

23. THESE GENERAL NOTES SHALL APPLY THROUGHOUT THE DRAWING SET.
24. ANY CONTRACTOR PREMIUM TIME DUE TO THE SCHEDULING OF SERVICE DISRUPTIONS SHALL BE INCLUDED IN THE BID.

PROJECT NOTES

1. ALL PLC/SCADA CONTROL DESIGN AND PROGRAMMING SHALL BE SUB CONTRACTED TO OWNER APPROVED INTEGRATOR AUTOMATIC SYSTEMS COMPANY. INTEGRATOR TO PROVIDE PLC/SCADA HARDWARE, CONTROL SYSTEM DESIGN, INSTRUMENTATION, FIELD DEVICES, MODIFICATIONS TO EXISTING CONTROL SYSTEM PANELS AND STARTUP/TESTING. ELECTRICAL CONTRACTOR TO PROVIDE DISTRIBUTION EQUIPMENT, GENERATORS, AUTOMATIC TRANSFER SWITCHES, CONDUIT, WIRE, TERMINATIONS AND STARTUP ASSISTANCE.
2. POWER TO GENERATOR ACCESSORIES (E.G. BLOCK HEATER, BATTERY CHARGER, ETC) ARE FED FROM UNIT DROP(S) TO GENERATOR MANUFACTURER PROVIDED POWER BLOCK AS NOTED ON DRAWINGS. FUSING AND DISTRIBUTION TO LOADS BY GENERATOR MANUFACTURER.
3. ALL PROJECT WORK REQUIRING MODIFICATIONS TO SWITCHBOARD BUSING, SWITCHES OR STRUCTURE(S) ARE TO BE SUBCONTRACTED TO EMI. CONTACT JIM JUHL AT (763) 546–5998. PROVIDE ALL CERTIFICATIONS (UL, NRTL OR 3RD PARTY) TO MAINTAIN UL LISTING/LABELING OF ANY MODIFIED EQUIPMENT.
4. INSTALL TKDA FURNISHED LAMINATED RISER FOR MAIN SWITCHBOARDS AND PROVIDE 2"X3" ORANGE NAMEPLATE WITH WHITE LETTERING FOR MCC, ATS, GENERATOR AND MAIN BREAKERS. SUBMIT NAMEPLATE PROOF PRINTOUT FOR APPROVAL BY ENGINEER.

ALL CONTRACTORS AND SUBCONTRACTORS SHALL VERIFY ALL DIMENSIONS BY MEASUREMENT AT THE BUILDING AND/OR SITE

NO.	DATE	BY	DESCRIPTION OF REVISIONS

DESIGNED RCS DRAWN RCS CHECKED RLY

I HEREBY CERTIFY THAT THIS PLAN WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MINNESOTA

ROBERT L. YOUNG

DATE 11/14/12 LIC. NO. 26746

TKDA
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Saint Paul, MN 55101
651.292.4400
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STANDBY GENERATOR PROJECT

MAPLE GROVE MINNESOTA

CLIENT NO. MSA NO.

ELECTRICAL SYMBOLS & ABBREVIATIONS

PROJECT NO. 15088.000

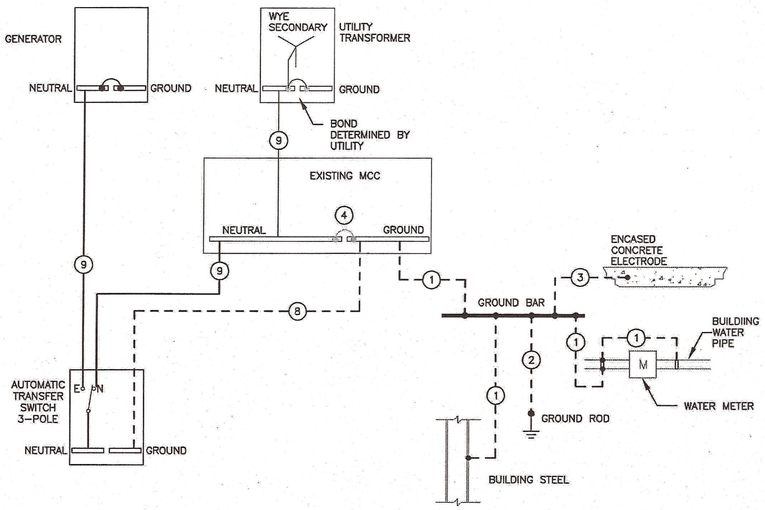
RECORD NO. SHEET NO. E000

FIELD BOOK:
Plot Date: 11/13/2012
Drawing Name: K:\Engrm\Maple Grove\15088000\elec\dwgs\E000-SYM.dwg
Xrefs:

GROUNDING TABLE

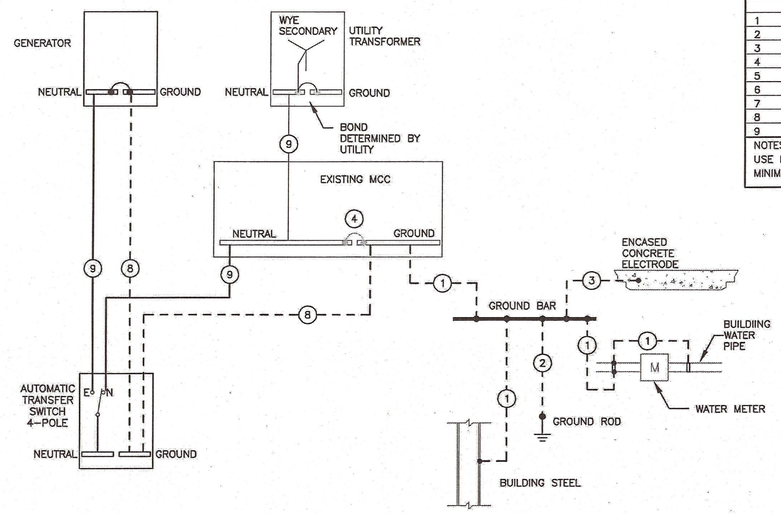
LEAD NO. (X)	DESCRIPTION	SIZE	REFERENCE	REMARKS
1	GROUNDING ELECTRODE CONDUCTOR	SEE DWG/SCHEDULE	PER NEC 250.66	TO BLDG STEEL, WTR MAIN
2	GROUNDING ELECTRODE CONDUCTOR	#6	PER NEC 250.66	TO GND ROD
3	GND RING/ENCASED CONDUCTOR ELECTRODE	#2/0	PER NEC 250.66	
4	MAIN BONDING JUMPER		PER NEC 250.28	
5	COMMON GROUNDING ELECTRODE CONDUCTOR	#3/0	PER NEC 250.30	DERIVED SOURCE COMMON GND
6	EQUIPMENT BONDING JUMPER	SEE DWG/SCHEDULE	PER NEC 250.122	
7	DERIVED SOURCE BONDING CONDUCTOR	SEE DWG/SCHEDULE	PER NEC 250.104	
8	EQUIPMENT GROUNDING CONDUCTOR	SEE DWG/SCHEDULE	PER NEC 250.122	
9	GROUNDED CONDUCTOR (NEUTRAL)	SEE DWG/SCHEDULE	PER NEC 220/250	

NOTES:
USE INSULATED, COPPER WIRE UNLESS GROUNDING WIRE IS BURIED OR ENCASED IN CONCRETE.
MINIMUM WIRE SIZE IS AS SHOWN AND NOT LESS THAN NEC 250.



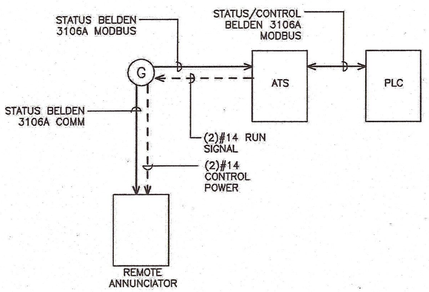
5 E100 TYPICAL 3-WIRE WELL BUILDING PARTIAL GROUNDING RISER DIAGRAM

SCALE: NONE



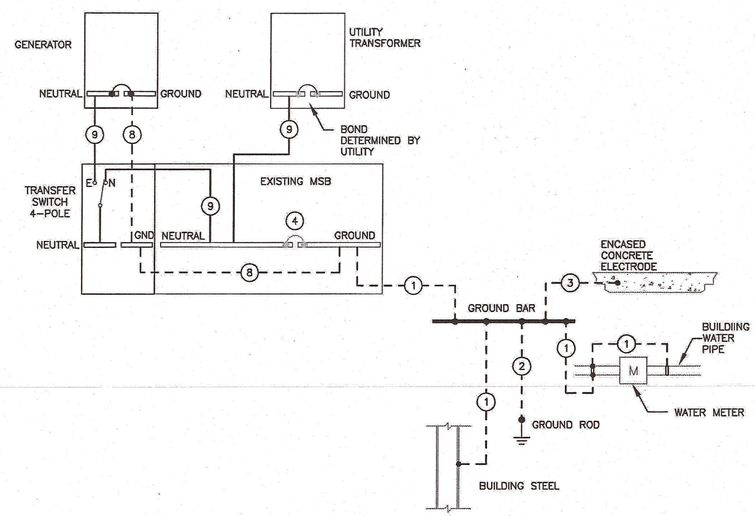
3 E100 TYPICAL 4-WIRE WELL BUILDING PARTIAL GROUNDING RISER DIAGRAM

SCALE: NONE



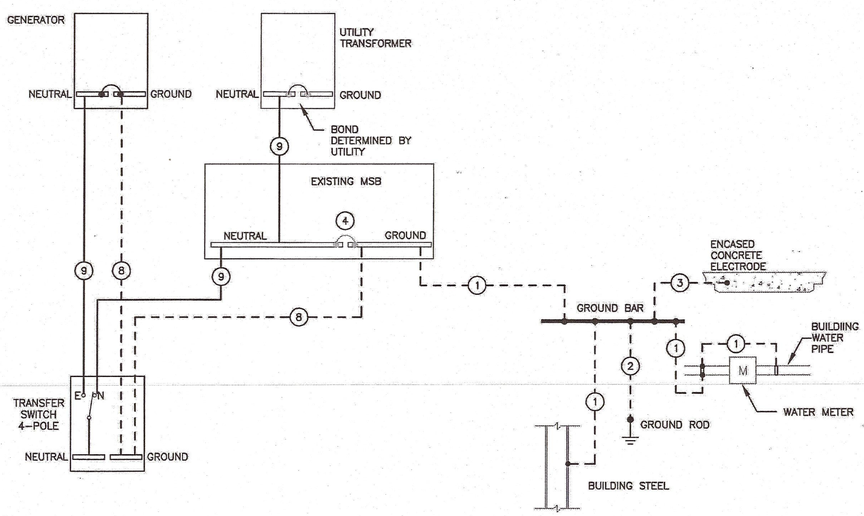
1 E100 SIMPLIFIED GENERATOR/ATS CONTROL/COMM WIRING DIAGRAM

SCALE: NONE



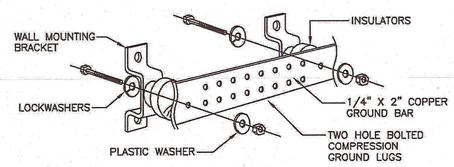
6 E100 GOVERNMENT CENTER PARTIAL GROUNDING RISER DIAGRAM

SCALE: NONE



4 E100 WATER TREATMENT PLANT PARTIAL GROUNDING RISER DIAGRAM

SCALE: NONE



2 E100 GROUND BAR DETAIL

SCALE: NONE

See new page

NO.	DATE	BY	DESCRIPTION OF REVISIONS

DESIGNED RCS DRAWN RCS CHECKED RLY

I HEREBY CERTIFY THAT THIS PLAN WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MINNESOTA

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STANDBY GENERATOR PROJECT
MAPLE GROVE MINNESOTA
CLIENT NO. MSA NO.

ELECTRICAL DETAILS

PROJECT NO. 15088.000
RECORD NO.
SHEET NO. E100

FIELD BOOK:
Plotted: 11/15/2012
Drawing name: K:\Eg-m\Maple Grove\150880000\elec\dwgs\E100-DETAILS.dwg
Xrefs:

STRUCTURAL NOTES:

A. DESIGN AND CONSTRUCTION:

1. DESIGN IN ACCORDANCE WITH THE FOLLOWING CODES:

   a. Minnesota Building Code 2007 (Incorporating 2006 IBC)
   b. A.C.I. 318-05

2. DESIGN LOADS:

   a. Generator and Housing (Estimated):

Well No. 4:	4.2k
Well No's. 6 & 10:	7.3k
Well No. 8:	7.3k
Well No. 9:	7.0k
Well No. 13:	7.0k
Sunny Slope Lift Station:	11.1k
Public Works Building:	28.6k
Water Treatment Plant:	38.5k

   b. Wind Load:
      Basic Wind Speed = 90 Mph
      Exposure = C
      Wind Load Importance Factor, Iw =1.0

3. SPECIAL INSPECTION:

   The Following Items Require Special Inspection (per Section 1704 of the IBC with MN Amendments):
   a. Concrete Cylinder (7 And 28 Day and Spare) And Slump
   b. Concrete Reinforcing

   Provide Minimum 24 Hour Notice To Inspector.

B. FOUNDATION

1. Foundations shall bear on undisturbed, unfrozen subgrade, except where compacted engineered-fill is specified on plans and in specification.

2. Allowable Soil Bearing Pressure = 1000 PSF (assumed).

C. DEMOLITION

1. Contractor shall verify all conditions and dimensions (including utilities and possible presence of hazardous materials) prior to starting work, and shall notify the engineer of any discrepancies or additional work that may be required.

2. Contractor is responsible for all damage to surrounding structures and equipment due to demolition work. This includes required cleanup due to spread of dust and debris. Contractor may wish to document condition of nearby structures prior to beginning demolition as a defense against damage claims.

3. Contractor is responsible for the safety of all workers, tenants, and the public during his work. Barricades, warning signs, traffic control, etc. shall be provided and maintained as required.

D. CONCRETE AND GROUT (CAST-IN-PLACE):

1. Concrete mix designs shall meet the following requirements:
   a. Concrete fc' = 4,000 PSI min. after 28 days.
   b. Maximum slump shall be 3 inches plus or minus 1 inch before addition of superplasticizer, max. 7 inches after.
   c. All concrete, shall be air entrained to 6% plus or minus 1.5%.
   d. Water:cement ratio shall not exceed 0.45.
   e. Admixtures shall not be used without pre-approval. Admixtures containing chlorides shall not be used.
   f. Maximum ratio of Fly-ash to total cementitious materials shall be 0.15.
   g. Concrete mix designs shall be provided to the Engineer for approval prior to concrete installation.

2. Concrete reinforcement shall be ASTM A615, Grade 60 fy = 60,000 PSI (except reinforcement to be welded shall be ASTM A706). All welding of reinforcement shall conform to the structural welding codes, Reinforcing Steel AWS D1.4, current edition.

3. All concrete shown on 's' series drawings shall be reinforced U.N.O. Sections and plans shown without reinforcement are intended to show dimensions and details of construction only. Reinforcement of these sections shall be provided in accordance with details showing reinforcement.

4. Reinforcement detailing requirements:
   a. All reinforcing steel shall have concrete cover as shown in section 1/S100.
   b. Place rebar per CRSI standards.

5. All bars and dowels shall be supported and wired in place. Dowels shall be wired in place, not pushed into fresh concrete. Welded wire fabric shall be positively supported, not pulled up after concrete placement. Bar supports in contact with exposed surfaces shall have plastic tips.

6. Concrete placement requirements:
   a. Concrete shall not be placed on frozen ground.
   b. Mechanically vibrate all concrete when placed.
   c. All exposed edges shall be finished with a standard edger.
   d. Newly placed concrete shall be kept damp on all surfaces for seven full days following the completion of each concrete placement. Curing operations shall commence as soon as it is practical without damaging the concrete surface. Curing methods used shall be appropriate to each situation. Curing shall not stain the concrete.
   e. If used, contractor shall submit curing compound to engineer for approval.

7. For details and location of electrical work, see E-Series and C-Series drawings.

8. Sizes and locations of equipment bases, sumps, and equipment anchor bolts shown on drawings are approximate; the general contractor shall verify all sizes and locations with electrical division.

9. Grout shall be non-shrink, non-metallic, Five Star Products, Inc., Five Star Grout; ASTM C-827, C-191, and C-109, or prior approved equal. Grout shall be mixed and installed per manufacturer's recommendation; minimum compressive strength shall be 5000 PSI in 7 days.

E. GENERAL CONSTRUCTION:

1. The Structural Engineer shall not have control or charge of, and shall not be responsible for, construction means, methods, techniques, sequences or procedures, for safety precautions and programs in connection with the work, for the acts or omissions of the contractor, subcontractors or any other persons performing any of the work in accordance with the contract documents.

2. Any temporary facilities placed by the contractor for construction operations, such as crane bases, trailers, etc., shall be located so as not to interfere with permanent building construction. If interference occurs, contractor shall remove or relocate his temporary facilities at his own expense.

3. The Contractor is responsible for the coordination of the work of all subcontractors.

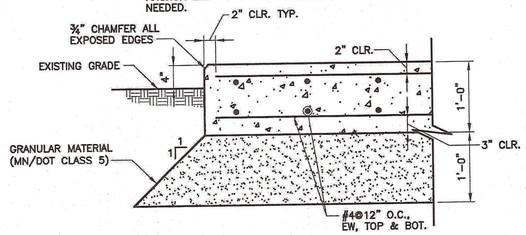
4. Contractor shall be aware of overhead power lines, subgrade utilities or tunnels at the site. Verify utility locations with Gopher State One-call (651-454-0002) and utility companies.

5. Any engineering design provided by others and submitted for review shall bear the signature and valid registration number of a professional engineer licensed in the state of Minnesota.

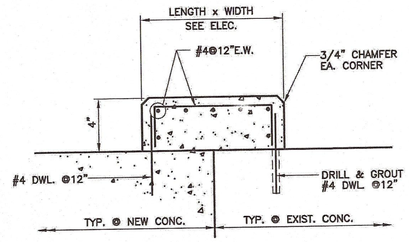
6. Unless otherwise noted, details on structural drawings are typical as indicated by cuts, references or titles.

7. Contractor shall field verify all existing conditions and dimensions at the buildings and/or sites.

NOTE:
ANCHORAGE OF NEW EQUIPMENT TO PAD BY SUPPLIER. ANCHORAGE OF RELOCATED EQUIPMENT TO MATCH EXISTING. MAINTAIN MINIMUM EDGE DISTANCE EQUAL TO ANCHOR EMBED. DEPTH. OVERSIZE PAD AS NEEDED.



1/S100 PAD SECTION
SCALE: 1" = 1'-0"



2/S100 EQUIPMENT BASE DETAIL
NO SCALE

Pad dimensions

LOCATION		PAD DIMENSIONS	REINFORCEMENT	DETAIL	CIVIL PLAN SHEET
WELL NO. 4	168 x 78 = 13,104	6'-6" x 14'-0" x 1'-0"	#4 @ 12" O.C., E.W., TOP & BOT	1/S100	C101-1
WELLS NOS. 6 & 10	170 x 80 = 13600	6'-8" x 14'-2" x 1'-0"	#4 @ 12" O.C., E.W., TOP & BOT	1/S100	C102-1
WELL NO. 8	254" x 87" = 22098	7'-3" x 21'-2" x 1'-0"	#4 @ 12" O.C., E.W., TOP & BOT	1/S100	C103-1
WELL NO. 9	140 x 68 = 9520	5'-8" x 11'-8" x 1'-0"	#4 @ 12" O.C., E.W., TOP & BOT	1/S100	C104-1
WELL NO. 13	206 x 77 = 15,862	6'-5" x 17'-2" x 1'-0"	#4 @ 12" O.C., E.W., TOP & BOT	1/S100	C105-1
SUNNY SLOPE LIFT STATION	156 x 102 = 15,912	8'-6" x 13'-0" x 1'-0"	#4 @ 12" O.C., E.W., TOP & BOT	1/S100	C106-1
PUBLIC WORKS BUILDING	255 x 99 = 25,245	8'-3" x 21'-3" x 1'-0"	#4 @ 12" O.C., E.W., TOP & BOT	1/S100	C107-1
WATER TREATMENT PLANT GENERATOR	438+150=588	12'-6" x 36'-6" x 1'-0"	#4 @ 12" O.C., E.W., TOP & BOT	1/S100	C108-1
WATER TREATMENT PLANT CONNECTION CABINET		3'-8" x 4'-10" x 1'-0"	#4 @ 12" O.C., E.W., TOP & BOT	1/S100	C108-1
NOTE: PAD DIMENSIONS FOR BIDDING PURPOSES ONLY, BASED ON SINGLE SUPPLIER ESTIMATES. CONTRACTOR TO VERIFY PAD DIMENSIONS FOLLOWING EQUIPMENT SELECTION.					

Well 13 - pad for connect, cab. & transfer sw

3/100 EQUIPMENT PAD SCHEDULE

Need
↓
119" x 68" = 8092
164" x 74" = 12,136
194" x 98" = 19,012
143" x 68" = 9724
178" x 90" = 15,390
144" x 90" = 12,960
194" x 104" = 20,176
411" x 191" = 57,957

39,998 yards = 155,441 sq ft

26,600 less 5%
÷ 144
177 cu' less
÷ 27
6.58 yards less

181,041 Sq"

46.6 Yards

1	11/14/12	FPB	ISSUED FOR BIDDING	
NO.	DATE	BY	DESCRIPTION OF REVISIONS	

DESIGNED FPB | DRAWN MDT | CHECKED TGA

I HEREBY CERTIFY THAT THIS PLAN WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MINNESOTA
FREDERICK P.R. BEUKEMA
DATE 11/14/12 LIC. NO. 47949

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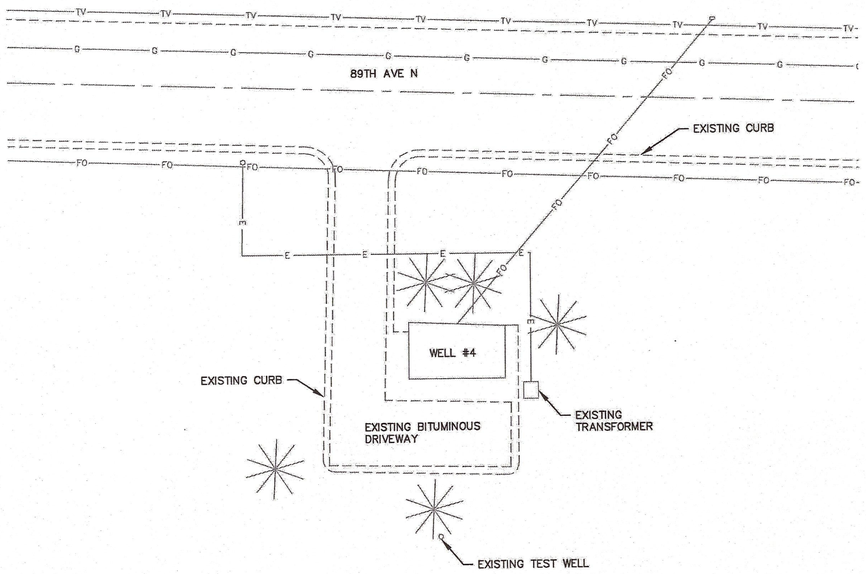
tkda.com

STANDBY GENERATOR PROJECT
MAPLE GROVE MINNESOTA
CLIENT NO. | MSA NO.

STRUCTURAL NOTES & DETAILS

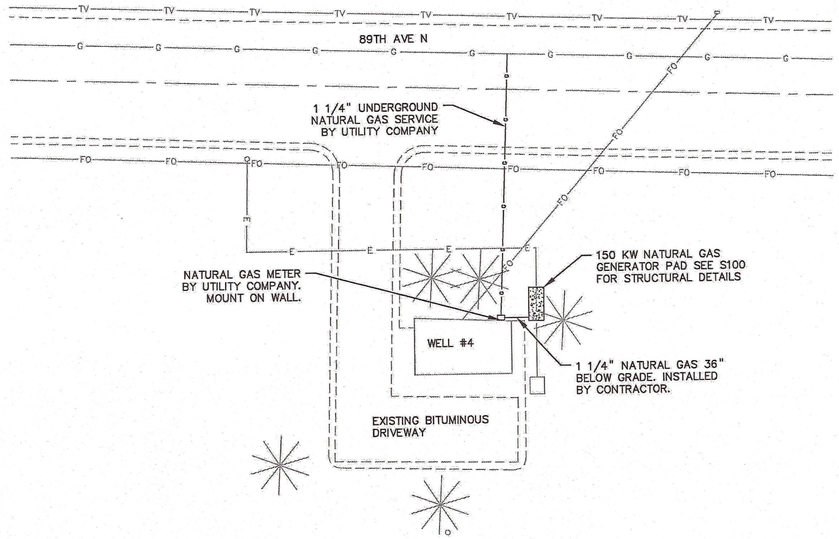
PROJECT NO. 15088.000
RECORD NO. | SHEET NO. S100

FIELD BOOK:
Plotted: 11/14/2012
Drawing name: K:\grp\MapleGrove\15088000\struct\dwgs\S100.dwg
Xrefs: FPB



1/101 EXISTING PLAN

SCALE: 1" = 30'



2/101 PROPOSED PLAN

SCALE: 1" = 30'

NOTES:

1. USE FLEXIBLE GAS CONNECTOR BETWEEN GAS SERVICE LINE AND GENERATOR.
2. GAS PRESSURE REQUIRED AT GENERATOR IS 11"–15" C.C.

LEGEND	
T	TELECOMMUNICATION
G	NATURAL GAS
E	ELECTRIC
TV	CABLE TELEVISION
FO	FIBER OPTIC

N

TKDA

NO.	DATE	BY	DESCRIPTION OF REVISIONS

DESIGNED JRN | DRAWN DCK | CHECKED VEJ

I HEREBY CERTIFY THAT THIS PLAN WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MINNESOTA

VERNE E. JACOBSEN

DATE 11/14/12 LIC. NO. 06988

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STANDBY GENERATOR PROJECT

MAPLE GROVE MINNESOTA

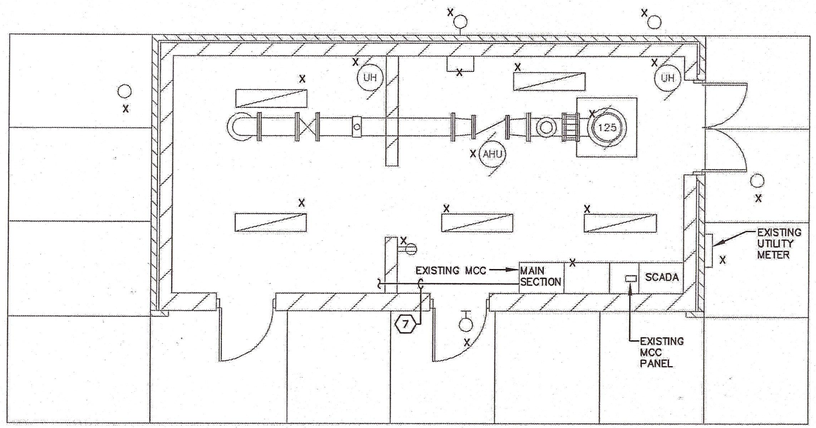
CLIENT NO. MSA NO.

WELL NO. 4

PROJECT NO. 15088.000

RECORD NO. SHEET NO. C101-1

FIELD BOOK:
Plot Date: 11/13/2012
Drawing name: K:\g-m\MapleGrove\15088000\mun\dwgs\PlanSet\C101.dwg
Xrefs: Well #4_ExistingBase



EXISTING UTILITY TRANSFORMER

1 E101 EXISTING PLAN

SCALE: 1/4" = 1'

0 2 4 8

SCALE IN FEET

GENERAL NOTES

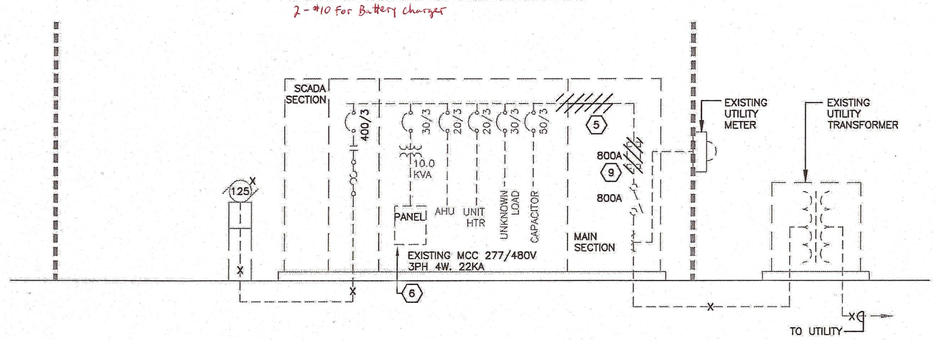
1. ALL RACEWAYS SHALL CONTAIN A GROUNDING CONDUCTOR.
2. CONDUIT ROUTING SHOWN FOR INFORMATION ONLY. VERIFY LOCATION OF CONDUIT STUB UPS WITH GENERATOR MANUFACTURER.
3. VERIFY LOCATION OF ALL EXISTING SITE UTILITIES PRIOR TO EXCAVATION.
4. PROVIDE PLASTIC LAMINATE NAMEPLATE ADJACENT TO THE GENERATOR CIRCUIT BREAKER WHICH READS: "THIS GENERATOR SERVES WELLHOUSE 4, PUMPS AND ALL ASSOCIATED BUILDING LOADS"

KEY NOTES

1. AUTOMATIC TRANSFER SWITCH TO BE 480 VOLT, 400 AMP, 4 POLE OPEN TRANSITION TYPE EQUAL TO ASCO 7000 SERIES WITH GROUP 17 ACCESSORY AND RATED AT 22,000 AIC MIN. IN NEMA 1 ENCLOSURE. SEE SPECIFICATIONS.
2. GENERATOR REMOTE MONITOR PANEL.
3. PROVIDE GENERATOR CONCRETE MOUNTING PAD. SEE STRUCTURAL DETAILS.
4. PROVIDE 3/4" CONDUIT WITH (1) BELDEN #3106A COMMUNICATION CABLE FOR MODBUS COMMUNICATIONS BETWEEN THE TRANSFER SWITCH AND THE PLC. SUBCONTRACT TO THE CITY'S SCADA SYSTEM INTEGRATOR. INTEGRATOR SHALL PROVIDE ALL PROGRAMMING AND HARDWARE REQUIRED TO MONITOR AND DISPLAY THE GENERATOR STATUS AND ALARMS. INTEGRATOR SHALL PROVIDE NEW SCADA SYSTEM SCREEN AT THE PUBLIC WORKS AND WATER TREATMENT PLANT BUILDING HMI TO DISPLAY THE GENERATOR AND TRANSFER SWITCH STATUS AND ALARMS. NEW SCREEN WILL ALLOW START-STOP OF THE GENERATOR AND REMOTE CONTROL OF THE AUTOMATIC TRANSFER SWITCH WITH PASSWORD PROTECTION. INTEGRATOR SHALL MODIFY OR REPLACE THE EXISTING PLC AT THE WELL TO PROVIDE MODBUS COMMUNICATIONS.
5. MODIFY EXISTING BUS TO ACCOMMODATE NEW FEEDERS TO AND FROM NEW AUTOMATIC TRANSFER SWITCH. PROVIDE REQUIRED NEW BUSSING, LUGS AND MODIFICATIONS FOR A COMPLETE INSTALLATION.
6. REMOVE (2) EXISTING SPARE CIRCUIT BREAKERS. PROVIDE 40/2 CIRCUIT BREAKER AT EXISTING PANEL. PANEL IS WESTINGHOUSE. BREAKER IS TYPE 'BA'.
7. REMOVE UNSUSED EXISTING ABANDONED CONDUIT
8. PROVIDE WEATHERPROOF PULLBOX. SIZE PER NEC.
9. REMOVE EXISTING 800A FUSES. REPLACE WITH 400A CURRENT LIMITING FUSES.
10. PROVIDE 1"PVC CONDUIT WITH (2) #14 FOR GENERATOR MONITOR PANEL POWER, (2) #14 FOR GENERATOR START SIGNAL, (2) BELDEN #3106A COMMUNICATION CABLE FOR MODBUS COMMUNICATIONS BETWEEN THE GENERATOR AND THE TRANSFER SWITCH AND BETWEEN THE GENERATOR AND REMOTE MONITOR PANEL.

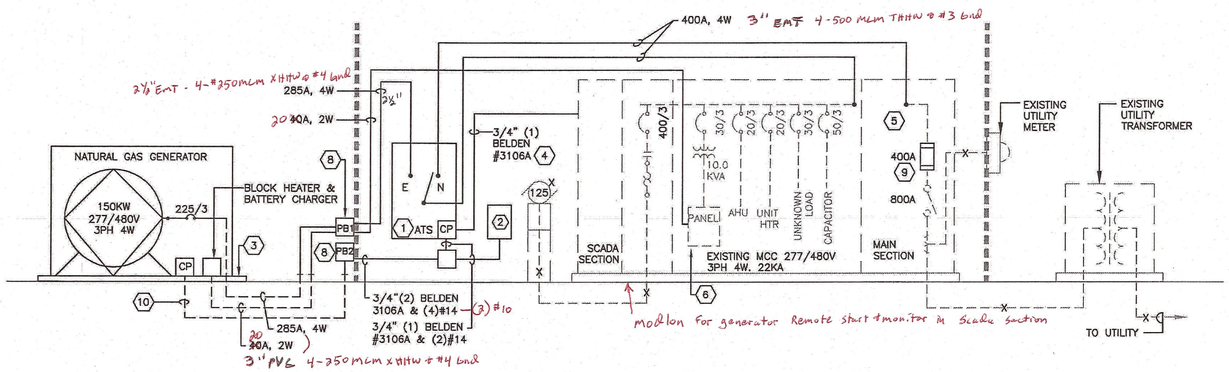
2 - #10 for Battery charger

N



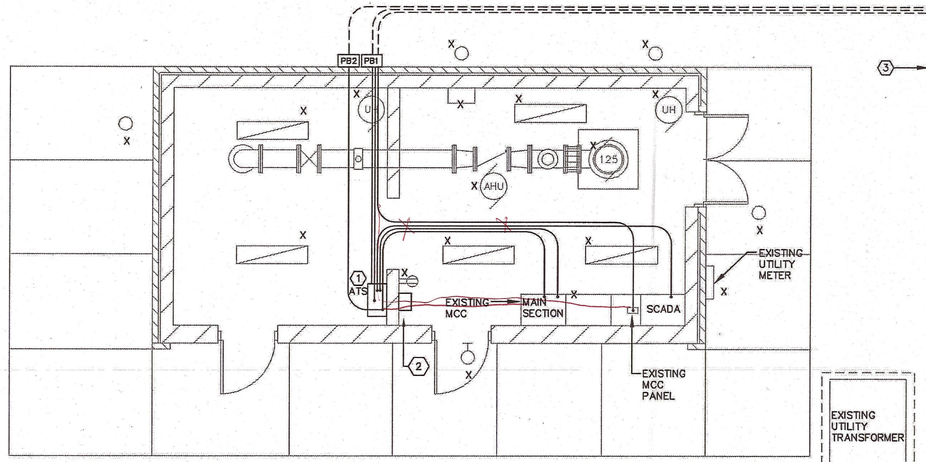
3 E101 EXISTING RISER DIAGRAM

SCALE: NONE



4 E101 NEW RISER DIAGRAM

SCALE: NONE



EXISTING UTILITY TRANSFORMER

2 E101 NEW PLAN

SCALE: 1/4" = 1'

0 2 4 8

SCALE IN FEET

NO.	DATE	BY	DESCRIPTION OF REVISIONS

DESIGNED RCS DRAWN RCS CHECKED RLY

I HEREBY CERTIFY THAT THIS PLAN WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MINNESOTA

ROBERT L. YOUNG

DATE 11/14/12 LIC. NO. 26746

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tkda.com

STANDBY GENERATOR PROJECT

MAPLE GROVE MINNESOTA

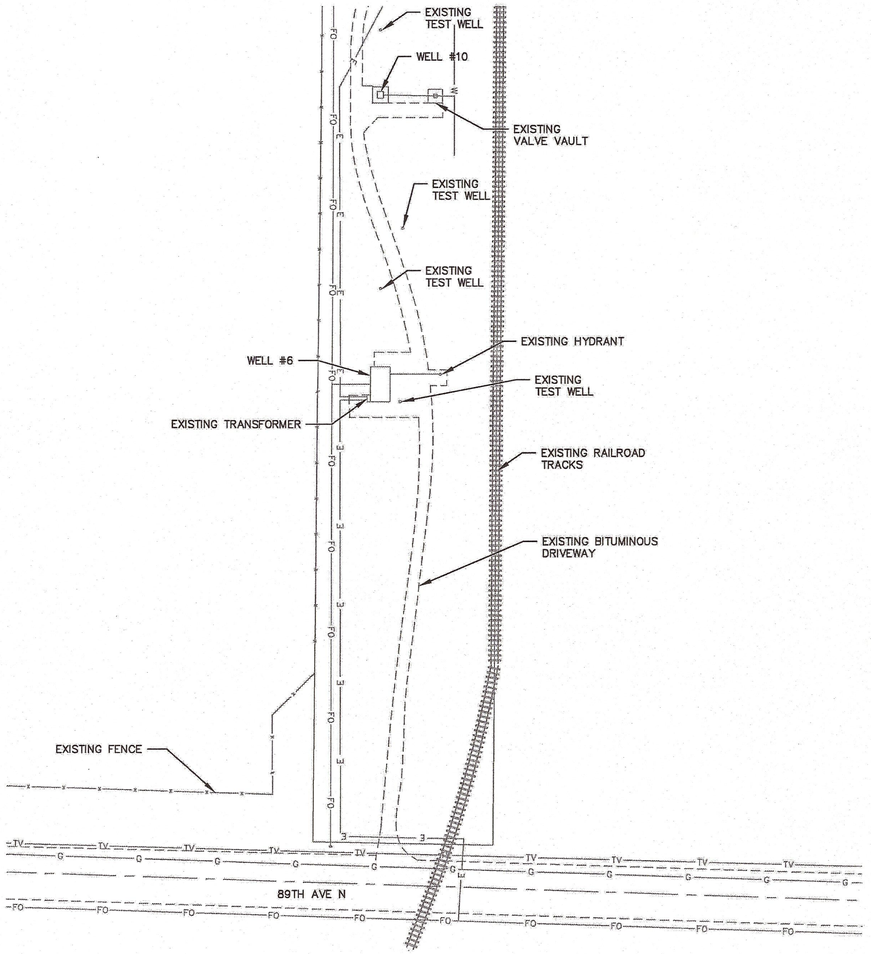
CLIENT NO. MSA NO.

WELL #4 ELECTRICAL PLAN

PROJECT NO. 15088.000

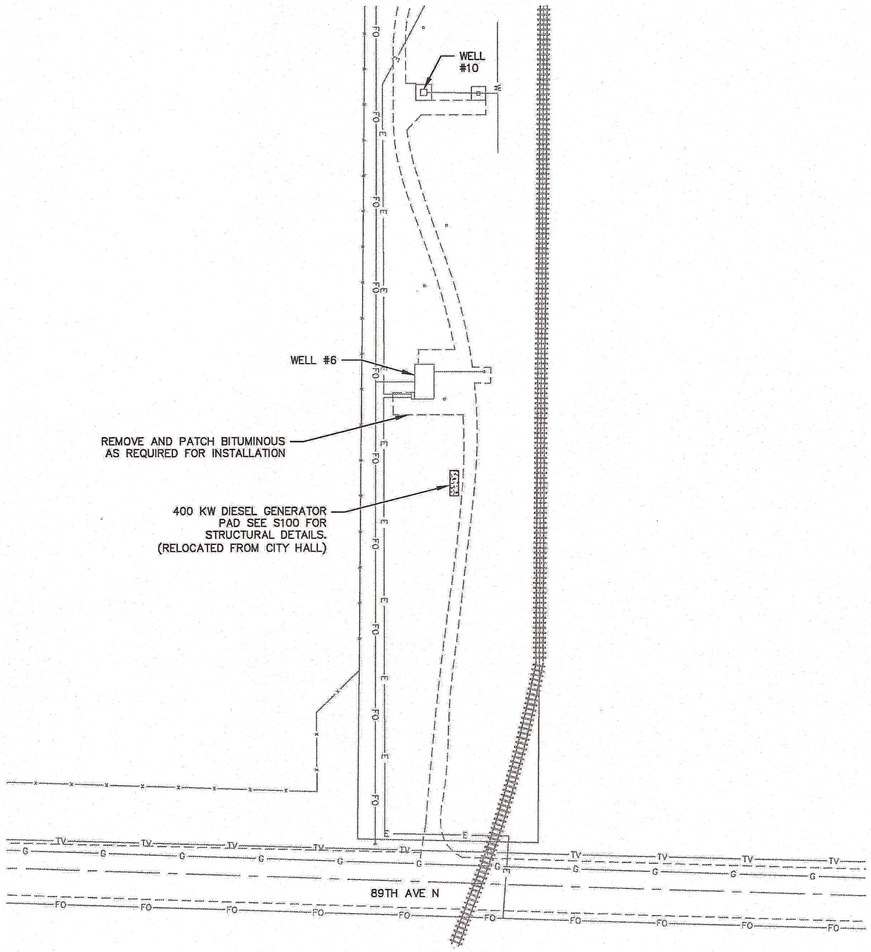
RECORD NO. SHEET NO. E101-1

FIELD BOOK:
Plot Date: 11/13/2012
Drawing name: K:\g-m\MapleGrove\15088000\elec\dwgs\E101-1-PH4.dwg
Xrefs:, Well #4_FLOORPLAN



1/103 EXISTING PLAN

SCALE: 1" = 60'



2/103 PROPOSED PLAN

SCALE: 1" = 60'

LEGEND

- T — TELECOMMUNICATION
- G — NATURAL GAS
- E — ELECTRIC
- TV — CABLE TELEVISION
- FO — FIBER OPTIC

N

TKDA

NO.	DATE	BY	DESCRIPTION OF REVISIONS

DESIGNED JRN | DRAWN DCK | CHECKED VEJ

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VERNE E. JACOBSEN

DATE 11/14/12 LIC. NO. 06988

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tkda.com

STANDBY GENERATOR PROJECT

MAPLE GROVE MINNESOTA

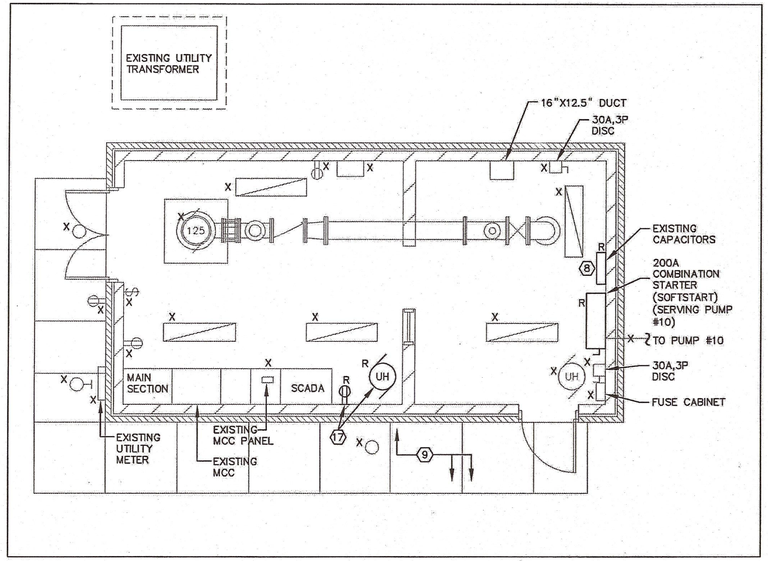
CLIENT NO. | MSA NO.

WELLS NOS. 6 & 10

PROJECT NO. 15088.000

RECORD NO. | SHEET NO. C102-1

FIELD BOOK:
Plot Date: 11/13/2012
Drawing name: K:\pw\mn\MapleGrove\15088000\mxm\dwgs\PlanSet\C102.dwg
Xrefs: Well #6-9-10_Exists@Base

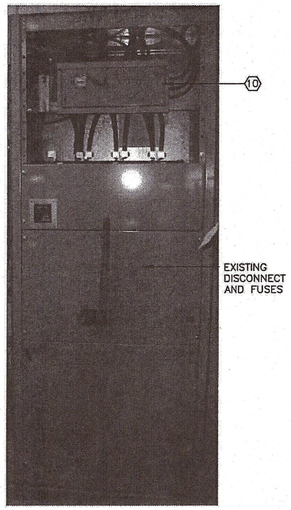


1 E102 EXISTING PLAN

SCALE: 1/4" = 1'

0 2 4 8

SCALE IN FEET



5 E102 EXISTING MCC–MAIN SECTION

SCALE: NONE

KEY NOTES

12 EXISTING 400KW GENERATOR RELOCATED BY ELECTRICAL CONTRACTOR FROM CITY HALL LOCATION. PROVIDE AUXILLIARY RELAY AS REQUIRED FOR BOTH PLC & REMOTE MONITOR.

13 PROVIDE MOTOR PROTECTION FILTER TCI, INC MODEL #V1K250A01 IN NEMA 1 ENCLOSURE OR EQUIVALENT.

14 PROVIDE NEW DISTRIBUTION PANEL EQUAL TO SQUARE D I-LINE 800A 277/480V 3PH 4W WITH DISTRIBUTION BREAKERS OR EQUIVALANT AS SHOWN. RECONNECT EXISTING LOADS AND MOTOR TO NEW DISTRIBUTION PANEL. PROVIDE ADDITIONAL LUGS AT EXISTING BUS IF REQUIRED. MATCH EXISTING SWITCHBOARD AIC RATING.

15 DISCONNECT BRANCH CIRCUIT SERVING EXISTING 200HP MOTOR.

16 RECONNECT BRANCH CIRCUIT TO NEW VFD AND FILTER.

17 REMOVE EXISTING ELECTRIC HEATER AND RECEPTACLE TO ACCOMMODATE NEW DISTRIBUTION PANEL. TURN OVER REMOVED EQUIPMENT TO OWNER.

18 PROVIDE (10) #14 WIRES FROM THE SCADA SYSTEM TO THE GENERATOR CONTROL PANEL. CONNECT GENERATOR RUN AND FAULT SIGNALS TO SCADA SYSTEM PLC INPUTS. SUBCONTRACT WITH THE CITY'S SCADA SYSTEM INTEGRATOR. REMAINING CONDUCTORS SHALL BE MARKED AS SPARE.

19 PROVIDE GENERATOR CONCRETE MOUNTING PAD. SEE STRUCTURAL DRAWINGS.

20 GENERATOR SUPPLY DISCONNECT. PROVIDE 480V, 600A 3 POLE CIRCUIT BREAKER IN NEMA 1 ENCLOSURE.

21 3/4" EMPTY CONDUIT FOR FUTURE USE.

22 REMOVE EXISTING 10KVA TRANSFORMER, PRIMARY AND SECONDARY CIRCUIT BREAKER.

23 PROVIDE NEW WALL MTD. 15KVA 480V TO 120/240V 1 PHASE DRY TYPE TRANSFORMER. PROVIDE NEW 40/2 PRIMARY AND 80/2 SECONDARY PROTECTION.

1 INSTALL OWNER FURNISHED 1000A ZENITH TRANSFER SWITCH RELOCATED BY ELECTRICAL CONTRACTOR FROM CITY HALL LOCATION.

2 INSTALL OWNER FURNISHED GENERATOR REMOTE MONITOR PANEL RELOCATED FROM CITY HALL LOCATION. PROVIDE (10)#14 WIRES FOR STATUS/FAULT SIGNALS.

3 PROVIDE CONCRETE EQUIPMENT PADS TO MATCH EXISTING HOUSEKEEPING PAD.

4 PROVIDE 3/4" CONDUIT WITH (1) BELDEN #3106A COMMUNICATION CABLE FOR MODBUS COMMUNICATIONS BETWEEN THE TRANSFER SWITCH AND THE PLC. SUBCONTRACT WITH THE CITY'S SCADA SYSTEM INTEGRATOR. INTEGRATOR SHALL PROVIDE ALL PROGRAMMING AND HARDWARE REQUIRED TO MONITOR AND DISPLAY THE GENERATOR STATUS AND ALARMS. INTEGRATOR SHALL PROVIDE NEW SCADA SYSTEM SCREEN AT THE PUBLIC WORKS AND WATER TREAMENT BUILDING HMI TO DISPLAY THE GENERATOR AND TRANSFER SWITCH STATUS AND ALARMS. NEW SCREEN WILL ALLOW START–STOP OF THE GENERATOR AND REMOTE CONTROL OF THE AUTOMATIC TRANSFER SWITCH WITH PASSWORD PROTECTION. INTEGRATOR SHALL MODIFY OR REPLACE THE EXISTING PLC AT THE WELL TO PROVIDE MODBUS COMMUNICATIONS.

5 REMOVE EXISTING FEEDER FROM LOAD SIDE OF MAIN FUSED SWITCH. MODIFY EXISTING BUS TO ACCOMMODATE NEW FEEDERS TO AND FROM NEW AUTOMATIC TRANSFER SWITCH, PROVIDE REQUIRED LUGS AND MODIFICATIONS FOR A COMPLETE INSTALLATION.

6 PROVIDE 200 HP VFD WITH ~~BY-PASS~~ AND ISOLATION SWITCH. MANUFACTURER SHALL BE DANFOSS. PROVIDE CONDUIT AND CONTROL WIRING BETWEEN NEW VFD AND AND EXISTING PLC.

7 INTEGRATOR SHALL PROVIDE HARDWARE AND SOFTWARE MODIFICATIONS TO EXISTING PLC FOR VFD CONTROL OF EXISTING PUMP #10.

8 EXISTING CAPACITOR BANK AND SOFT START TO BE REMOVED TO ACCOMMODATE NEW VFD. MAINTAIN EXISTING WIRING TO MOTOR FOR RECONNECTION AND REUSE. TURN OVER REMOVED EQUIPMENT TO OWNER.

9 REMOVE A PARTIAL SECTION OF EXISTING CONCRETE SIDEWALK TO ACCOMMODATE TRENCHING FOR NEW GENERATOR CONDUITS. REPLACE CONCRETE AFTER WORK HAS BEEN COMPLETED.

10 REMOVE EXISTING 100A DISCONNECT SWITCH AND FEEDER SERVING MISCELLANEOUS LOADS AND PANEL.

11 PROVIDE 40/2 CIRCUIT BREAKER AT EXISTING PANEL. FOR GENERATOR. IN ADDITION REUSE EXISTING 30/2 CIRCUIT BREAKER FOR GENERATOR AND PROVIDE BRANCH CIRCUITS AS NOTED.

GENERAL NOTES

1. ALL RACEWAYS SHALL CONTAIN A GROUNDING CONDUCTOR.
2. CONDUIT ROUTING SHOWN FOR INFORMATION ONLY. VERIFY LOCATION OF CONDUIT STUB UPS WITH GENERATOR MANUFACTURER.
3. VERIFY LOCATION OF ALL EXISTING SITE UTILITIES PRIOR TO EXCAVATION.
4. PROVIDE PLASTIC LAMINATE NAMEPLATE ADJACENT TO THE GENERATOR CIRCUIT BREAKER WHICH READS: "THIS GENERATOR SERVES WELLHOUSE 6, PUMPS 6 & 10 AND ALL ASSOCIATED BUILDING LOADS"

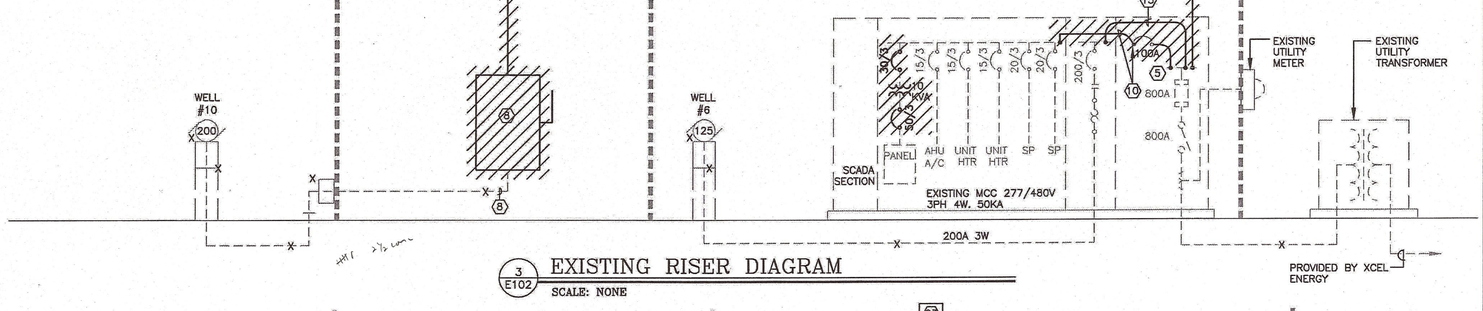
N

2 E102 NEW PLAN

SCALE: 1/4" = 1'

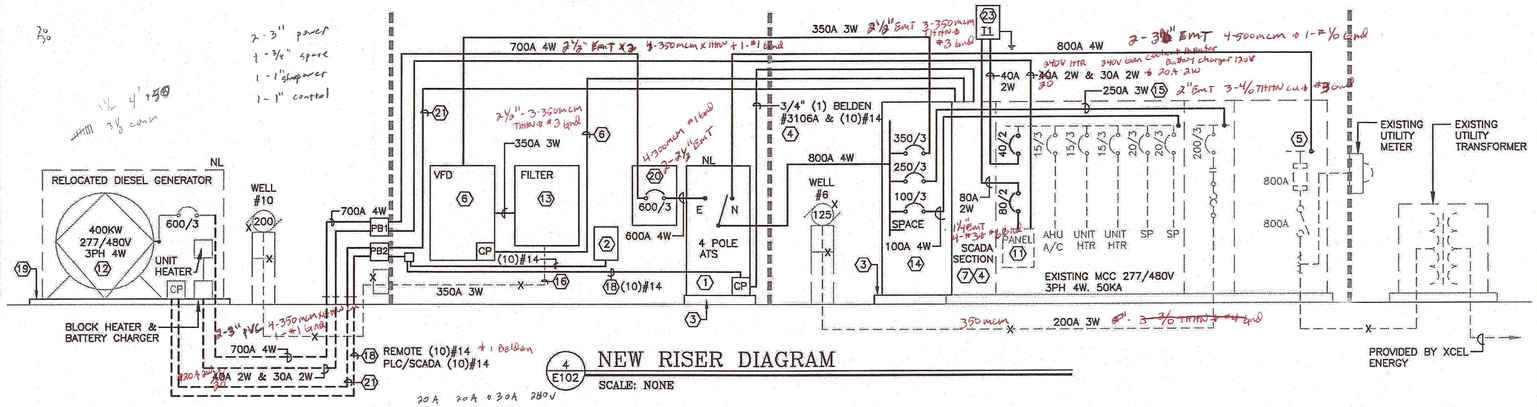
0 2 4 8

SCALE IN FEET



3 E102 EXISTING RISER DIAGRAM

SCALE: NONE



4 E102 NEW RISER DIAGRAM

SCALE: NONE

NO.	DATE	BY	DESCRIPTION OF REVISIONS

DESIGNED RCS DRAWN RCS CHECKED RLY

I HEREBY CERTIFY THAT THIS PLAN WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MINNESOTA

ROBERT L. YOUNG

DATE 11/14/12 LIC. NO. 26746

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STANDBY GENERATOR PROJECT

MAPLE GROVE MINNESOTA

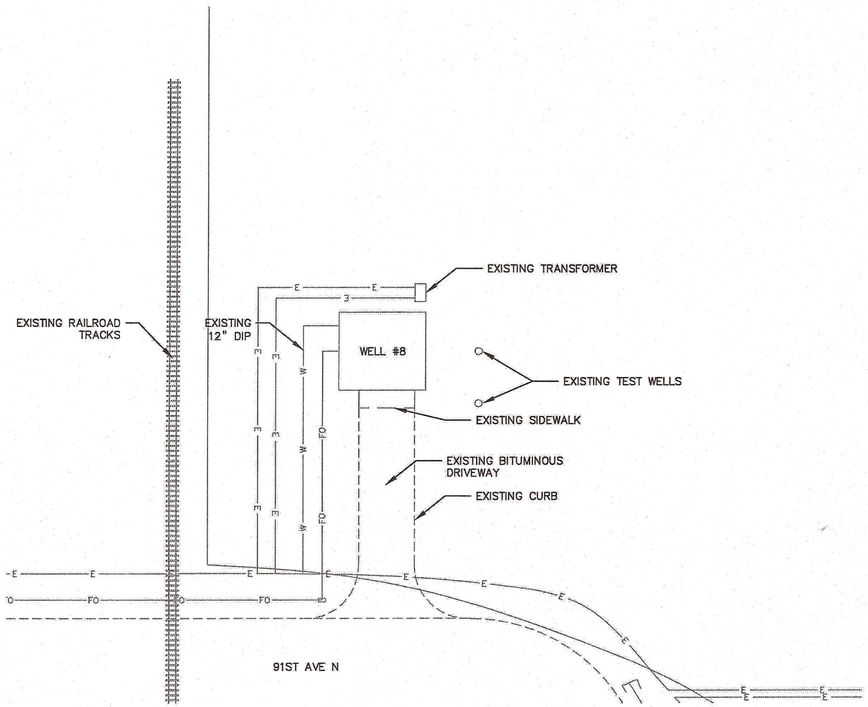
CLIENT NO. MSA NO.

WELL #6 & 10 ELECTRICAL PLAN

PROJECT NO. 15088.000

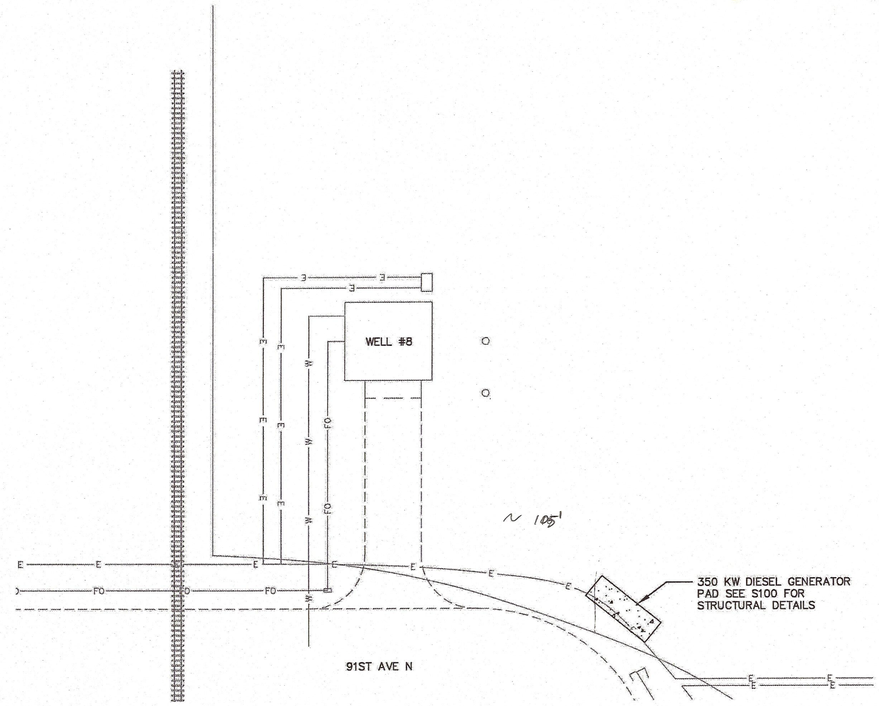
RECORD NO. SHEET NO. E102-1

FIELD BOOK:
PLOT: Nov 13, 2012
Drawing name: K:\jobs\MapleGrove\15088000\elec\dwgs\E102-1-PHB.dwg
Xrefs: Well #6-10_FLOORPLAN



1/105 EXISTING PLAN

SCALE: 1" = 20'



2/105 PROPOSED PLAN

SCALE: 1" = 20'

LEGEND	
T	TELECOMMUNICATION
G	NATURAL GAS
E	ELECTRIC
TV	CABLE TELEVISION
FO	FIBER OPTIC

N

LDKA

NO.	DATE	BY	DESCRIPTION OF REVISIONS

DESIGNED JRN | DRAWN DCK | CHECKED VEJ

I HEREBY CERTIFY THAT THIS PLAN WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MINNESOTA

VERNE E. JACOBSEN

DATE 11/14/12 LIC. NO. 06988

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Saint Paul, MN 55101
651.292.4400
tkda.com

STANDBY GENERATOR PROJECT

MAPLE GROVE MINNESOTA

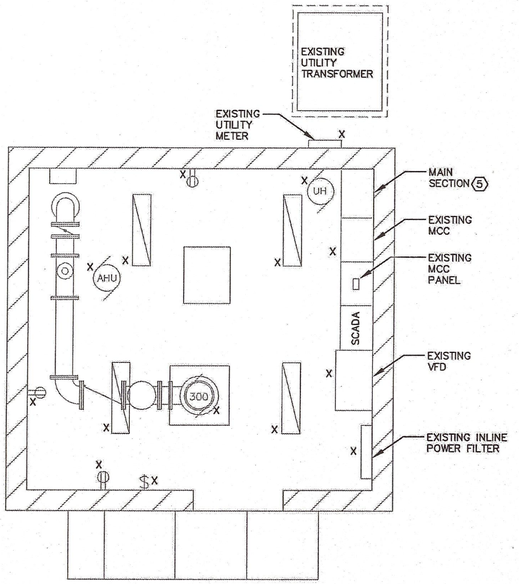
CLIENT NO. MSA NO.

WELL NO. 8

PROJECT NO. 15088.000

RECORD NO. SHEET NO. C103-1

FIELD BOOK:
Plot Date: 11/13/2012
Drawing Name: K:\p-m\MapleGrove\15088000\mun\dwgs\PlanSet\C103.dwg
Xrefs: Veh #2_ExistingBase

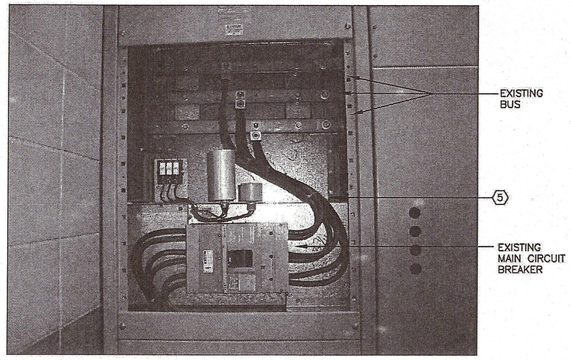


1 E103 EXISTING PLAN

SCALE: 1/4" = 1'

0 2 4 8

SCALE IN FEET



5 E103 EXISTING MCC – MAIN SECTION

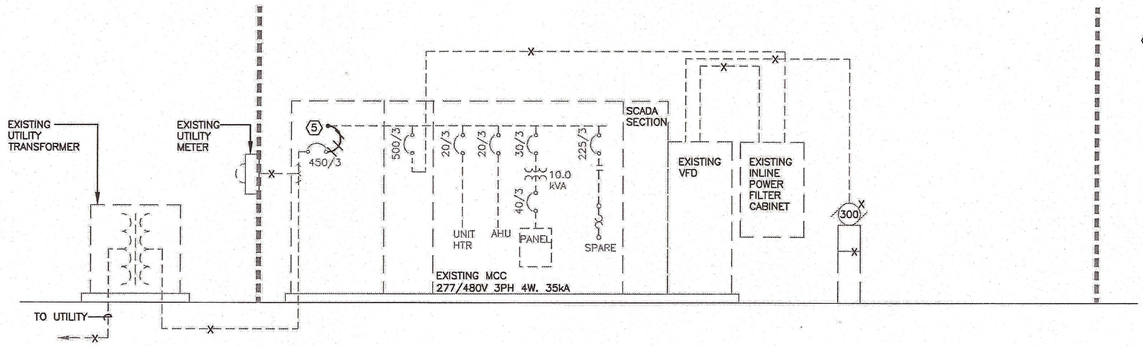
SCALE: NONE

GENERAL NOTES

1. ALL RACEWAYS SHALL CONTAIN A GROUNDING CONDUCTOR.
2. CONDUIT ROUTING SHOWN FOR INFORMATION ONLY. VERIFY LOCATION OF CONDUIT STUB UPS WITH GENERATOR MANUFACTURER.
3. VERIFY LOCATION OF ALL EXISTING SITE UTILITIES PRIOR TO EXCAVATION.
4. PROVIDE PLASTIC LAMINATE NAMEPLATE ADJACENT GENERATOR TO THE CIRCUIT BREAKER WHICH READS: "THIS GENERATOR SERVES WELLHOUSE 8, PUMPS AND ALL ASSOCIATED BUILDING LOADS"

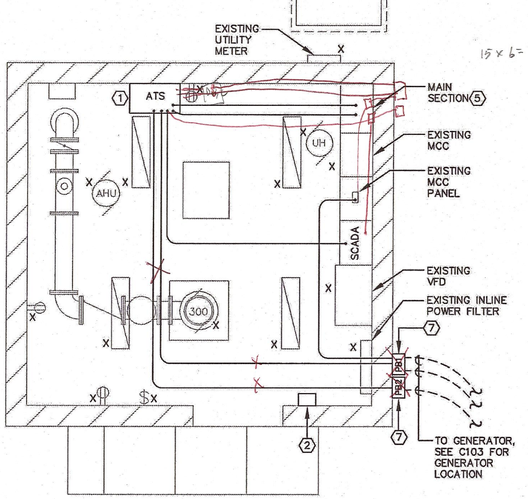
KEY NOTES

1. AUTOMATIC TRANSFER SWITCH TO BE 480 VOLT, 600 AMP, 4 POLE OPEN TRANSITION TYPE EQUAL TO ASCO 7000 SERIES WITH GROUP 17 ACCESSORY AND RATED AT 35,000 AIC MIN. IN NEMA 1 ENCLOSURE. SEE SPECIFICATIONS.
2. GENERATOR REMOTE MONITOR PANEL.
3. PROVIDE GENERATOR CONCRETE MOUNTING PAD. SEE STRUCTURAL DETAILS.
4. PROVIDE 3/4" CONDUIT WITH (1) BELDEN #3106A COMMUNICATION CABLE FOR MODBUS COMMUNICATIONS BETWEEN THE TRANSFER SWITCH AND THE PLC. SUBCONTRACT TO THE CITY'S SCADA SYSTEM INTEGRATOR. INTEGRATOR SHALL PROVIDE ALL PROGRAMMING AND HARDWARE REQUIRED TO MONITOR AND SCREEN THE GENERATOR STATUS AND ALARMS. INTEGRATOR SHALL PROVIDE NEW SCADA SYSTEM DISPLAY AT THE PUBLIC WORKS AND WATER TREATMENT BUILDING HMI TO DISPLAY THE GENERATOR AND TRANSFER SWITCH STATUS AND ALARMS. NEW SCREEN WILL ALLOW START-STOP OF THE GENERATOR AND REMOTE CONTROL OF THE AUTOMATIC TRANSFER SWITCH WITH PASSWORD PROTECTION. INTEGRATOR SHALL MODIFY OR REPLACE THE EXISTING PLC AT THE WELL TO PROVIDE MODBUS COMMUNICATIONS.
5. REMOVE EXISTING FEEDERS FROM LOAD SIDE MAIN CIRCUIT BREAKER TO EXISTING BUS. MODIFY EXISTING BUS TO ACCOMMODATE NEW FEEDERS TO AND FROM NEW AUTOMATIC TRANSFER SWITCH, PROVIDE REQUIRED NEW LUGS AND MODIFICATIONS FOR A COMPLETE INSTALLATION.
6. PROVIDE 40/2 CIRCUIT BREAKER AT EXISTING PANEL. PANEL IS SIEMENS I-T-E. RECONFIGURE EXISTING CIRCUIT BREAKERS TO ACCOMMODATE NEW 40/2 CIRCUIT BREAKER.
7. PROVIDE WEATHERPROOF PULLBOX. SIZE PER NEC.
8. PROVIDE 1"PVC CONDUIT WITH (2) #14 FOR GENERATOR MONITOR PANEL POWER, (2) #14 FOR GENERATOR START SIGNAL, (2) BELDEN #3106A COMMUNICATION CABLE FOR MODBUS COMMUNICATIONS BETWEEN THE GENERATOR AND THE TRANSFER SWITCH AND BETWEEN THE GENERATOR AND REMOTE MONITOR PANEL. 2-#14 Spares
9. GENERATOR SUPPLY DISCONNECT. PROVIDE 480V, 500A 3 POLE CIRCUIT BREAKER IN NEMA 1 ENCLOSURE.



3 E103 EXISTING RISER DIAGRAM

SCALE: NONE

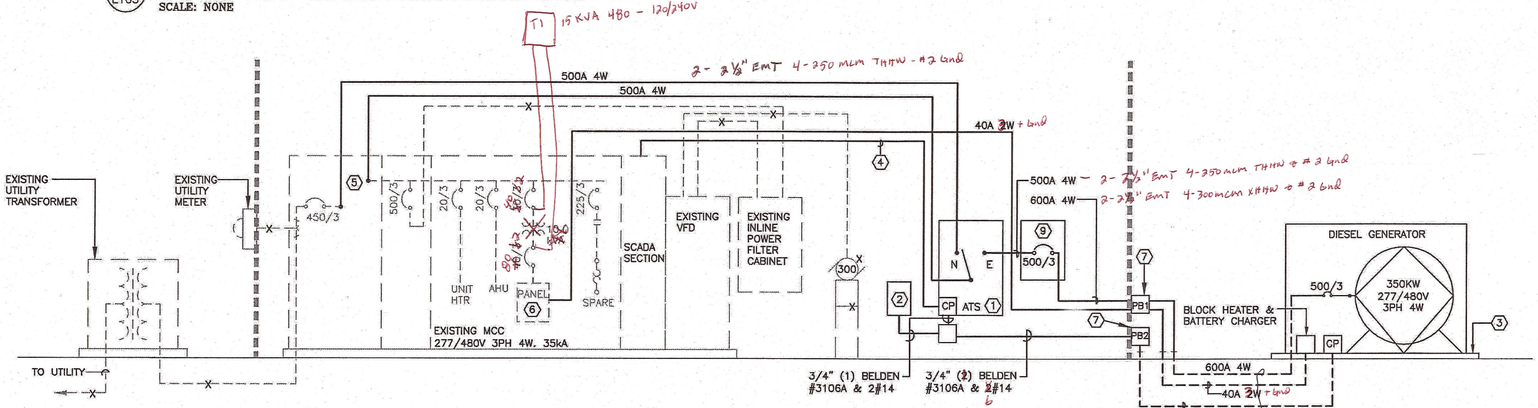


2 E103 NEW PLAN

SCALE: 1/4" = 1'

0 2 4 8

SCALE IN FEET



4 E103 NEW RISER DIAGRAM

SCALE: NONE

N

FIELD BOOK:
Plot Date: 1/13/2012
Drawing name: K:\p-m\MapleGrove\15088000\elec\dwgs\E103-1-PH8.dwg
Xrefs: Well #8_FLOORPLAN

NO.	DATE	BY	DESCRIPTION OF REVISIONS

DESIGNED RCS DRAWN RCS CHECKED RLY

I HEREBY CERTIFY THAT THIS PLAN WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MINNESOTA

ROBERT L. YOUNG

DATE 11/14/12 LIC. NO. 26746

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tkda.com

STANDBY GENERATOR PROJECT

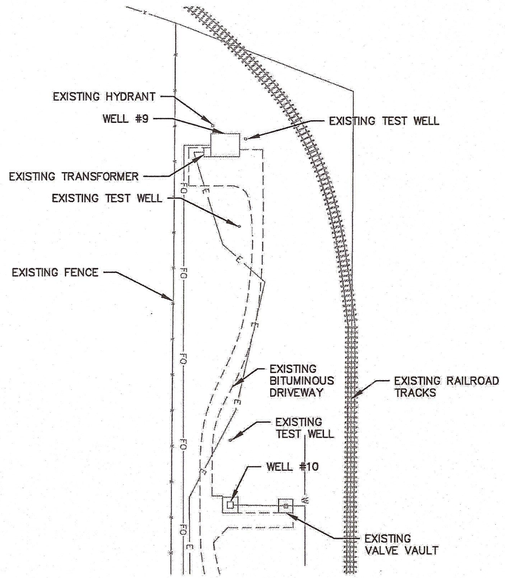
MAPLE GROVE MINNESOTA

CLIENT NO. MSA NO.

WELL #8 ELECTRICAL PLAN

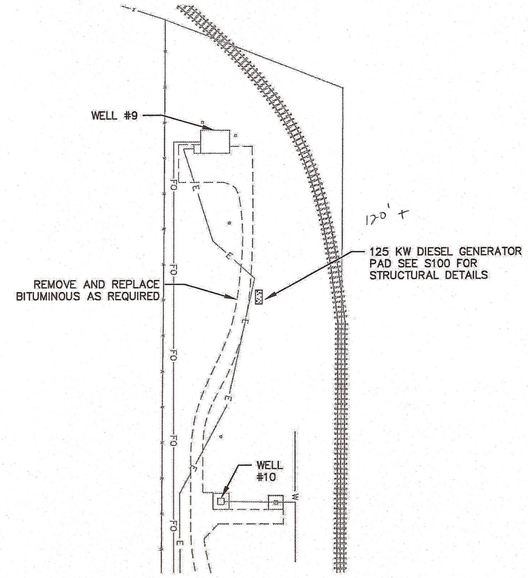
PROJECT NO. 15088.000

RECORD NO. SHEET NO. E103-1



1/104 EXISTING PLAN

SCALE: 1" = 60'



2/104 PROPOSED PLAN

SCALE: 1" = 60'

LEGEND

- T — TELECOMMUNICATION
- G — NATURAL GAS
- E — ELECTRIC
- TV — CABLE TELEVISION
- FO — FIBER OPTIC

N

T K D A

NO.	DATE	BY	DESCRIPTION OF REVISIONS

DESIGNED JRN DRAWN DCK

CHECKED VEJ

I HEREBY CERTIFY THAT THIS PLAN WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MINNESOTA

VERNE E. JACOBSEN

DATE 11/14/12 LIC. NO. 06988

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STANDBY GENERATOR PROJECT

MAPLE GROVE MINNESOTA

CLIENT NO. MSA NO.

WELLS NO. 9
(ALTERNATE DEDUCT BID)

PROJECT NO. 15088.000

RECORD NO.

SHEET NO. C104-1

FIELD BOOK
Plot Date: 11/13/2012
Plotted by: mmun
Xrefs: Wall #8-9-10_ExistingBase

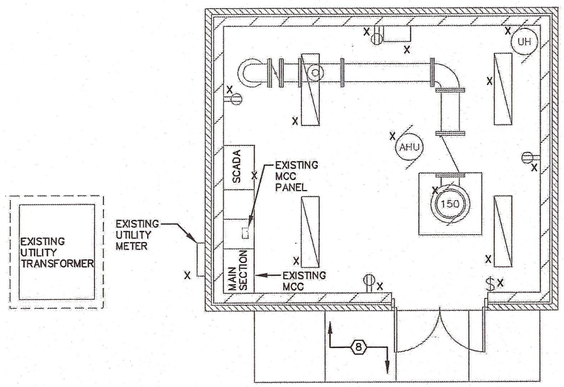
P:\Projects\MapleGrove\15088000\mmun\dwgs\PlanSet\C104.dwg

GENERAL NOTES

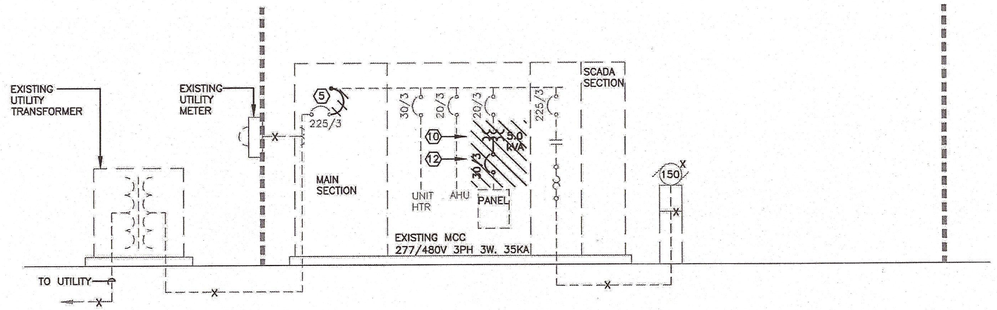
1. ALL RACEWAYS SHALL CONTAIN A GROUNDING CONDUCTOR.
2. CONDUIT ROUTING SHOWN FOR INFORMATION ONLY. VERIFY LOCATION OF CONDUIT STUB UPS WITH GENERATOR MANUFACTURER.
3. VERIFY LOCATION OF ALL EXISTING SITE UTILITIES PRIOR TO EXCAVATION.
4. PROVIDE PLASTIC LAMINATE NAMEPLATE ADJACENT TO THE GENERATOR CIRCUIT BREAKER WHICH READS: "THIS GENERATOR SERVES WELLHOUSE 9, PUMPS AND ALL ASSOCIATED BUILDING LOADS"

KEY NOTES

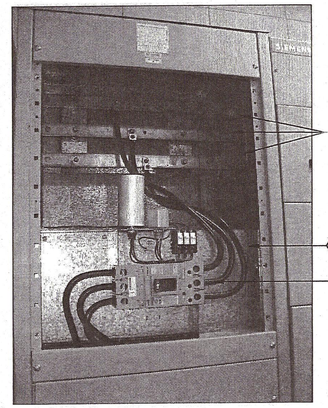
1. AUTOMATIC TRANSFER SWITCH TO BE 480 VOLT, 260 AMP, 3 POLE OPEN TRANSITION TYPE EQUAL TO ASCO 7000 SERIES WITH GROUP 17 ACCESSORY AND RATED AT 35KA MIN. AIC IN NEMA 1 ENCLOSURE. SEE SPECIFICATIONS.
2. GENERATOR REMOTE MONITOR PANEL.
3. PROVIDE GENERATOR CONCRETE MOUNTING PAD. SEE STRUCTURAL DETAILS.
4. PROVIDE 3/4" CONDUIT WITH (1) BELDEN #3106A COMMUNICATION CABLE FOR MODBUS COMMUNICATIONS BETWEEN THE TRANSFER SWITCH AND THE PLC. SUBCONTRACT TO THE CITY'S SCADA SYSTEM INTEGRATOR. INTEGRATOR SHALL PROVIDE ALL PROGRAMMING AND HARDWARE REQUIRED TO MONITOR AND DISPLAY THE GENERATOR STATUS AND ALARMS. INTEGRATOR SHALL PROVIDE NEW SCADA SYSTEM SCREEN AT THE PUBLIC WORKS AND WATER TREATMENT BUILDING HMI TO DISPLAY THE GENERATOR AND TRANSFER SWITCH STATUS AND ALARMS. NEW SCREEN WILL ALLOW START-STOP OF THE GENERATOR AND REMOTE CONTROL OF THE AUTOMATIC TRANSFER SWITCH WITH PASSWORD PROTECTION. INTEGRATOR SHALL MODIFY OR REPLACE THE EXISTING PLC AT THE WELL TO PROVIDE MODBUS COMMUNICATIONS.
5. REMOVE EXISTING FEEDERS FROM LOAD SIDE MAIN CIRCUIT BREAKER TO EXISTING BUS. MODIFY EXISTING BUS TO ACCOMMODATE NEW FEEDERS TO AND FROM NEW AUTOMATIC TRANSFER SWITCH, PROVIDE REQUIRED NEW LUGS AND MODIFICATIONS FOR A COMPLETE INSTALLATION.
6. PROVIDE 40/2 CIRCUIT BREAKER AT EXISTING PANEL. PANEL IS SIEMENS I-T-E.
7. PROVIDE WEATHERPROOF PULLBOX. SIZE PER NEC.
8. REMOVE A PARTIAL SECTION OF EXISTING CONCRETE SIDEWALK TO ACCOMMODATE TRENCHING FOR NEW GENERATOR CONDUITS. REPLACE CONCRETE AFTER WORK HAS BEEN COMPLETED.
9. PROVIDE 1"PVC CONDUIT WITH (2) #14 FOR GENERATOR MONITOR PANEL POWER, (2) #14 GENERATOR START SIGNAL, (2) BELDEN #3106A COMMUNICATION CABLE FOR MODBUS COMMUNICATIONS BETWEEN THE GENERATOR AND THE TRANSFER SWITCH AND BETWEEN THE GENERATOR AND REMOTE MONITOR PANEL. 3-#14 spares
10. DISCONNECT EXISTING 5KVA TRANSFORMER SERVING EXISTING MCC PANEL.
11. PROVIDE NEW 10KVA 480V TO 120/240 1PH 3W DRY TYPE TRANSFORMER.
12. REMOVE EXISTING PRIMARY AND SECONDARY CIRCUIT BREAKER AND WIRING SERVING EXISTING MCC PANEL. PROVIDE NEW 40/2 SECONDARY AND 30/2 PRIMARY CIRCUIT BREAKER TO SERVE NEW 10KVA TRANSFORMER.
13. GENERATOR SUPPLY DISCONNECT. PROVIDE 480V, 225A 3 POLE CIRCUIT BREAKER IN NEMA 1 ENCLOSURE.



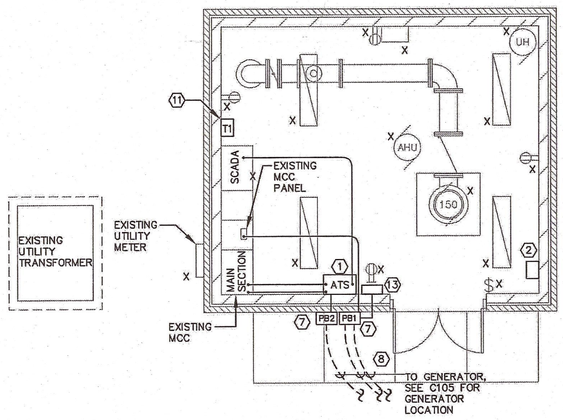
1 E104 EXISTING PLAN
SCALE: 1/4" = 1'
0 2 4 8
SCALE IN FEET



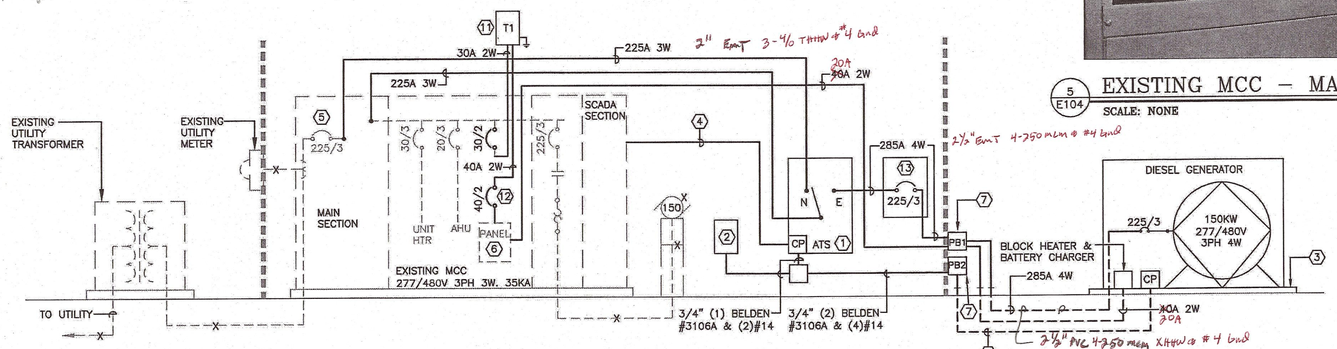
3 E104 EXISTING RISER DIAGRAM
SCALE: NONE



5 E104 EXISTING MCC – MAIN SECTION
SCALE: NONE



2 E104 NEW PLAN
SCALE: 1/4" = 1'
0 2 4 8
SCALE IN FEET



4 E104 NEW RISER DIAGRAM
SCALE: NONE

N

NO.	DATE	BY	DESCRIPTION OF REVISIONS

DESIGNED RCS DRAWN RCS
CHECKED RLY

I HEREBY CERTIFY THAT THIS PLAN WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MINNESOTA
Robert L. Young
ROBERT L. YOUNG
DATE 11/14/12 LIC. NO. 26746

TKDA
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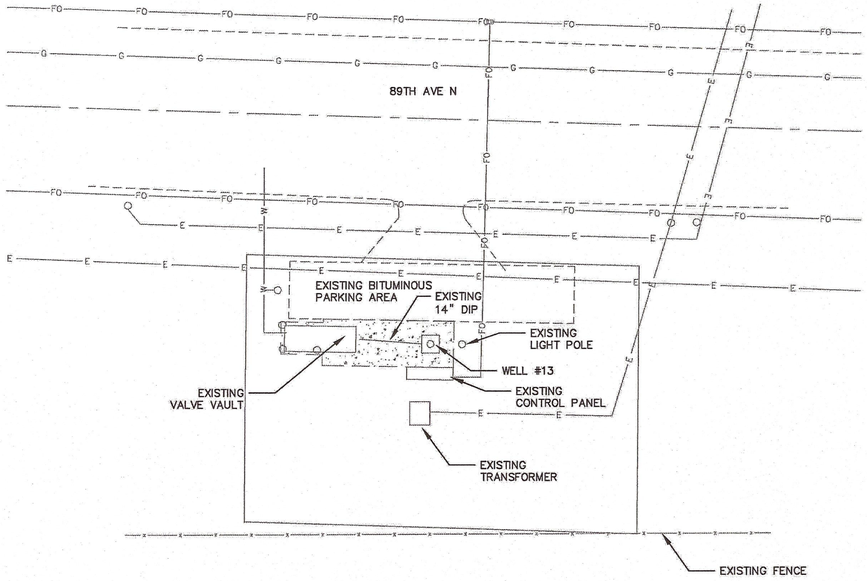
tkda.com

STANDBY GENERATOR PROJECT
MAPLE GROVE MINNESOTA
CLIENT NO. MSA NO.

WELL #9 ELECTRICAL PLAN
DEDUCT ALTERNATE

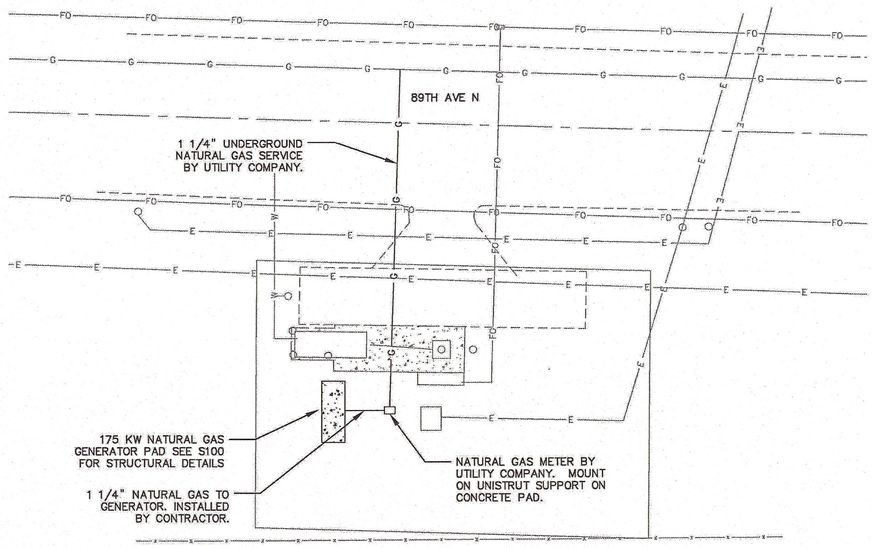
PROJECT NO. 15088.000
RECORD NO. SHEET NO. E104-1

FIELD BOOK:
Plot Date: 11/13/2012
Drawing Name: C:\Projects\Maple Grove\15088000\Sheet\dwg\E104-1-PH9.dwg
Xrefs: Name #@_FLOORPLAN



1/106 EXISTING PLAN

SCALE: 1" = 20'



2/106 PROPOSED PLAN

SCALE: 1" = 20'

NOTES:

1. USE FLEXIBLE GAS CONNECTOR BETWEEN GAS SERVICE LINE AND GENERATOR.
2. GAS PRESSURE REQUIRED AT GENERATOR IS 11"–15" C.C.

LEGEND	
T	TELECOMMUNICATION
G	NATURAL GAS
E	ELECTRIC
TV	CABLE TELEVISION
FO	FIBER OPTIC

N

TKDA

NO.	DATE	BY	DESCRIPTION OF REVISIONS

DESIGNED JRN DRAWN DCK CHECKED VEJ

I HEREBY CERTIFY THAT THIS PLAN WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MINNESOTA

VERNE E. JACOBSEN

DATE 11/14/12 LIC. NO. 06988

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Saint Paul, MN 55101
651.292.4400
tkda.com

STANDBY GENERATOR PROJECT

MAPLE GROVE MINNESOTA

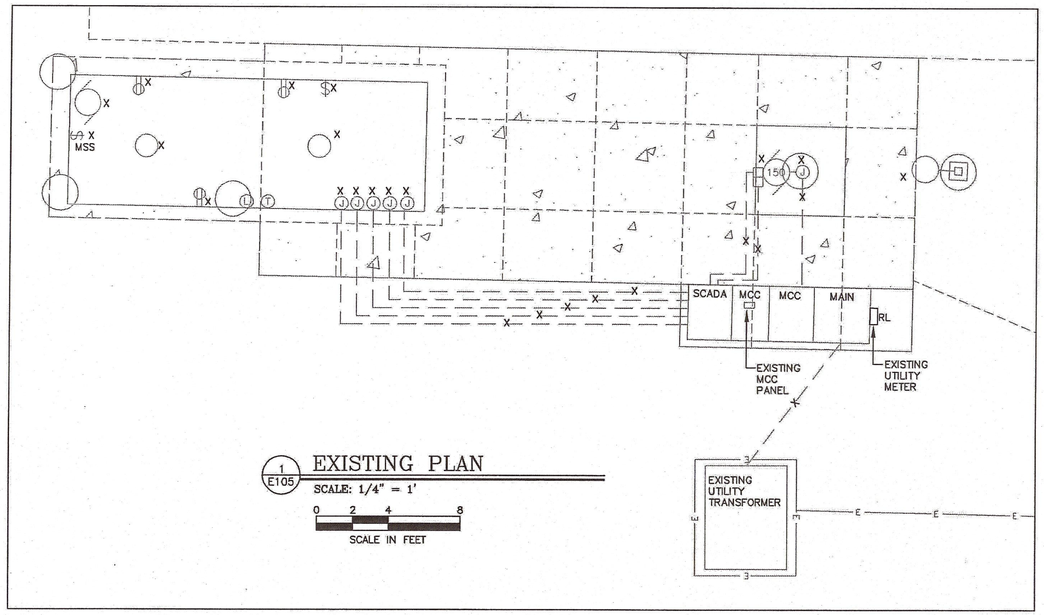
CLIENT NO. MSA NO.

WELL NO. 13

PROJECT NO. 15088.000

RECORD NO. SHEET NO. C105-1

FIELD BOOK:
Plot Date: 11/14/2012
Drawing name: K:\Grnn\MapleGrove\15088000\mun\dwgs\PlanSet\C105.dwg
Xrefs: Well #13_ExistingBase

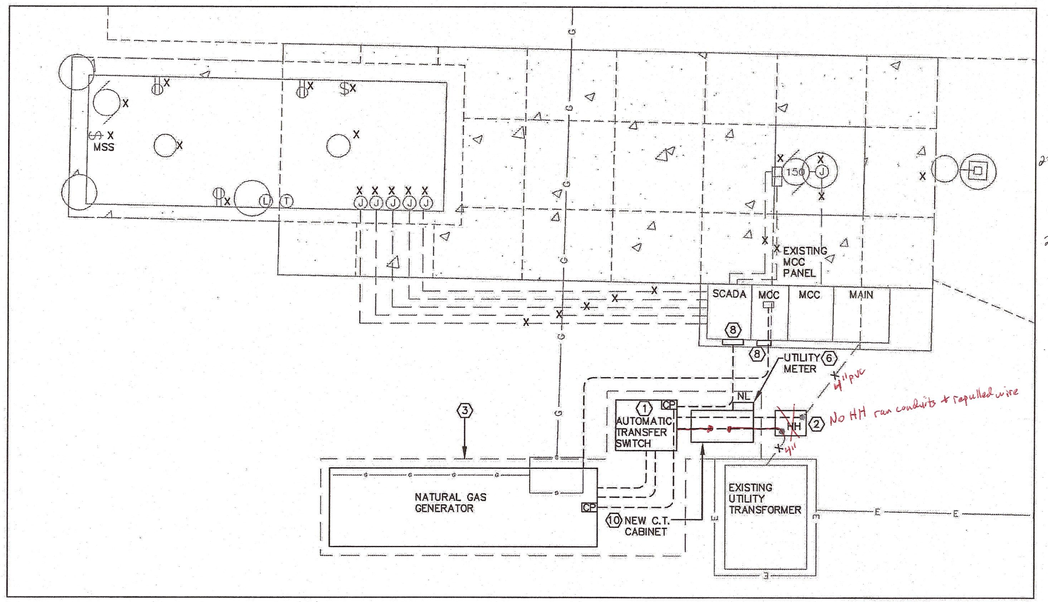


1 E105 EXISTING PLAN

SCALE: 1/4" = 1'

0 2 4 8

SCALE IN FEET



2 E105 NEW PLAN

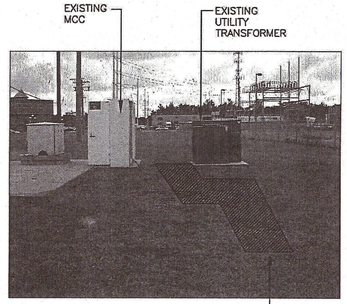
SCALE: 1/4" = 1'

0 2 4 8

SCALE IN FEET

GENERAL NOTES

1. ALL RACEWAYS SHALL CONTAIN A GROUNDING CONDUCTOR.
2. CONDUIT ROUTING SHOWN FOR INFORMATION ONLY. VERIFY LOCATION OF CONDUIT STUB UPS WITH GENERATOR MANUFACTURER.
3. VERIFY LOCATION OF ALL EXISTING SITE UTILITIES PRIOR TO EXCAVATION.
4. PROVIDE PLASTIC LAMINATE NAMEPLATE ADJACENT TO THE GENERATOR CIRCUIT BREAKER WHICH READS: "THIS GENERATOR SERVES WELL 13, PUMPS AND ALL ASSOCIATED SITE LOADS"



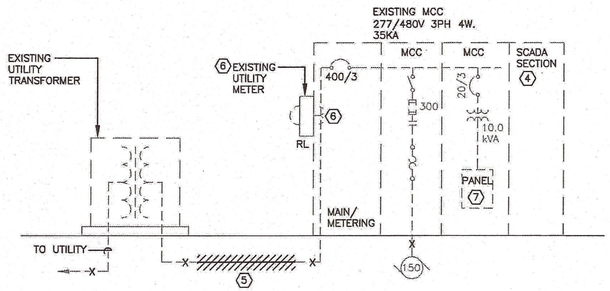
3 E105 EXISTING MCC ENCLOSURE

SCALE: NONE

KEY NOTES

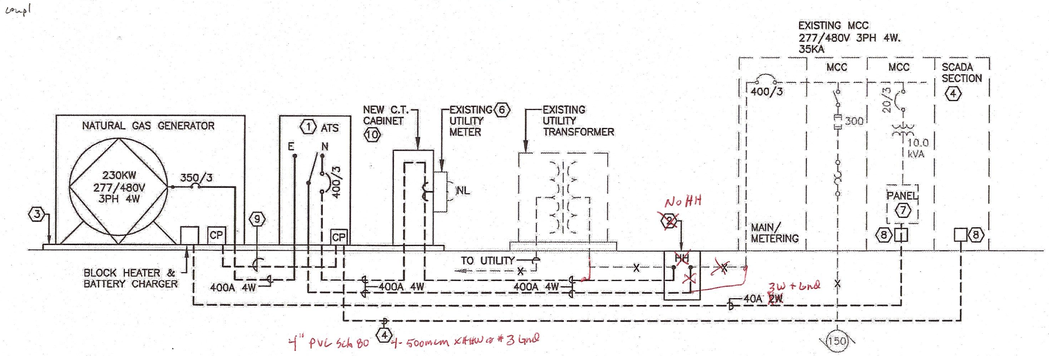
7. REMOVE (2) EXISTING SPARE 20/1 CIRCUIT BREAKERS AT EXISTING CUTLER HAMMER PRL1A PANEL. PROVIDE NEW 40/2 CIRCUIT BREAKER TO SERVE GENERATOR BLOCK HEATER & BATTERY CHARGER.
8. PROVIDE REQUIRED PULLBOX AT REAR OF EXISTING MCC ENCLOSURE TO ACCOMMODATE TRANSITION FROM BELOW GRADE TO REAR OF ENCLOSURE.
9. PROVIDE 1"PVC CONDUIT WITH (2) #14 FOR GENERATOR MONITOR PANEL POWER, (2) #14 FOR GENERATOR START SIGNAL, (2) BELDEN #3106A COMMUNICATION CABLE FOR MODBUS COMMUNICATIONS BETWEEN THE GENERATOR AND THE TRANSFER SWITCH.
10. PROVIDE NEW C.T CABINET. COORDINATE SIZE AND LOCATION WITH UTILITY.

1. AUTOMATIC TRANSFER SWITCH TO BE 480 VOLT, 400 AMP, SERVICE ENTRANCE RATED, 4 POLE OPEN TRANSITION, NEMA 3R TYPE, EQUAL TO ASCO 7000 SERIES WITH GROUP 17 ACCESSORY, GROUP 44G ACCESSORY (HEATER) AND RATED AT 35,000 AIC MIN. SEE SPECIFICATIONS.
2. PROVIDE HANDHOLE SIZED PER N.E.C. FOR SPLICING OF EXISTING AND NEW CONDUCTORS.
3. PROVIDE GENERATOR & ATS CONCRETE MOUNTING PAD. SEE STRUCTURAL DETAILS.
4. PROVIDE 3/4" CONDUIT WITH (1) BELDEN #3106A COMMUNICATION CABLE FOR MODBUS COMMUNICATIONS BETWEEN THE TRANSFER SWITCH AND THE PLC. SUBCONTRACT TO THE CITY'S SCADA SYSTEM INTEGRATOR. INTEGRATOR SHALL PROVIDE ALL PROGRAMMING AND HARDWARE REQUIRED TO MONITOR AND DISPLAY THE GENERATOR STATUS AND ALARMS. INTEGRATOR SHALL PROVIDE NEW SCADA SYSTEM SCREEN AT THE PUBLIC WORKS AND WATER TREATMENT PLANT BUILDING HMI TO DISPLAY THE GENERATOR AND TRANSFER SWITCH STATUS AND ALARMS. NEW SCREEN WILL ALLOW START–STOP OF THE GENERATOR AND REMOTE CONTROL OF THE AUTOMATIC TRANSFER SWITCH WITH PASSWORD PROTECTION. INTEGRATOR SHALL MODIFY OR REPLACE THE EXISTING PLC AT THE WELL TO PROVIDE MODBUS COMMUNICATIONS.
5. INTERCEPT EXISTING SERVICE CONDUIT AND CONDUCTORS AND RE–ROUTE TO NEW CONNECTION CABINET. EXTEND EXISTING FEEDERS FROM NEW ATS TO EXISTING MCC.
6. RELOCATE EXISTING METER OR INSTALL NEW AND BASE TO NEW CONNECTION CABINET. REMOVE C.T'S AND MODIFY MCC BUS AS REQUIRED. PROVIDE COVERPLATE OVER EXISTING HOLE IN MCC ENCLOSURE. PAINT TO MATCH EXISTING ENCLOSURE. COORDINATE WITH ELECTRICAL UTILITY.



4 E105 EXISTING RISER DIAGRAM

SCALE: NONE



5 E105 NEW RISER DIAGRAM

SCALE: NONE

20' 4" PVC
4" PVC 90 (1)
8 4" PVC coupl
20' 3½" PVC
5 3½" PVC 90°
40' 1" PVC
8 1" PVC 90
1" coupl

4" PVC Existing

N

T.D.K.A.

NO.	DATE	BY	DESCRIPTION OF REVISIONS

DESIGNED RCS DRAWN RCS CHECKED RLY

I HEREBY CERTIFY THAT THIS PLAN WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MINNESOTA

ROBERT L. YOUNG

DATE 11/14/12 LIC. NO. 26746

TKDA

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tkda.com

STANDBY GENERATOR PROJECT

MAPLE GROVE MINNESOTA

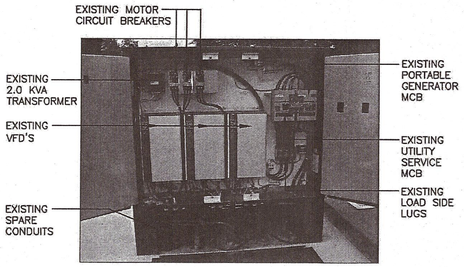
CLIENT NO. MSA NO.

WELL #13 ELECTRICAL PLAN

PROJECT NO. 15088.000

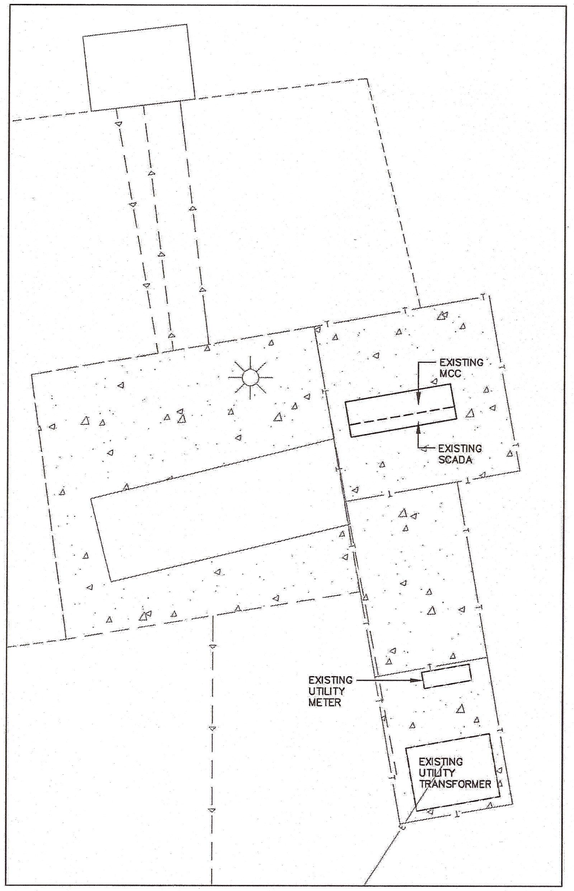
RECORD NO. SHEET NO. E105-1

FIELD BOOK:
Plot Date: 11/13/2012
Drawing name: K:\gr-mn\MapleGrove\15088000\elec\dwgs\E105-1-PH13.dwg
Xrefs: Well #13_ExistingBase



5 E106 EXISTING MCC

SCALE: NONE

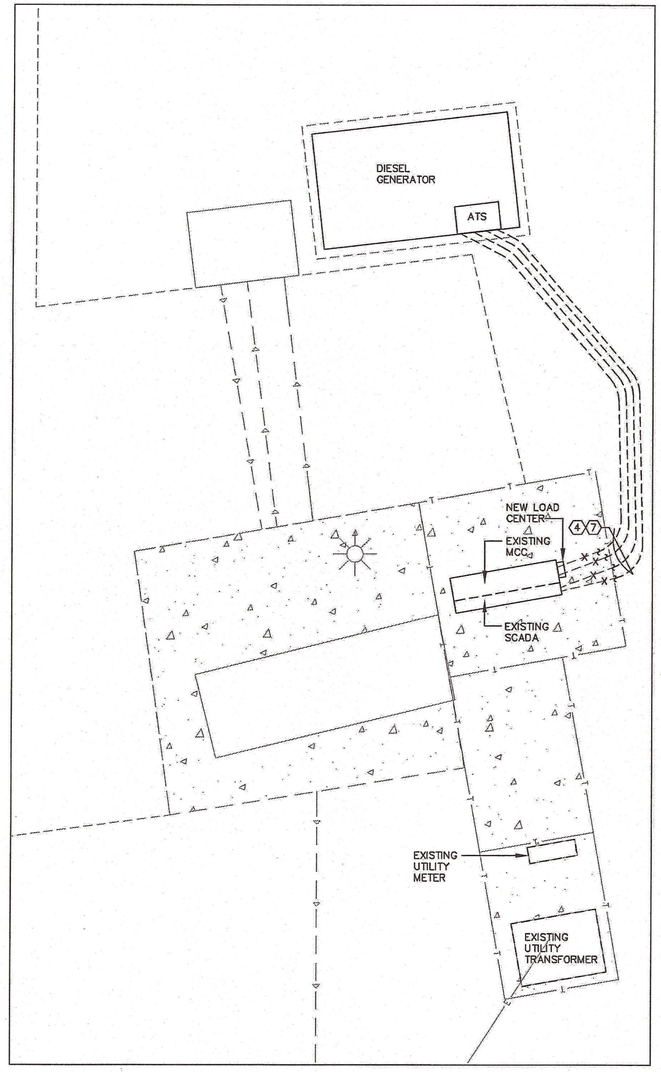


1 E106 EXISTING PLAN

SCALE: 1/4" = 1'

0 2 4 8

SCALE IN FEET



2 E106 NEW PLAN

SCALE: 1/4" = 1'

0 2 4 8

SCALE IN FEET

GENERAL NOTES

1. ALL RACEWAYS SHALL CONTAIN A GROUNDING CONDUCTOR.
2. CONDUIT ROUTING SHOWN FOR INFORMATION ONLY. VERIFY LOCATION OF CONDUIT STUB UPS WITH GENERATOR MANUFACTURER.
3. VERIFY LOCATION OF ALL EXISTING SITE UTILITIES PRIOR TO EXCAVATION.
4. PROVIDE PLASTIC LAMINATE NAMEPLATE ADJACENT TO THE GENERATOR CIRCUIT BREAKER WHICH READS: "THIS GENERATOR SERVES THE LIFT STATION, PUMPS AND ALL ASSOCIATED SITE LOADS"
5. SUNNYSLOPE SYSTEM IS SIZED SUCH ONLY (2) PUMPS WILL OPERATE SIMULTANEOUSLY, THE THIRD PUMP IS FOR BACK-UP ONLY.

N

T.D.K.A

KEY NOTES

(1) AUTOMATIC TRANSFER SWITCH TO BE 480 VOLT, 400 AMP, 4 POLE OPEN TRANSITION TYPE EQUAL TO ASCO 7000 SERIES WITH GROUP 17 ACCESSORY AND RATED AT 35,000 AIC MIN. INTEGRAL TO GENERATOR ENCLOSURE. SEE SPECIFICATIONS.

(2) PROVIDE 10KVA 480V PRIMARY TO 120/240 1PH SECONDARY MOUNTED WITHIN EXISTING ENCLOSURE. PROVIDE CIRCUIT BREAKER MOUNTED WITHIN EXISTING ENCLOSURE FOR TRANSFORMER PRIMARY PROTECTION.

(3) PROVIDE GENERATOR/ATS CONCRETE MOUNTING PAD. SEE STRUCTURAL DETAILS.

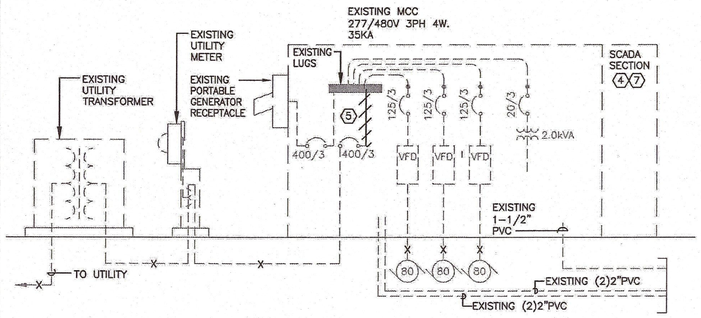
(4) PROVIDE 3/4" CONDUIT WITH (1) BELDEN #3106A COMMUNICATION CABLE FOR MODBUS COMMUNICATIONS BETWEEN THE TRANSFER SWITCH AND THE PLC. SUBCONTRACT TO THE CITY'S SCADA SYSTEM INTEGRATOR. INTEGRATOR SHALL PROVIDE ALL PROGRAMMING AND HARDWARE REQUIRED TO MONITOR AND DISPLAY THE GENERATOR STATUS AND ALARMS. INTEGRATOR SHALL PROVIDE NEW SCADA SYSTEM SCREEN AT THE PUBLIC WORKS BUILDING AND WATER TREATMENT PLANT HMI TO DISPLAY THE GENERATOR AND TRANSFER SWITCH STATUS AND ALARMS. NEW SCREEN WILL ALLOW START-STOP OF THE GENERATOR AND REMOTE CONTROL OF THE AUTOMATIC TRANSFER SWITCH WITH PASSWORD PROTECTION. INTEGRATOR SHALL MODIFY OR REPLACE THE EXISTING PLC AT THE WELL TO PROVIDE MODBUS COMMUNICATIONS.

(5) REMOVE EXISTING FEEDERS FROM LOAD SIDE OF MAIN CIRCUIT BREAKER TO EXISTING LUGS. PROVIDE ADDITIONAL LUGS TO ACCOMMODATE NEW FEEDERS TO AND FROM NEW AUTOMATIC TRANSFER SWITCH, PROVIDE REQUIRED NEW BUSSING, LUGS AND COMPONENTS FOR A COMPLETE INSTALLATION.

(6) PROVIDE 12 CIRCUIT LOAD CENTER, 120/240V 1PH 3W, 50/2 MCB IN NEMA 3R ENCLOSURE TO SERVE NEW GENERATOR LOADS. LOAD CENTER SHALL BE MOUNTED EXTERNAL TO EXISTING ENCLOSURE. PROVIDE (4) 20/1 CIRCUIT BREAKERS FOR FUTURE USE AND (1) 40/2 CIRCUIT BREAKER FOR GENERATOR BLOCK HEATER AND BATTERY CHARGER.

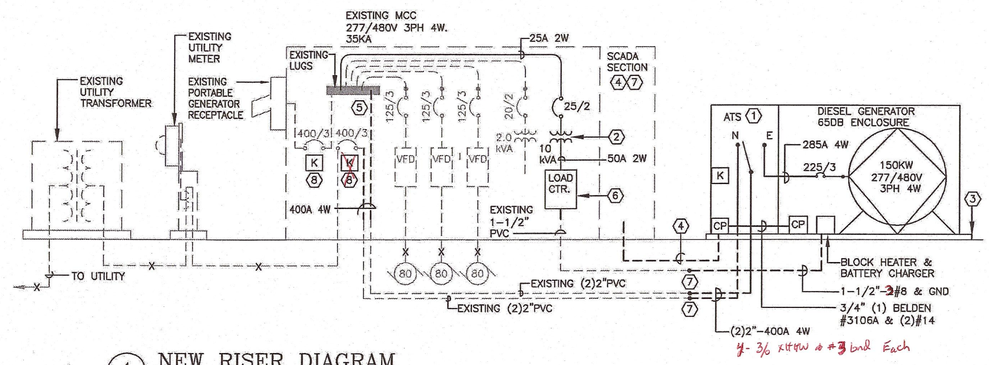
(7) EXTEND EXISTING CONDUIT TO NEW GENERATOR LOCATION. PROVIDE NEW WIRING AS SHOWN.

(8) PROVIDE KIRK KEY FOR EXISTING MCB AND PROTABLE GENERATOR C.B. TP PREVENT OPERATING BOTH SOURCES OR FIXED GENERATOR SIMULTANEOUSLY. ~~MODIFY~~ USE THE EXISITNG MECHANICAL INTERLOCK.



3 E106 EXISTING RISER DIAGRAM

SCALE: NONE



4 E106 NEW RISER DIAGRAM

SCALE: NONE

NO.	DATE	BY	DESCRIPTION OF REVISIONS

DESIGNED RCS DRAWN RCS

CHECKED RLY

I HEREBY CERTIFY THAT THIS PLAN WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MINNESOTA

ROBERT L. YOUNG

DATE 11/14/12 LIC. NO. 26746

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Saint Paul, MN 55101
651.292.4400
tkda.com

STANDBY GENERATOR PROJECT

MAPLE GROVE MINNESOTA

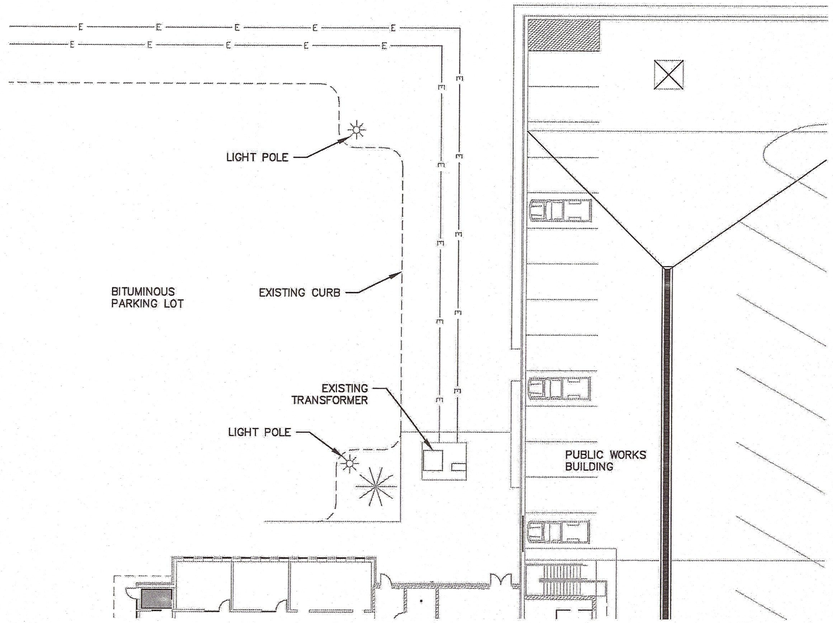
CLIENT NO. MSA NO.

SUNNY SLOPE LIFT STATION
ELECTRICAL PLAN

PROJECT NO. 15088.000

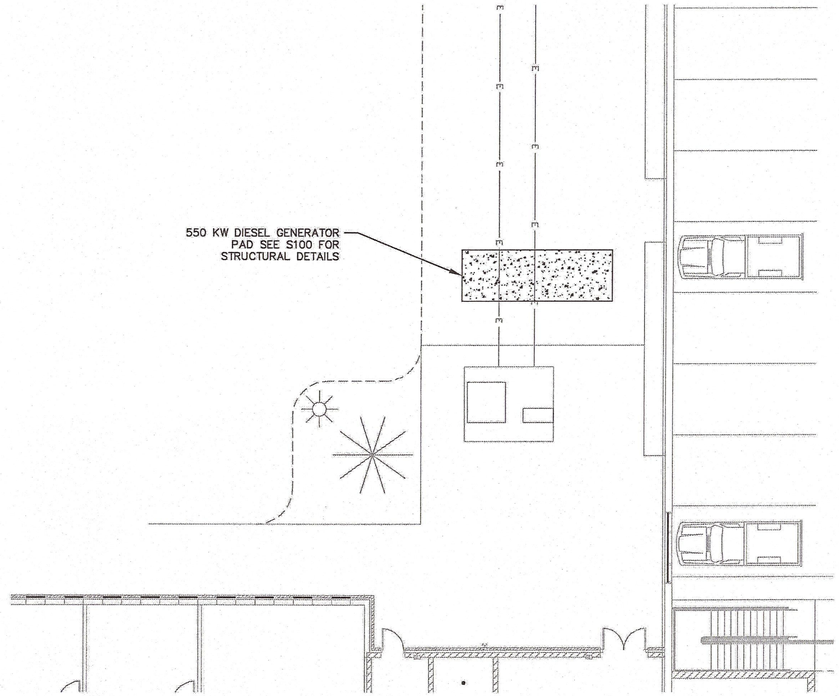
RECORD NO. SHEET NO. E106-1

FIELD BOOK:
Plot Date: 11/13/2012
Drawing name: K:\lg-m\MapleGrove\15088000\elec\dwgs\E106-1-Sunny Slope.dwg
Xrefs:, Sunny Slope_ExistingBase



1/107 EXISTING PLAN

SCALE: 1" = 30'



2/107 PROPOSED PLAN

SCALE: 1" = 10'

LEGEND	
T	TELECOMMUNICATION
G	NATURAL GAS
E	ELECTRIC
TV	CABLE TELEVISION
FO	FIBER OPTIC

N

L D K A

NO.	DATE	BY	DESCRIPTION OF REVISIONS

DESIGNED JRN DRAWN DCK

CHECKED VEJ

I HEREBY CERTIFY THAT THIS PLAN WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MINNESOTA

VERNE E. JACOBSEN

DATE 11/14/12 LIC. NO. 06988

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tkda.com

P.W. EMERGENCY POWER

MAPLE GROVE MINNESOTA

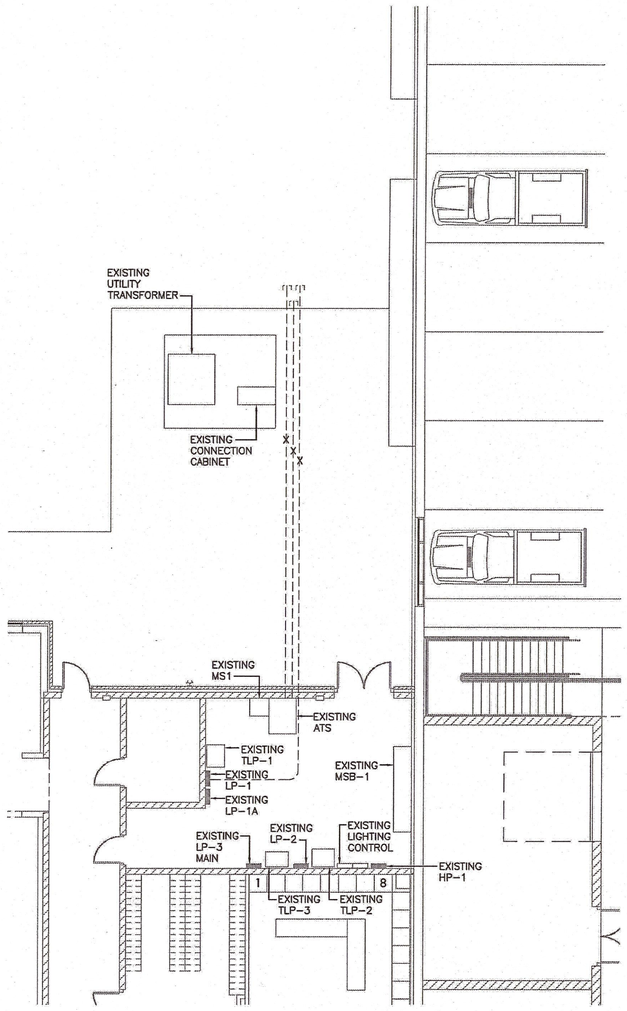
CLIENT NO. MSA NO.

PUBLIC WORKS BUILDING

PROJECT NO. 15088.000

RECORD NO. SHEET NO. C107-1

FIELD BOOK:
Plot Date: 11/13/2012
Drawing name: K:\p-m\MapleGrove\15088000\m...\dwgs\PlanSet\C107.dwg
Xrefs: PLCase

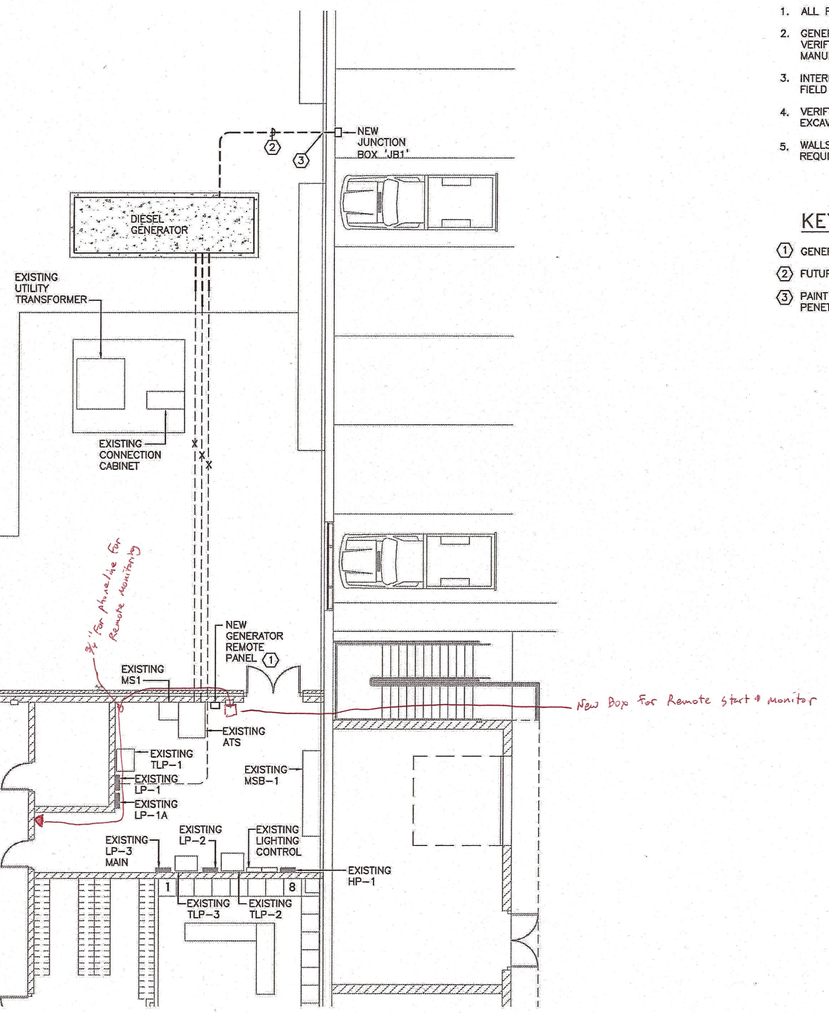


1 E107 EXISTING PLAN – MAIN ELECTRICAL ROOM

SCALE: 1/8" = 1'

0 4 8 16

SCALE IN FEET



2 E107 NEW PLAN – MAIN ELECTRICAL ROOM

SCALE: 1/8" = 1'

0 4 8 16

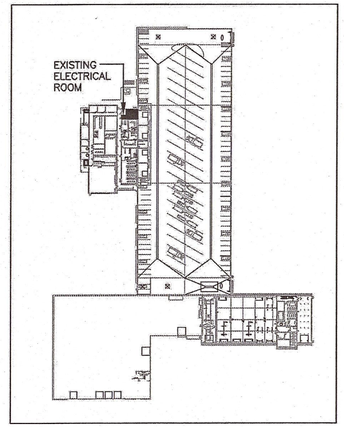
SCALE IN FEET

GENERAL NOTES

1. ALL RACEWAYS SHALL CONTAIN A GROUNDING CONDUCTOR.
2. GENERATOR CONDUIT ROUTING SHOWN FOR INFORMATION ONLY. VERIFY LOCATION OF CONDUIT STUB UPS WITH GENERATOR MANUFACTURER.
3. INTERIOR CONDUIT ROUTING SHOWN FOR INFORMATION ONLY. FIELD COORDINATE WITH OWNER.
4. VERIFY LOCATION OF ALL EXISTING SITE UTILITIES PRIOR TO EXCAVATION.
5. WALLS AND FLOOR ARE CONCRETE. CORE DRILLING WILL BE REQUIRED. FIELD COORDINATE.

KEY NOTES

1. GENERATOR REMOTE MONITOR PANEL.
2. FUTURE LIFE SAFETY CONDUIT.
3. PAINT EXPOSED CONDUIT, ETC TO MATCH ADJACENT SURFACE AT WALL PENETRATION.



KEY PLAN

N

T.D.K.A.

NO.	DATE	BY	DESCRIPTION OF REVISIONS

DESIGNED RCS DRAWN RCS

CHECKED RLY

I HEREBY CERTIFY THAT THIS PLAN WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MINNESOTA

Robert L. Young

ROBERT L. YOUNG

DATE 11/14/12 LIC. NO. 26746

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STANDBY GENERATOR PROJECT

MAPLE GROVE MINNESOTA

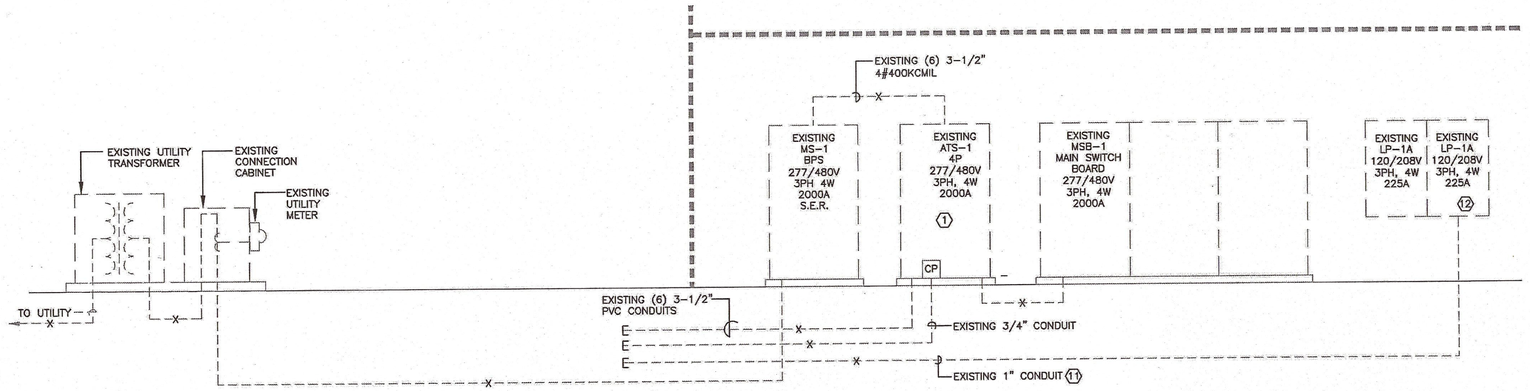
CLIENT NO. MSA NO.

PUBLIC WORKS BUILDING
ELECTRICAL FLOOR PLAN

PROJECT NO. 15088.000

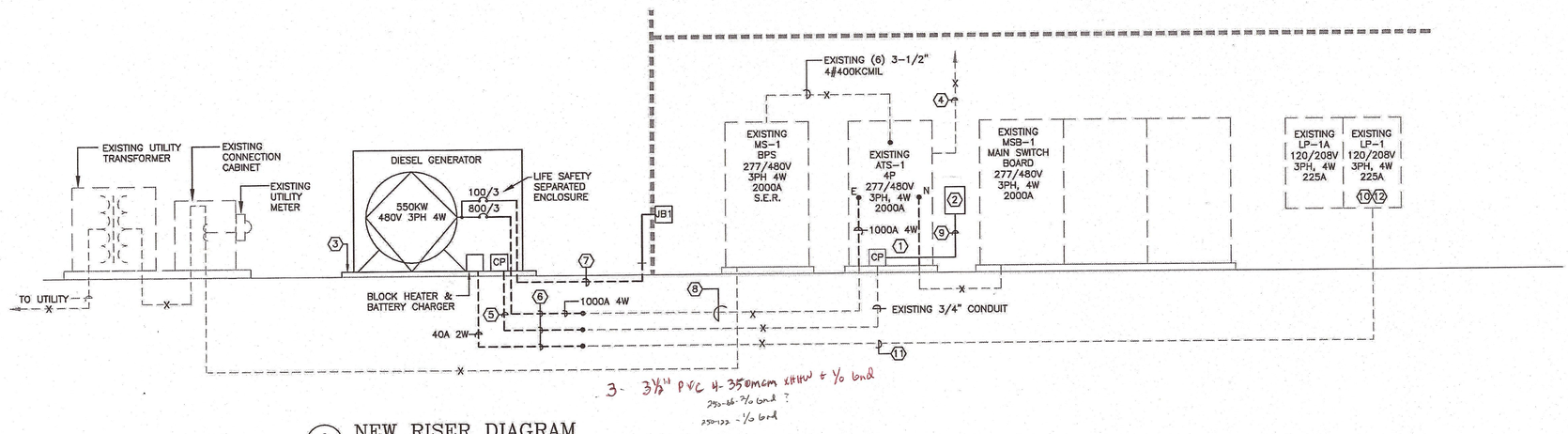
RECORD NO. SHEET NO. E107-1

FIELD BOOK:
Plot Date: 11/13/2012
Drawing name: K:\p-m\MapleGrove\15088000\electric\wga\E107-1-PW.dwg
Xrefs: PW_base



1 / 107-2 EXISTING RISER DIAGRAM

SCALE: NONE



2 / 107-2 NEW RISER DIAGRAM

SCALE: NONE

GENERAL NOTES

1. ALL RACEWAYS SHALL CONTAIN A GROUNDING CONDUCTOR.
2. CONDUIT ROUTING SHOWN FOR INFORMATION ONLY. VERIFY LOCATION OF CONDUIT STUB UPS WITH GENERATOR MANUFACTURER.
3. VERIFY LOCATION OF ALL EXISTING SITE UTILITIES PRIOR TO EXCAVATION.
4. PROVIDE PLASTIC LAMINATE NAMEPLATE ADJACENT TO THE GENERATOR CIRCUIT BREAKER WHICH READS: "THIS GENERATOR SERVES THE PUBLIC WORKS BUILDING AND ALL ASSOCIATED BUILDING LOADS"

KEY NOTES

(1) EXISTING 2000A 4 POLE ZENITH AUTOMATIC TRANSFER SWITCH.

(2) GENERATOR REMOTE MONITOR PANEL.

(3) PROVIDE GENERATOR CONCRETE MOUNTING PAD. SEE STRUCTURAL DETAILS.

(4) PROVIDE 3/4" CONDUIT WITH (1) BELDEN #3106A COMMUNICATION CABLE FOR MODBUS COMMUNICATIONS BETWEEN THE TRANSFER SWITCH AND THE PLC. SUBCONTRACT TO THE CITY'S SCADA SYSTEM INTEGRATOR. INTEGRATOR SHALL PROVIDE ALL PROGRAMMING AND HARDWARE REQUIRED TO MONITOR AND DISPLAY THE GENERATOR STATUS AND ALARMS. INTEGRATOR SHALL PROVIDE NEW SCADA SYSTEM SCREEN AT THE PUBLIC WORKS BUILDING AND WATER TREATMENT PLANT HMI TO DISPLAY THE GENERATOR AND TRANSFER SWITCH STATUS AND ALARMS. NEW SCREEN WILL ALLOW START-STOP OF THE GENERATOR AND REMOTE CONTROL OF THE AUTOMATIC TRANSFER SWITCH WITH PASSWORD PROTECTION.

(5) EXTEND 3/4" PVC CONDUIT AND PROVIDE WITH NEW (2) #14 - for Lock-up Relay FOR GENERATOR MONITOR PANEL POWER, (2) #14 FOR GENERATOR START STOP, (2) BELDEN #3106A COMMUNICATION CABLE FOR MODBUS COMMUNICATIONS BETWEEN THE GENERATOR AND THE TRANSFER SWITCH. 2-#14 ~~#3106~~ to shunt trip From transfer Switch

(6) EXTEND EXISTING CONDUIT AND PROVIDE NEW GENERATOR LOCATION. PROVIDE NEW WIRING AS SHOWN.

(7) PROVIDE 2" EMPTY PVC CONDUIT FOR FUTURE USE.

(8) EXTEND EXISTING (3) 3-1/2" CONDUITS. (USE 3 OF EXISTING 6 CONDUITS) PROVIDE NEW WIRING FROM NEW GENERATOR TO EXISTING ATS.

(9) PROVIDE (2) #14 AND (1) BELDEN #3106A FOR MODBUS CONNECTION BETWEEN GENERATOR AND REMOTE MONITORING PANEL.

(10) PROVIDE NEW SQUARE D 'NQ' 40/2 CIRCUIT BREAKER AT EXISTING PANEL.

(11) EXISTING 1" CONDUIT. PROVIDE NEW WIRING FROM GENERATOR TO NEW CIRCUIT BREAKER. (CIRCUIT 33,35)

(12) REMOVE (2) SPARE 20/1 CIRCUIT BREAKERS TO ACCOMMODATE NEW 2-POLE CIRCUIT BREAKER. RETURN REMOVED BREAKER TO OWNER.

N

LDKA

NO.	DATE	BY	DESCRIPTION OF REVISIONS

DESIGNED RCS DRAWN RCS CHECKED RLY

I HEREBY CERTIFY THAT THIS PLAN WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MINNESOTA

ROBERT L. YOUNG

DATE 11/14/12 LIC. NO. 26746

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STANDBY GENERATOR PROJECT

MAPLE GROVE MINNESOTA

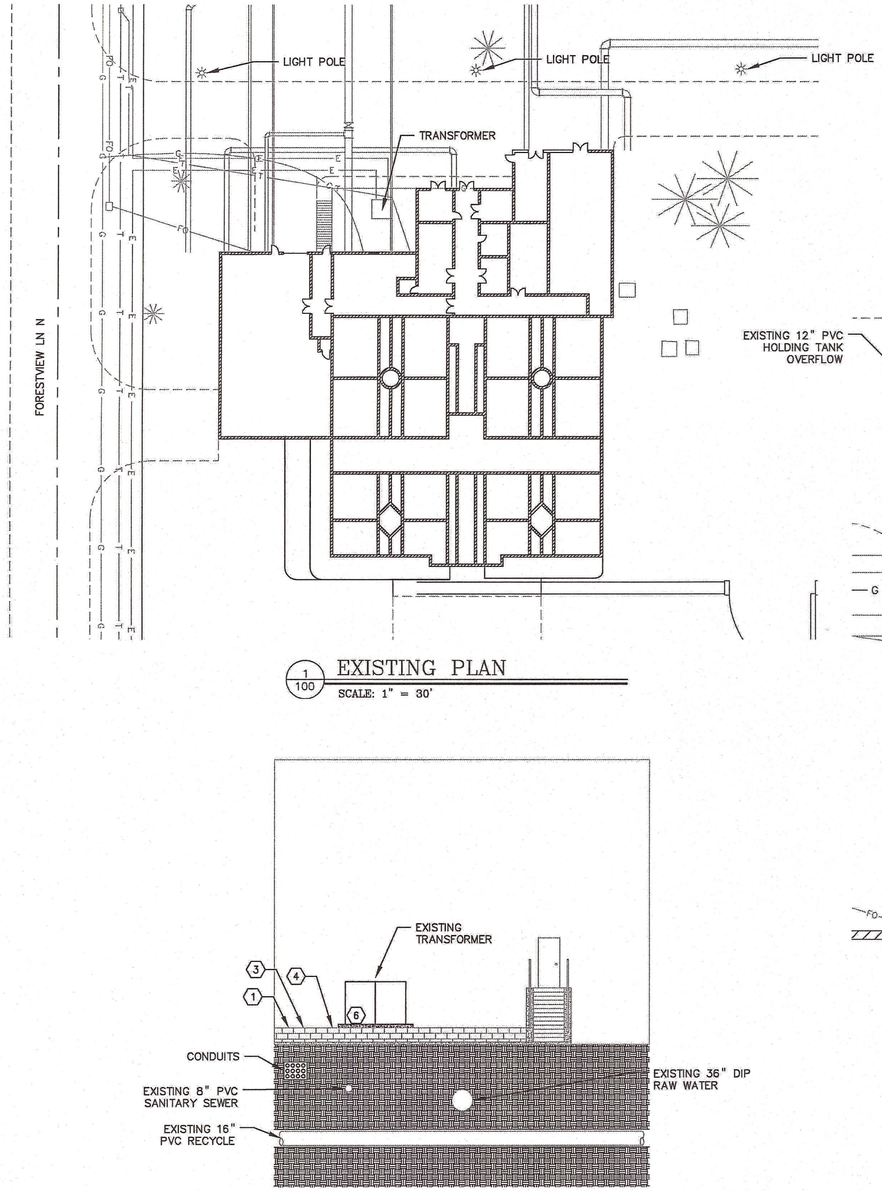
CLIENT NO. MSA NO.

PUBLIC WORKS BUILDING
PARTIAL RISER DIAGRAM

PROJECT NO. 15088.000

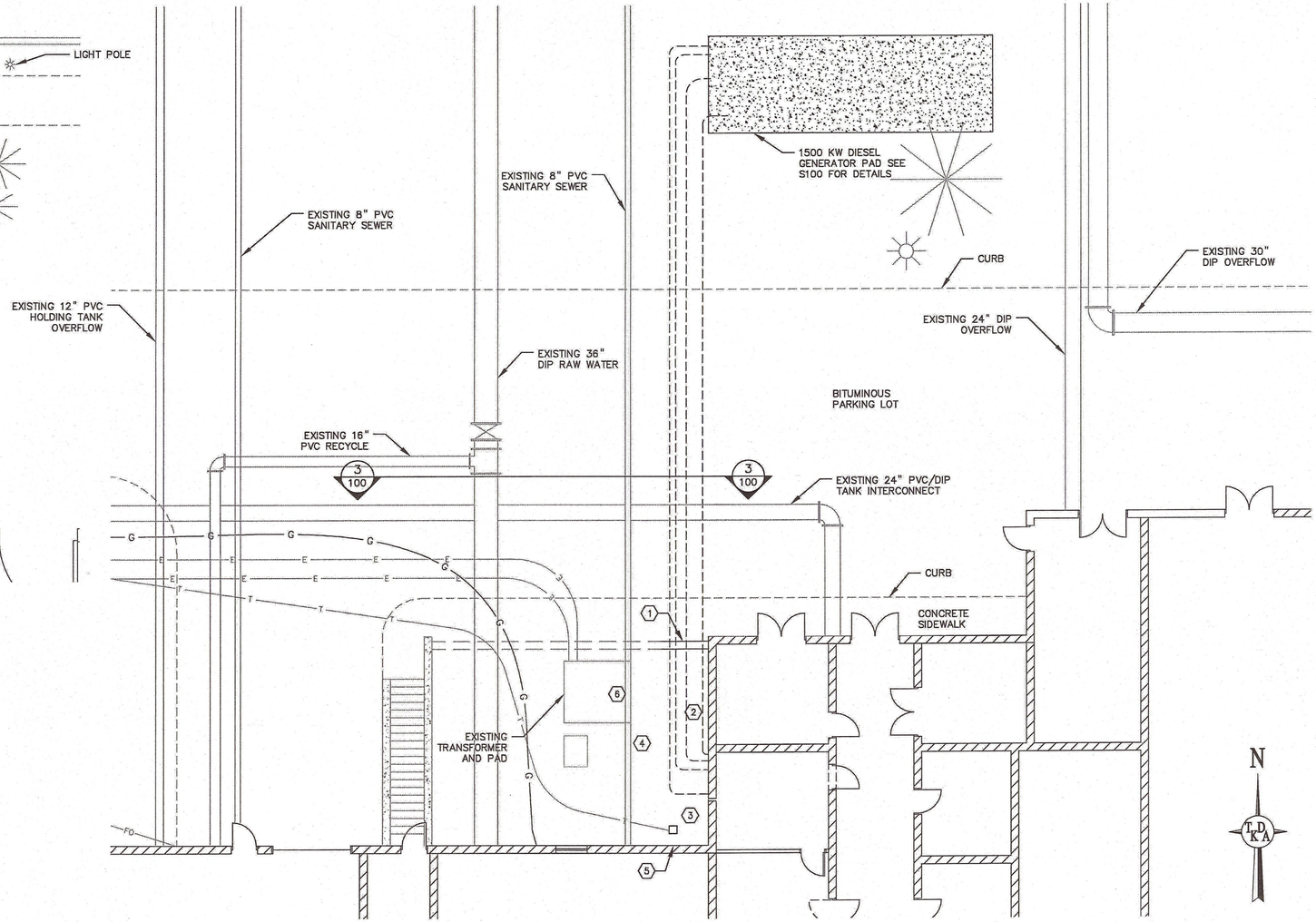
RECORD NO. SHEET NO. E107-2

FIELD BOOK:
Plot Date: 11/13/2012
Drawing name: K:\gn\Maple Grove\15088000\elec\dwgs\E107-2-PW RISER.dwg
Xrefs:



1/100 EXISTING PLAN
SCALE: 1" = 30'

3/100 SECTION
SCALE: 1" = 10'



2/100 PROPOSED PLAN
SCALE: 1" = 10'

GENERAL NOTES:

1. THE CONTRACTOR SHALL LOCATE ALL EXISTING UTILITIES. ALL UTILITIES SHALL BE MARKED PRIOR TO BEGINNING CONSTRUCTION.
2. ALL EXISTING UTILITIES MAY NOT BE SHOWN ON THIS PLAN AND THOSE THAT ARE SHOWN MAY NOT BE EXACT LOCATIONS.
3. EXISTING TRANSFORMER SHALL REMAIN IN SERVICE. PROTECT DURING CONSTRUCTION. WATER TREATMENT PLANT CAN BE SHUT DOWN FOR NO MORE THAN 12 HOURS.

KEY NOTES:

1. REMOVE EXISTING WALL AS SHOWN. SALVAGE AND REINSTALL BLOCKS FOR WALL INSTALLATION.
2. SALVAGE LANDSCAPING ROCK AND FILL FOR REINSTALLATION.
3. PLACE SALVAGED ROCK OVER ELECTRICAL CONDUITS FROM BLOCK WALL SOUTH TO BUILDING WALL AT ELEVATION EQUAL TO EXISTING BLOCK WALL WHEN ELECTRICAL WORK IS COMPLETE.
4. SLOPE SALVAGED ROCK FROM FLAT GRADE OVER CONDUITS TO ELEVATION OF EXISTING TRANSFORMER AND GRADE OF EXISTING LANDSCAPING ROCK.
5. PROVIDE A MINIMUM OF 1'-6" OF FILL ABOVE BUILDING FOUNDATION.
6. PROTECT EXISTING TRANSFORMER IN PLACE DURING EXCAVATION.

LEGEND

T	TELECOMMUNICATION
G	NATURAL GAS
E	ELECTRIC

NO.	DATE	BY	DESCRIPTION OF REVISIONS

DESIGNED JRN | DRAWN DCK | CHECKED VEJ

I HEREBY CERTIFY THAT THIS PLAN WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MINNESOTA
VERNE E. JACOBSEN
DATE 11/14/12 LIC. NO. 06988

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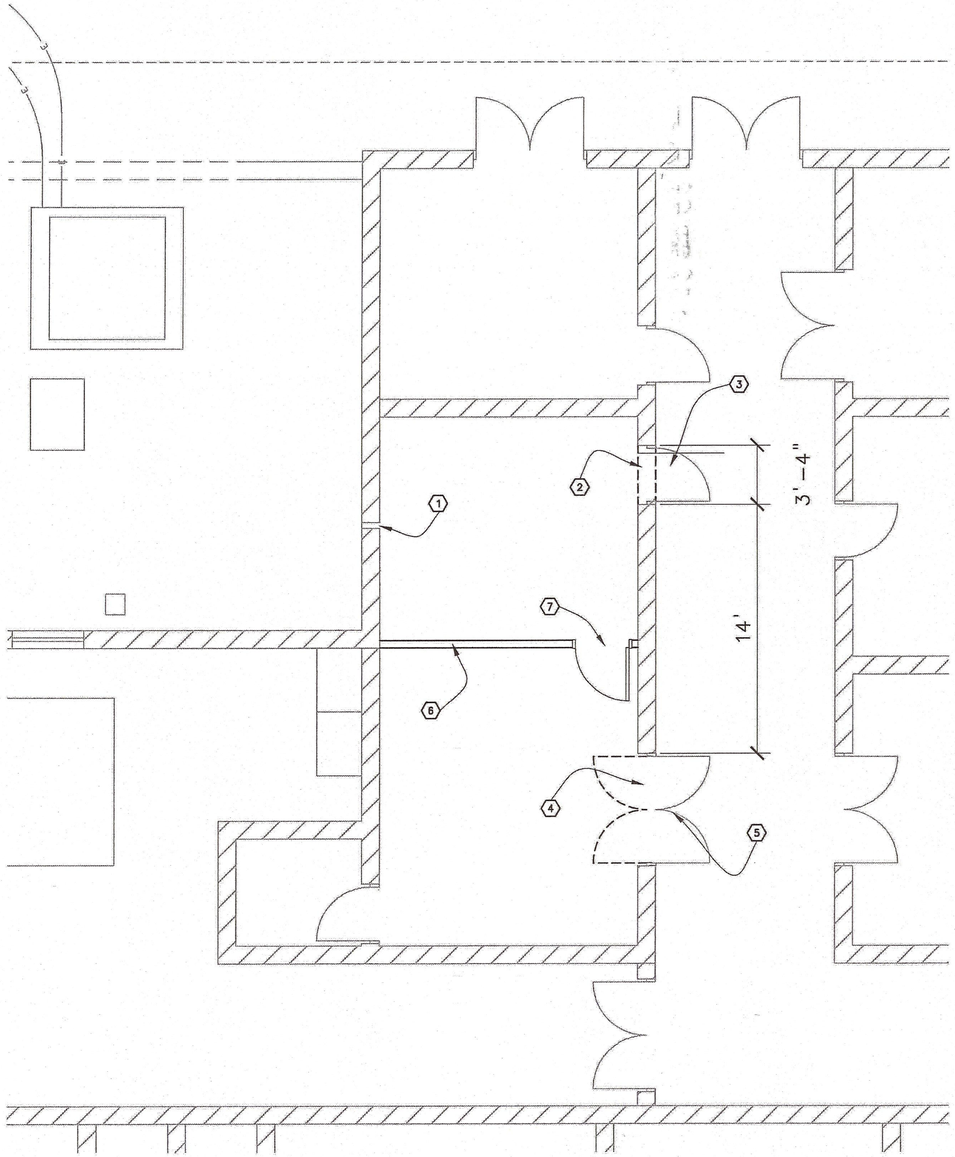
tkda.com

STANDBY GENERATOR PROJECT
MAPLE GROVE MINNESOTA
CLIENT NO. | MSA NO.

WATER TREATMENT PLANT

PROJECT NO. 15088.000
RECORD NO. | SHEET NO. C108-1

FIELD BOOK:
Plot Date: 11/13/2012
Drawing name: K:\grn\MapleGrove\15088000\mun\dngs\PlanSet\C108.dwg
Xrefs: WTP_ExistingBase



1 A 1.1 PLAN – PARTIAL LOWER LEVEL

SCALE: 1/4" = 1'

0 2 4 8

SCALE IN FEET

GENERAL NOTES

1. PROVIDE TEMPORARY ENCLOSURE TO CONTROL DUST WHEN CUTTING WALL FOR NEW DOOR OPENING..
2. SET DOORS PLUMB IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS AND TEMPLATES
3. VERIFY LOCATION OF ALL EXISTING OUTLETS, SWITCHES OR CONDUIT. SINGLE LEAF DOOR MAY BE MOVED TO EITHER SIDE TO AVOID CONDUIT, OUTLETS OR SWITCHES.
4. SIGNAGE OVER THE EXISTING DOUBLE DOOR TO BE REMOVED AND NEW SIGNAGE INSTALLED. NEW SIGNAGE TO MATCH EXISTING IN TYPE, COLOR AND SIZE. COORDINATE SIGNAGE WITH ELECTRICAL AND OWNER.
5. HARDWARE NOTED IS FOR BIDDING PURPOSES ONLY. VERIFY HARDWARE TYPE AND FINISH MATCHES EXISTING BUILDING HARDWARE PRIOR TO INSTALLATION.

KEY NOTES

(1) EXISTING 4" THRU WALL EXHAUST VENT PIPE TO BE REMOVED BY OTHERS. PATCH EXTERIOR WALL WITH BRICK TO MATCH EXISTING, INSTALL CAVITY WALL INSULATION TO MATCH EXISTING, PATCH INTERIOR AND PAINT TO MATCH EXISTING.

(2) CUT NEW DOOR OPENING IN EXISTING 12" CMU WALL, PROVIDE NEW LINTEL– PAIR OF 3 1/2"x3 1/2"x5/16" STL. ANGLES W/ MIN. 4" BEARING, INFILL CORES OF THE ENDS OF THE CUT CMU WITH CONCRETE AND PAINT TO MATCH EXISTING WALLS. AT THE NEW OPENING, FILL FLOOR WITH CONCRETE AND FINISH TO MATCH EXISTING.

(3) PROVIDE AND INSTALL NEW 3'–0" X 7'–0" STANDARD 1 HOUR RATED INDUSTRIAL HOLLOW METAL DOOR AND FRAME. FRAME 16 GA. MIN. DOOR 18 GA. MIN. CECO OR STEELCRAFT OR EQUAL. SECURE TO JAMBS WITH MIN. OF 3 EXISTING OPENING ANCHORS EACH SIDE. PAINT DOOR AND FRAME TO MATCH ADJACENT DOORS AND FRAMES. PROVIDE THE FOLLOWING HARDWARE OR EQUAL:
- HINGES: 3/ DOOR –BUTT HINGES– 5 KNUCKLE TYPE– STANLEY FBB168 OR EQUAL
- LOCK: RIM PANIC EXIT DEVICE , VON DUPRIN SERIES 99 ENTRANCE FUNCTION ONE RIM CYLINDER
- CLOSER: LCN 4041 OR EQUAL
- SMOKE GASKET SET

HARDWARE FINISH TO MATCH EXISTING FINISH.

(4) REMOVE COMPLETELY EXISTING HM DOOR AND FRAME. FILL ANY HOLES IN CMU WALL, PAINT TO MATCH EXISTING WALLS.

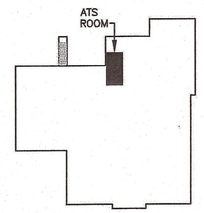
(5) PROVIDE AND INSTALL NEW 6'–0" X 7'–0" STANDARD 1 HOUR RATED INDUSTRIAL GAGE HOLLOW METAL DOOR AND FRAME INTO EXISTING OPENING. FRAME 16 GA. MIN. DOOR 18 GA. MIN. CECO OR STEELCRAFT OR EQUAL. SECURE TO JAMBS WITH MIN. OF 3 EXISTING OPENING ANCHORS EACH SIDE. PAINT DOOR AND FRAME TO MATCH ADJACENT DOORS AND FRAMES. PROVIDE THE FOLLOWING HARDWARE OR EQUAL:
- HINGES: 3/ DOOR –BUTT HINGES– 5 KNUCKLE TYPE– STANLEY FBB168 OR EQUAL
- PANIC HARDWARE– ONE 9947 LDR VON DUPRIN–PASSAGE
- MANUAL FLUSH BOLT AT HEAD
- CLOSER: LCN 4041 OR EQUAL
- ASTRAGAL– INTEGRAL PART OF THE INSIDE FACE OF THE INACTIVE LEAF.
- SMOKE GASKET SET

HARDWARE FINISH TO MATCH EXISTING FINISH.

(6) PROVIDE AND INSTALL NEW 3⅝" STL. STUD WALL W/ ⅝" TYPE "X" G.W.B EA. SIDE. TAPE AND PAINT TO MATCH EXISTING WALLS.

(7) PROVIDE AND INSTALL NEW 3'–0" X 7'–0" DOOR AND FRAME. PROVIDE THE FOLLOWING HARDWARE OR EQUAL:
- HINGES: 3/ DOOR –BUTT HINGES– 5 KNUCKLE TYPE– STANLEY FBB168 OR EQUAL
- LOCK: RIM PANIC EXIT DEVICE , VON DUPRIN SERIES 99 ENTRANCE FUNCTION ONE RIM CYLINDER
- CLOSER: LCN 4041 OR EQUAL
- SMOKE GASKET SET

HARDWARE FINISH TO MATCH EXISTING FINISH.



KEY PLAN

N

T.D.K.A.

NO.	DATE	BY	DESCRIPTION OF REVISIONS

DESIGNED PMB DRAWN PMB CHECKED PMB

I HEREBY CERTIFY THAT THIS PLAN WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY LICENSED ARCHITECT UNDER THE LAWS OF THE STATE OF MINNESOTA.
PETER M. BROZEK
DATE 11/14/12 LIC. NO. 17597

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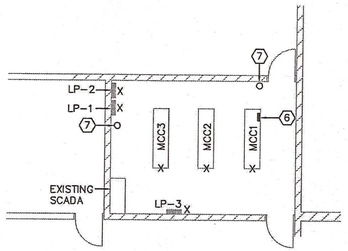
tkda.com

STANDBY GENERATOR PROJECT
MAPLE GROVE MINNESOTA
CLIENT NO. MSA NO.

WATER TREATMENT PLANT
FLOOR PLAN AND DETAILS

PROJECT NO. 15088.000
RECORD NO. SHEET NO. A 1.1

FIELD BOOK:
Plot Date: 11/13/2012
Drawing name: K:\gm\MapleGrove\15088000\Arch\A1.1-WTP.dwg
Xrefs: WTP_ExistingBase



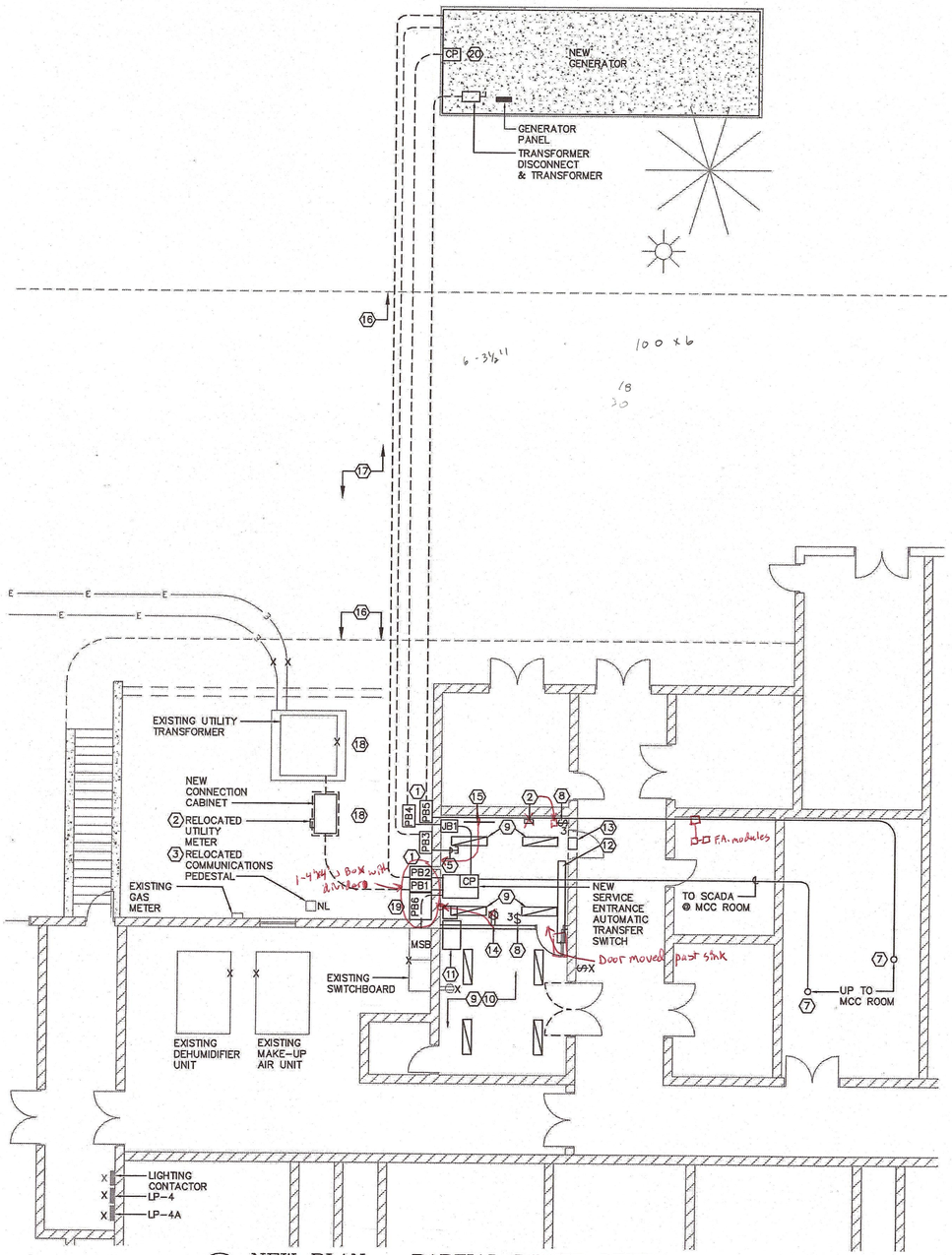
EXISTING PLAN MCC ROOM PARTIAL UPPER LEVEL

3 / 108-1

SCALE: 1/8" = 1'

0 4 8 16

SCALE IN FEET



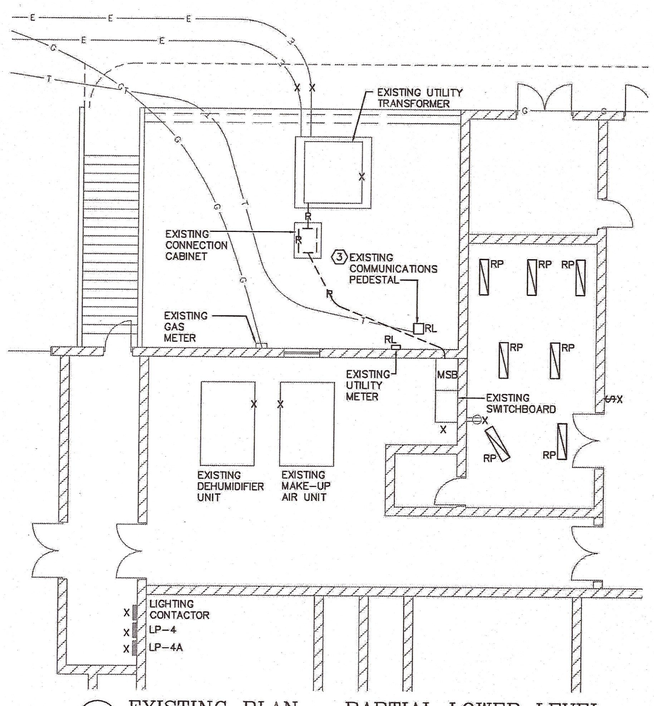
NEW PLAN – PARTIAL LOWER LEVEL

2 / 108-1

SCALE: 1/8" = 1'

0 4 8 16

SCALE IN FEET



EXISTING PLAN – PARTIAL LOWER LEVEL

1 / 108-1

SCALE: 1/8" = 1'

0 4 8 16

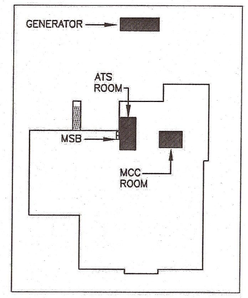
SCALE IN FEET

GENERAL NOTES

1. ALL RACEWAYS SHALL CONTAIN A GROUNDING CONDUCTOR.
2. GENERATOR CONDUIT ROUTING SHOWN FOR INFORMATION ONLY. VERIFY LOCATION OF CONDUIT STUB UPS WITH GENERATOR MANUFACTURER.
3. INTERIOR CONDUIT ROUTING SHOWN FOR INFORMATION ONLY. FIELD COORDINATE WITH OWNER.
4. VERIFY LOCATION OF ALL EXISTING SITE UTILITIES PRIOR TO EXCAVATION.
5. WALLS AND FLOOR ARE CONCRETE. CORE DRILLING AND SAW CUTTING WILL BE REQUIRED. FIELD COORDINATE.

KEY NOTES

1. PAINT ALL NEW PULLBOXES TO MATCH ADJACENT SURFACE.
2. RELOCATE EXISTING METER AND BASE TO NEW CONNECTION CABINET. PATCH EXISTING HOLE IN BLOCK WALL. COORDINATE WITH ELECTRICAL UTILITY.
3. RELOCATE EXISTING TELEPHONE SERVICE. EXTEND EXISTING CONDUIT AND WIRING TO NEW LOCATION.
4. GENERATOR REMOTE MONITOR PANEL.
5. FUTURE LIFE SAFETY CONDUIT.
6. LOCATION OF EXISTING CIRCUIT BREAKER TO SERVE GENERATOR PANEL, ETC.
7. PROPOSED CONDUIT STUB-UP THROUGH FLOOR.
8. PROVIDE NEW SWITCHING FOR CONTROL OF NEW LIGHT FIXTURES.
9. PROVIDE NEW FLUORESCENT FIXTURE EQUAL TO LITHONIA AF10-2-32-MVOLT-GEB CHAIN HUNG. FIELD COORDINATE LOCATION WITH EXISTING PIPING, DUCTWORK ETC. CONNECT TO EXISTING LIGHTING CIRCUIT.
10. CONNECT NEW LIGHTING TO EXISTING LIGHTING CIRCUIT AND SWITCHING.
11. EXISTING SANDBLASTER AND ASSOCIATED DUST COLLECTOR TO BE REMOVED BY OWNER. DISCONNECT, CAP AND REMOVE 15' OF EXISTING COMPRESSED AIR PIPING. DISCONNECT AND REMOVE EXISTING EXHAUST DUCT DOWNSTREAM OF DUST COLLECTOR. FILL IN WALL PENETRATION WITH LIKE MATERIAL AND PAINT TO MATCH.
12. EXISTING 8x10 SUPPLY DUCT. REMOVE 4' SECTION OF DUCTWORK BETWEEN EXISTING SUPPLY DIFFUSER ON BOTTOM OF THE DUCT AND THE NEW WALL. RELOCATED SUPPLY DIFFUSER SHALL BE LOCATED IN THE NEW ROOM.
13. EXISTING TRANSFER DUCT TO REMAIN.
14. PROVIDE NEW DUPLEX RECEPTACLE. CONNECT TO NEAREST EXISTING RECEPTACLE CIRCUIT.
15. GROUND BAR LOCATION. SEE DETAIL DRAWING E100.
16. REMOVE A PARTIAL SECTION OF EXISTING SIDEWALK AND CURBING TO ACCOMMODATE TRENCHING FOR NEW GENERATOR CONDUITS. REPLACE CONCRETE AFTER WORK HAS BEEN COMPLETED.
17. SEE CIVIL DRAWINGS FOR BITUMINOUS WORK THIS AREA.
18. SEE STRUCTURAL AND CIVIL DRAWINGS FOR GRADING AND TEMPORARY SHORING REQUIRED AT THIS AREA.
19. CONTRACTOR SHALL PROVIDE MINIMUM 18" OF COVER FOR FROST PROTECTION TO THE TOP OF THE EXISTING FOOTING THIS AREA.
20. PROVIDE FIRE ALARM CONNECTION FROM GENERATOR ENCLOSURE CONTACT TO EXISTING BUILDING FIRE ALARM SYSTEM. PROVIDE ALL PROGRAMMING AND REQUIRED EQUIPMENT.



KEY PLAN

N

NO.	DATE	BY	DESCRIPTION OF REVISIONS

DESIGNED RCS | DRAWN RCS | CHECKED RLY

I HEREBY CERTIFY THAT THIS PLAN WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MINNESOTA

ROBERT L. YOUNG

DATE 11/14/12 LIC. NO. 26746

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651.292.4400
tkda.com

STANDBY GENERATOR PROJECT

MAPLE GROVE MINNESOTA

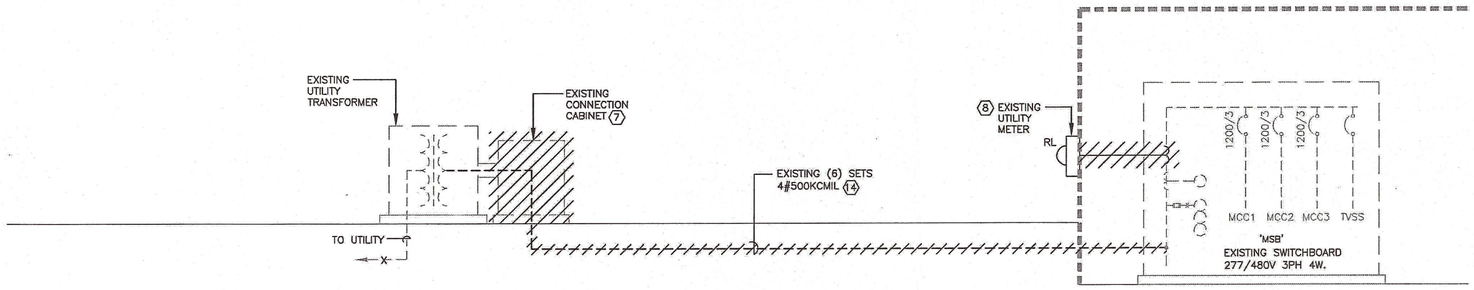
CLIENT NO. MSA NO.

WATER TREATMENT PLANT ELECTRICAL FLOOR PLAN

PROJECT NO. 15088.000

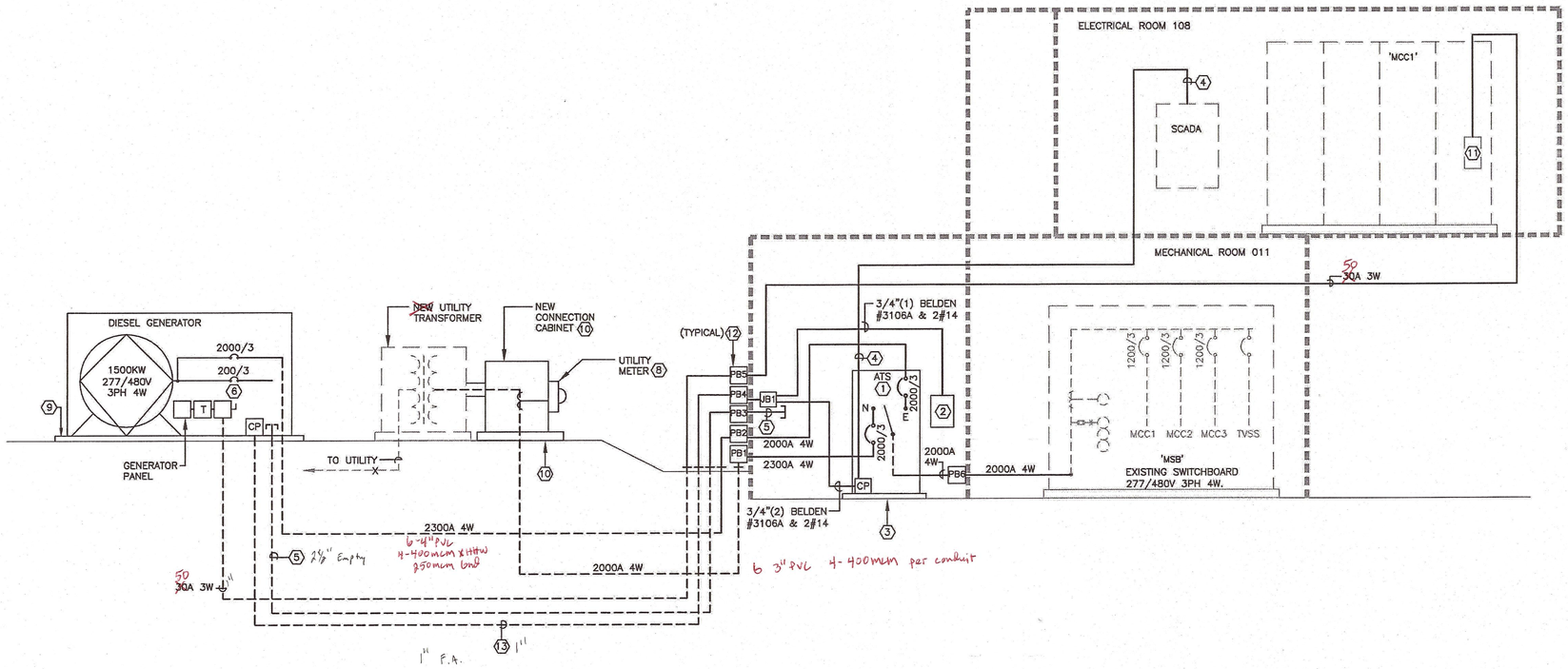
RECORD NO. SHEET NO. E108-1

FIELD BOOK:
PLOT DATE: 11/15/2012
Drawing name: K:\p\n\Maple Grove\15088000\elec\dwgs\E108-1-WTP.dwg
Xrefs: WTP_ExistingBase



1/108-2 EXISTING PARTIAL RISER DIAGRAM

SCALE: NONE



2/108-2 NEW PARTIAL RISER DIAGRAM

SCALE: NONE

KEY NOTES

(1) AUTOMATIC TRANSFER SWITCH TO BE 480 VOLT, 2000 AMP, SERVICE ENTRANCE RATED, 4 POLE OPEN TRANSITION, NEMA 1 TYPE, WITH (2) POWER CIRCUIT BREAKERS. EQUAL TO EATON DS, 100,000 AIC. SEE SPECIFICATIONS. UTILITY AND GENERATOR FEEDER ENTRY ROUTING FROM BUILDING EXTERIOR TO WITHIN ATS ENCLOSURE IS ANTICIPATED BEING THROUGH THE REAR OF THE EQUIPMENT. LOAD FEEDER ROUTING IS ANTICIPATED BEING INSTALLED VIA CONDUIT AT REAR OF ATS.

(2) GENERATOR REMOTE MONITOR PANEL.

(3) PROVIDE 4" HOUSEKEEPING PAD FOR NEW ATS. SEE STRUCTURAL DETAILS.

(4) PROVIDE 3/4" CONDUIT WITH (1) BELDEN #3106A COMMUNICATION CABLE FOR MODBUS COMMUNICATIONS BETWEEN THE TRANSFER SWITCH AND THE PLC. SUBCONTRACT TO THE CITY'S SCADA SYSTEM INTEGRATOR. INTEGRATOR SHALL PROVIDE ALL PROGRAMMING AND HARDWARE REQUIRED TO MONITOR AND DISPLAY THE GENERATOR STATUS AND ALARMS. INTEGRATOR SHALL PROVIDE NEW SCADA SYSTEM SCREEN AT THE PUBLIC WORKS AND THE WATER TREATMENT PLANT BUILDING HMI TO DISPLAY THE GENERATOR AND TRANSFER SWITCH STATUS AND ALARMS. NEW SCREEN WILL ALLOW START-STOP OF THE GENERATOR AND REMOTE CONTROL OF THE AUTOMATIC TRANSFER SWITCH WITH PASSWORD PROTECTION. IN ADDITION THIS SCREEN SHALL ALLOW SELECTION OF WHICH HIGH SERVICE PUMPS ARE INHIBITED WHEN THE PLANT IS OPERATING ON GENERATOR POWER. A TOTAL OF 2 OF THE 7 HIGH SERVICE PUMPS WILL BE INHIBITED WHEN OPERATING ON GENERATOR POWER. PROVIDE PLC LOGIC TO ACCOMPLISH THIS TASK. ANY HIGH SERVICE PUMP NOT IN THE 'AUTOMATIC' MODE WILL COUNT AS ONE OF THE INHIBITED PUMPS. IF ONE OR TWO HIGH SERVICE PUMPS ARE IN THE 'OFF' MODE THE PUMP(S) SELECTED FOR INHIBITION SHALL BE OVER-RIDDEN. LOGIC SHALL ALLOW A MAXIMUM OF 5 HIGH SERVICE PMPS TO RUN IF AT LEAST 5 PUMPS ARE IN THE 'AUTO' MODE. PUMP STARTING SHALL BE STAGGERED. INTEGRATOR SHALL MODIFY THE EXISTING PLC TO PROVIDE MODBUS COMMUNICATIONS.

(5) EMPTY 2-1/2" CONDUIT FOR CONNECTION TO FUTURE LIFE SAFETY PANEL. PROVIDE PULLSTRING, CAP AND LABEL ENDS.

(6) PROVIDE LIFE SAFETY CIRCUIT BREAKER FOR FUTURE USE. PROVIDE SEPARATE LIFE SAFETY ENCLOSURE.

(7) REMOVE EXISTING CONNECTION CABINET ASSOCIATED WIRING AND CONCRETE PAD.

(8) RELOCATE EXISTING UTILITY METER. PROVIDE CONDUIT WIRING AND CT'S TO NEW CONNECTION CABINET. COORDINATE WITH XCEL ENERGY. REMOVE EXISTING UTILITY CT'S FROM MSB AND MODIFY BUS AS REQUIRED. RETURN CT'S TO UTILITY.

(9) PROVIDE GENERATOR CONCRETE MOUNTING PAD, SEE STRUCTURAL DRAWINGS FOR FURTHER INFORMATION.

(10) NEW CONNECTION CABINET. 2500A, 480V 3PH 4W, NEMA 3R, 85KA WITH UTILITY C.T. PROVISIONS. PROVIDE CONCRETE MOUNTING PAD.

(11) EXISTING 20/3 CIRCUIT BREAKER AT EXISTING MCC1 TO SERVE GENERATOR PANEL ETC.

(12) PROVIDE WEATHERPROOF PULLBOX. SIZE PER NEC.

(13) PROVIDE 1"PVC CONDUIT WITH (2) #14 FOR GENERATOR MONITOR PANEL POWER, (2) #14 FOR GENERATOR START SIGNAL, (2) BELDEN #3106A COMMUNICATION CABLE FOR MODBUS COMMUNICATIONS BETWEEN THE GENERATOR AND THE TRANSFER SWITCH AND BETWEEN THE GENERATOR AND REMOTE MONITOR PANEL. 2 #14 Spares

(14) REMOVE EXISTING SERVICE CONDUIT AND FEEDERS.

GENERAL NOTES

1. ALL RACEWAYS SHALL CONTAIN A GROUNDING CONDUCTOR.
2. CONDUIT ROUTING SHOWN FOR INFORMATION ONLY.
3. VERIFY LOCATION OF ALL EXISTING SITE UTILITIES PRIOR TO EXCAVATION.
4. PROVIDE PLASTIC LAMINATE NAMEPLATE ADJACENT TO THE GENERATOR CIRCUIT BREAKER WHICH READS: "THIS GENERATOR SERVES THE WATER TREATMENT PLANT AND ALL ASSOCIATED BUILDING LOADS"

N

LDKA

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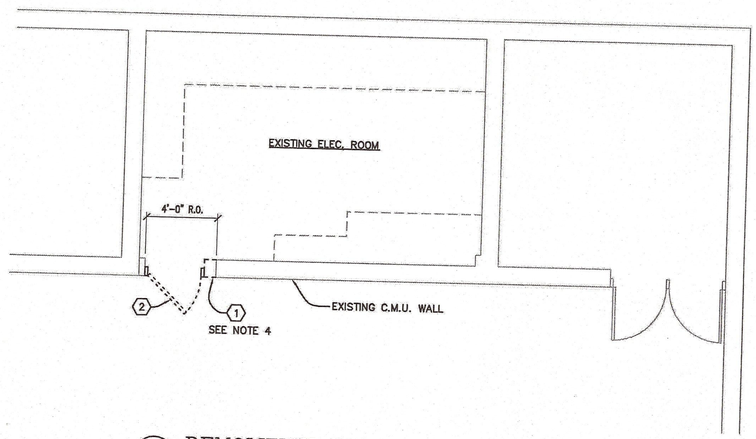
STANDBY GENERATOR PROJECT
MAPLE GROVE MINNESOTA
CLIENT NO. MSA NO.

WATER TREATMENT PLANT
PARTIAL RISER DIAGRAM

PROJECT NO. 15088.000
RECORD NO. SHEET NO. E108-2

FIELD BOOK:
Plot Date: 11/13/2012
Drawing Name: K:\gm\Maple Grove\15088000\eled\dwgs\E108-2-WTP RISER.dwg
Xrefs:



1 / A 1.2 DEMOLITION PLAN

SCALE: 1/4" = 1'

0 2 4 8

SCALE IN FEET

GENERAL NOTES

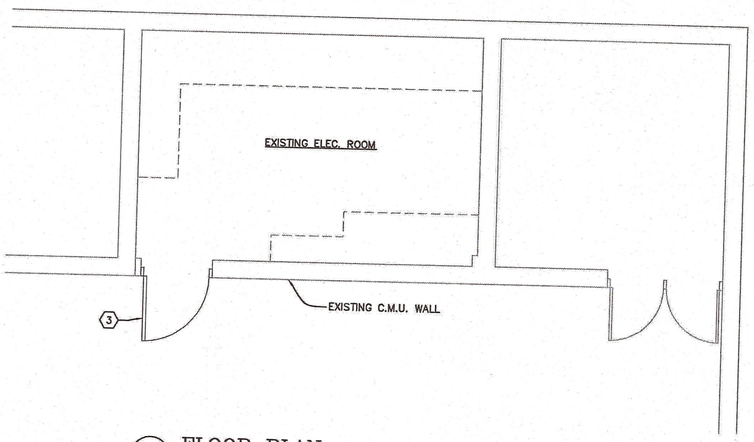
1. PROVIDE TEMPORARY ENCLOSURE TO CONTROL DUST WHEN CUTTING WALL FOR NEW DOOR OPENING..
2. SET DOORS PLUMB IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS AND TEMPLATES
3. COORDINATE REMOVAL & RELOCATION OF ELECTRICAL DEVICES OR WIRING ON OR IN WALL PRIOR TO DEMOLITION.
4. PREPARE NEW OPENING IN WALL AND PROVIDE TEMPORARY PLYWOOD DOOR W/ PADLOCK UNTIL NEW ELECTRICAL EQUIPMENT IS INSTALLED IN ROOM. INSTALLATION OF DOOR AND FRAME SHALL OCCUR ONLY AFTER ELECTRICAL EQUIPMENT HAS BEEN INSTALLED INSIDE ROOM. WALL DEMOLITION IS REQUIRED TO INSTALL EQUIPMENT.
5. HARDWARE NOTED IS FOR BIDDING PURPOSES ONLY. VERIFY HARDWARE TYPE AND FINISH MATCHES EXISTING BUILDING HARDWARE PRIOR TO INSTALLATION.

KEY NOTES

(1) DEMOLISH PORTION OF EXISTING C.M.U. WALL (FIELD VERIFY WALL THICKNESS) TO CREATE 4'-0"W x 7'-4"H R.O. FOR NEW DOOR & FRAME, PROVIDE NEW LINTEL – PAIR OF 3 1/2"x3 1/2"x5/16" STL. ANGLES W/ MIN. 4" BEARING EA. SIDE – PAINT EXPOSED PORTION OF ANGLES TO MATCH DOOR & FRAME, INFILL CORES OF THE ENDS OF THE CUT C.M.U. WITH CONCRETE. PATCH FLOOR AT DEMOLISHED WALL WITH CONCRETE AND FINISH TO MATCH EXISTING.

(2) REMOVE COMPLETELY EXISTING H.M. DOOR AND FRAME. FILL ANY HOLES IN C.M.U. WALL

(3) PROVIDE AND INSTALL NEW 3'-8" X 7'-0" STANDARD INDUSTRIAL HOLLOW METAL DOOR AND FRAME. FRAME 16 GA. MIN. DOOR 18 GA. MIN. CECO OR STEELCRAFT OR EQUAL. SECURE TO JAMBS WITH MIN. OF 3 EXISTING OPENING ANCHORS EACH SIDE. PAINT DOOR AND FRAME TO MATCH ADJACENT DOORS AND FRAMES. PROVIDE THE FOLLOWING HARDWARE OR EQUAL:
HINGES: 4/ DOOR –BUTT HINGES– 5 KNUCKLE TYPE– STANLEY FBB168 OR EQUAL
LOCK: RIM PANIC EXIT DEVICE , VON DUPRIN SERIES 99 ENTRANCE FUNCTION ONE RIM CYLINDER
CLOSER: LCN 4041 OR EQUAL
HARDWARE FINISH TO MATCH EXISTING FINISH.



2 / A 1.2 FLOOR PLAN

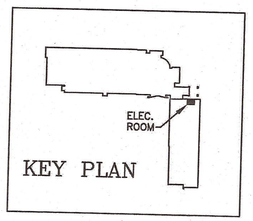
SCALE: 1/4" = 1'

0 2 4 8

SCALE IN FEET

N

T.D. K.A.



FIELD BOOK:
Plot Date: 11/13/2012
Drawing name: K:\p-m\MapleGrove\15088000\Arch\A-1-2-CITY HALL.dwg
Xrefs: Wall #5-10_FLOORPLAN

NO.	DATE	BY	DESCRIPTION OF REVISIONS

DESIGNED PMB DRAWN DNC
CHECKED PMB

I HEREBY CERTIFY THAT THIS PLAN WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY LICENSED ARCHITECT UNDER THE LAWS OF THE STATE OF MINNESOTA
PETER M. BROZEK
DATE 11/14/12 LIC. NO. 17597

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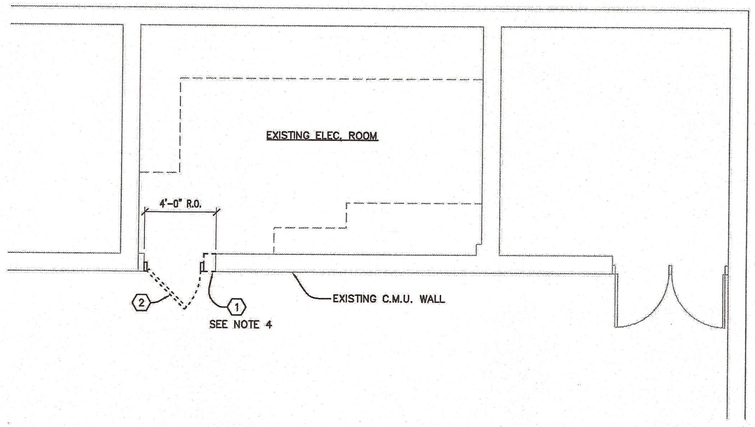
tkda.com

STANDBY GENERATOR PROJECT
MAPLE GROVE MINNESOTA
CLIENT NO.

GOVERNMENT CENTER
PUBLIC SAFETY
DEMOLITION AND FLOOR PLANS

PROJECT NO. 15088.000
RECORD NO.
SHEET NO.



1 / A 1.2 DEMOLITION PLAN

SCALE: 1/4" = 1'

0 2 4 8

SCALE IN FEET

GENERAL NOTES

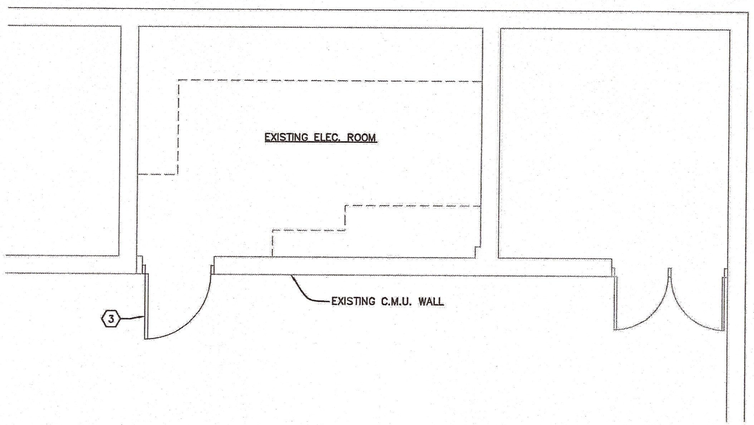
1. PROVIDE TEMPORARY ENCLOSURE TO CONTROL DUST WHEN CUTTING WALL FOR NEW DOOR OPENING..
2. SET DOORS PLUMB IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS AND TEMPLATES
3. COORDINATE REMOVAL & RELOCATION OF ELECTRICAL DEVICES OR WIRING ON OR IN WALL PRIOR TO DEMOLITION.
4. PREPARE NEW OPENING IN WALL AND PROVIDE TEMPORARY PLYWOOD DOOR W/ PADLOCK UNTIL NEW ELECTRICAL EQUIPMENT IS INSTALLED IN ROOM. INSTALLATION OF DOOR AND FRAME SHALL OCCUR ONLY AFTER ELECTRICAL EQUIPMENT HAS BEEN INSTALLED INSIDE ROOM. WALL DEMOLITION IS REQUIRED TO INSTALL EQUIPMENT.
5. HARDWARE NOTED IS FOR BIDDING PURPOSES ONLY. VERIFY HARDWARE TYPE AND FINISH MATCHES EXISTING BUILDING HARDWARE PRIOR TO INSTALLATION.

KEY NOTES

(1) DEMOLISH PORTION OF EXISTING C.M.U. WALL (FIELD VERIFY WALL THICKNESS) TO CREATE 4'-0"Wx7'-4"H R.O. FOR NEW DOOR & FRAME, PROVIDE NEW LINTEL – PAIR OF 3 1/2"x3 1/2"x5/16" STL. ANGLES W/ MIN. 4" BEARING EA. SIDE – PAINT EXPOSED PORTION OF ANGLES TO MATCH DOOR & FRAME, INFILL CORES OF THE ENDS OF THE CUT C.M.U. WITH CONCRETE. PATCH FLOOR AT DEMOLISHED WALL WITH CONCRETE AND FINISH TO MATCH EXISTING.

(2) REMOVE COMPLETELY EXISTING H.M. DOOR AND FRAME. FILL ANY HOLES IN C.M.U. WALL.

(3) PROVIDE AND INSTALL NEW 3'-8" X 7'-0" STANDARD INDUSTRIAL HOLLOW METAL DOOR AND FRAME. FRAME 16 GA. MIN. DOOR 18 GA. MIN. CECO OR STEELCRAFT OR EQUAL. SECURE TO JAMBS WITH MIN. OF 3 EXISTING OPENING ANCHORS EACH SIDE. PAINT DOOR AND FRAME TO MATCH ADJACENT DOORS AND FRAMES. PROVIDE THE FOLLOWING HARDWARE OR EQUAL:
HINGES: 4/ DOOR –BUTT HINGES– 5 KNUCKLE TYPE– STANLEY FBB168 OR EQUAL
LOCK: RIM PANIC EXIT DEVICE , VON DUPRIN SERIES 99 ENTRANCE FUNCTION ONE RIM CYLINDER
CLOSER: LCN 4041 OR EQUAL
HARDWARE FINISH TO MATCH EXISTING FINISH.



2 / A 1.2 FLOOR PLAN

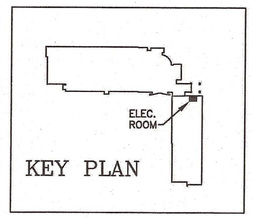
SCALE: 1/4" = 1'

0 2 4 8

SCALE IN FEET

N

TKDA



FIELD BOOK:
Plot Date: 11/13/2012
Drawing name: K:\gm\Maple\Grove\15088000\Arch\A-1.2-CITY HALL.dwg
Xrefs: Wall #s-10_FLOORPLAN

NO.	DATE	BY	DESCRIPTION OF REVISIONS

DESIGNED PMB DRAWN DNC
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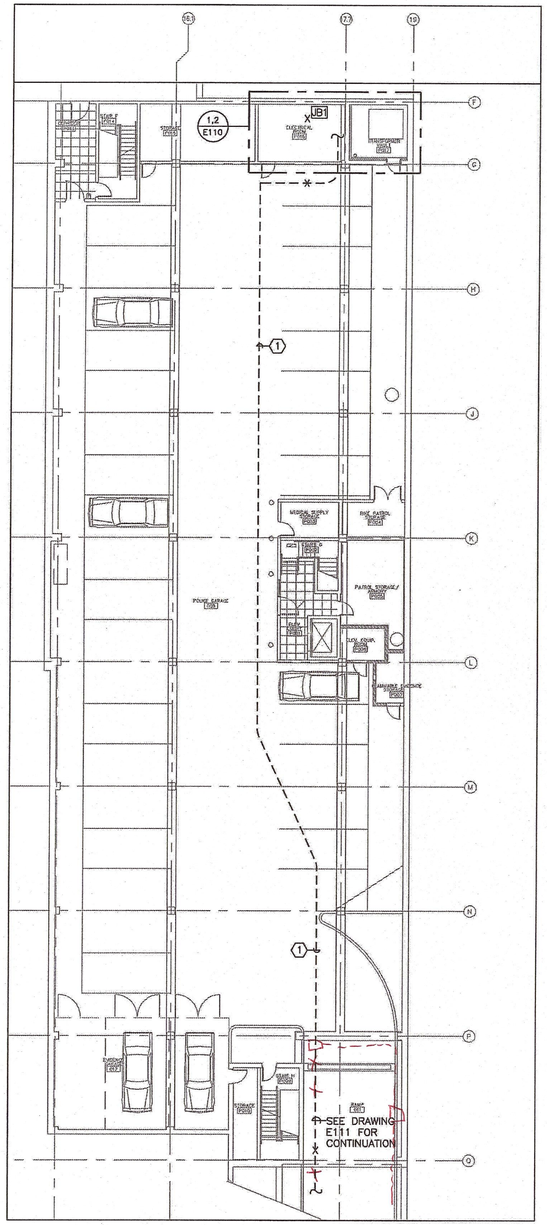
STANDBY GENERATOR PROJECT
MAPLE GROVE MINNESOTA
CLIENT NO. MSA NO.

GOVERNMENT CENTER
PUBLIC SAFETY
DEMOLITION AND FLOOR PLANS

PROJECT NO. 15088.000
RECORD NO.
SHEET NO. A 1.2

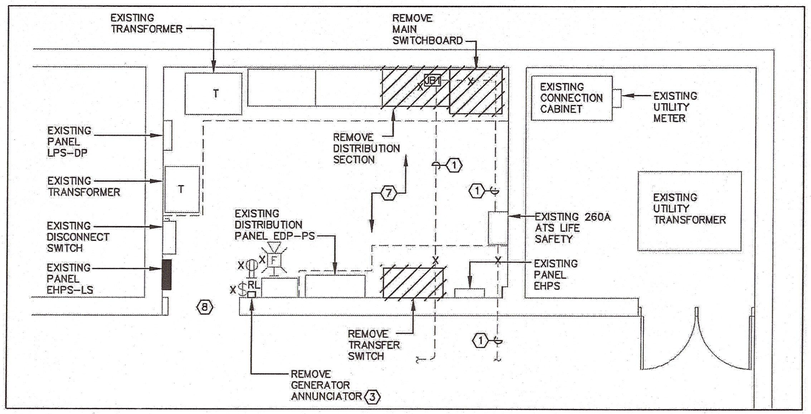


3 E110 GROUND FLOOR ELECTRICAL PLAN

SCALE: 1/4" = 1'

0 8 16 32

SCALE IN FEET

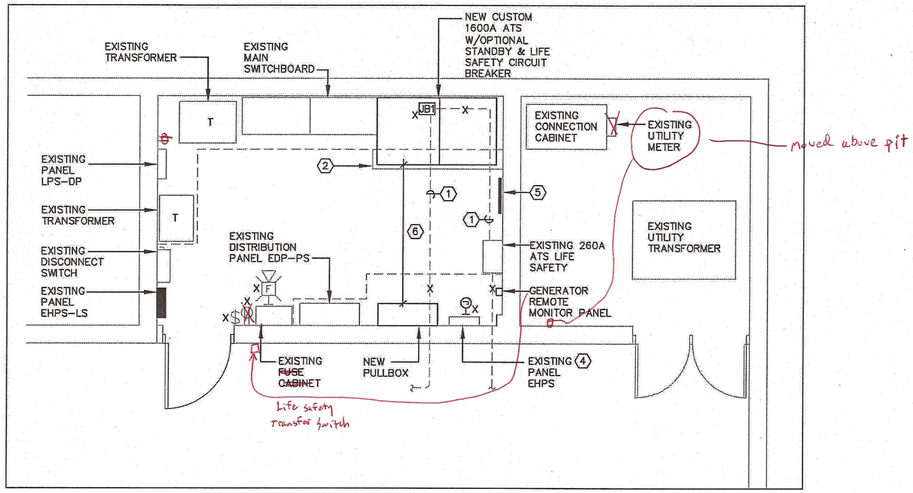


1 E110 EXISTING PLAN

SCALE: 1/4" = 1'

0 2 4 8

SCALE IN FEET



2 E110 NEW ELECTRICAL ROOM PLAN

SCALE: 1/4" = 1'

0 2 4 8

SCALE IN FEET

GENERAL NOTES

A. SEE RISER DIAGRAM FOR FURTHER INFORMATION.

KEY NOTES

1. ROUTING OF EXISTING GENERATOR POWER AND CONTROL LIFE SAFETY CONDUITS.
2. PROVIDE NEW CONCRETE HOUSEKEEPING PAD BENEATH NEW EQUIPMENT TO MATCH EXISTING SLAB HEIGHT.
3. REMOVE EXISTING GENERATOR ANNUNCIATOR. RETURN TO OWNER FOR USE AT WELL 6 & 10.
4. INTERCEPT AND EXTEND EXISTING CIRCUITS ~~8~~,10,12,~~14~~,16,~~18~~ TO LIFE SAFETY PANEL 'EHPS–LS'.
5. GROUND BAR LOCATION SEE DETAIL E100.
6. MAINTAIN MIN 8'–0" CLEARANCE TO MAINTAIN NEC REQUIRED WORKING CLEARANCE.
7. EXISTING UNDERGROUND CONDUITS AND CONDUCTORS TO REMAIN BETWEEN REMOVED DISTRIBUTION SECTION AND REMOVE TRANSFER SWITCH. PROTECT IN PLACE. COORDINATE QUANTITY AND SIZE PRIOR TO REMOVAL AND FABRICATION OF NEW CUSTOM ATS AND PULLBOX ENCLOSURES.
8. DOOR OPENING TO BE MODIFIED TO ALLOW INSTALLATION OF NEW CUSTOM 1600A ATS.

N

KEY PLAN

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STANDBY GENERATOR PROJECT

MAPLE GROVE MINNESOTA

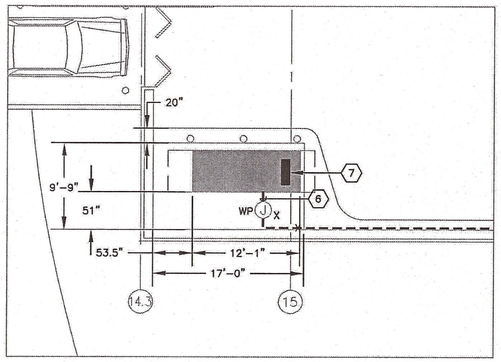
CLIENT NO. MSA NO.

GOVERNMENT CENTER
PUBLIC SAFETY
ELECTRICAL ROOM PLAN

PROJECT NO. 15088.000

RECORD NO. SHEET NO. E110

FIELD DOCK
Plot Date: 11/13/2012
Drawing name: K:\gm\MapleGrove\15088000\electric\dgn\E110-city hall.dwg
Xrefs: Well #6-10_FLOORPLAN_PS_BA00

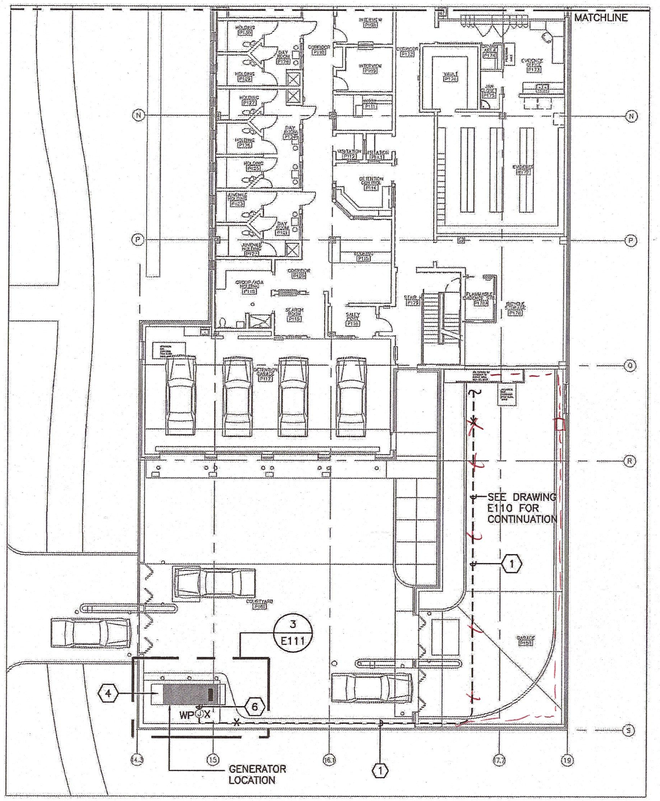


3 E111 ENLARGED GENERATOR PLAN

SCALE: 1/4" = 1'

0 4 8 16

SCALE IN FEET

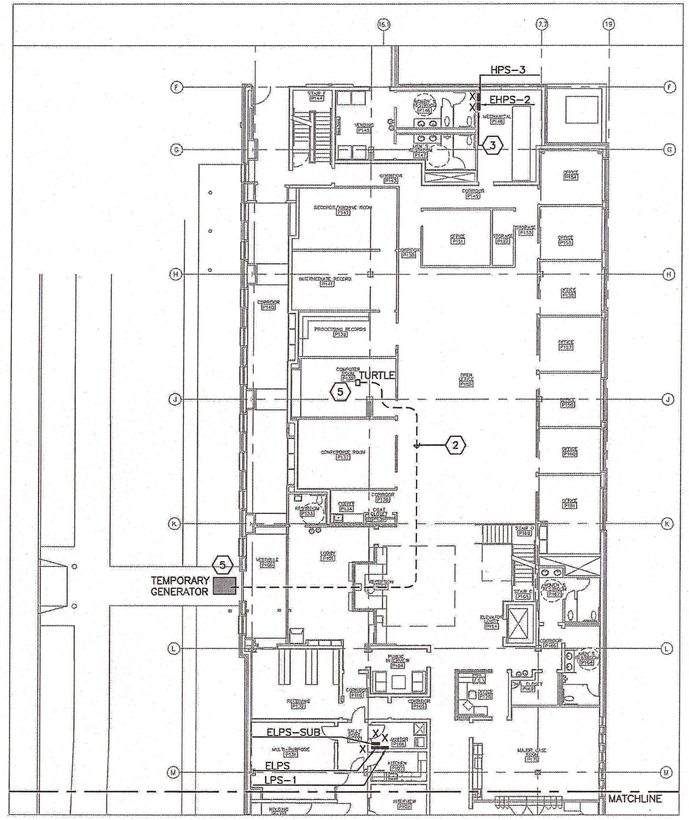


2 E111 FIRST FLOOR ELECTRICAL PLAN

SCALE: 1/16" = 1'

0 8 16 32

SCALE IN FEET



1 E111 FIRST FLOOR ELECTRICAL PLAN

SCALE: 1/16" = 1'

0 8 16 32

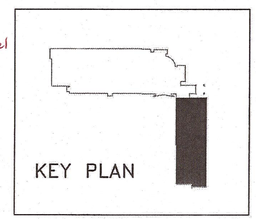
SCALE IN FEET

GENERAL NOTES

A. SEE RISER DIAGRAM FOR FURTHER INFORMATION.

KEY NOTES

1. ROUTING OF EXISTING LIFE SAFETY POWER & CONTROL CONDUIT. WIRING SHALL BE PROVIDED UNDER THIS CONTRACT.
2. PROPOSED ELECTRICAL FEEDER ROUTE TO MAINTAIN BUILDING SECURITY DURING OUTAGE. TEMPORARY FEEDER SHALL BE ROUTED THROUGH RECEPTION PASS-THRU.
3. INTERCEPT AND EXTEND EXISTING PANEL 'EHPS-2' CIRCUITS 36,38,40 TO EXISTING LIFE SAFETY PANEL 'EHPS-LS'. UTILIZE EXISTING CONDUIT. Left spares in new J Box by panel
4. REMOVE AND REPLACE EXISTING GENERATOR WITH NEW CUSTOM BUILT GENERATOR TO FIT EXISTING SLAB. VERIFY EXACT SIZE OF GENERATOR PAD PRIOR PURCHASE OF GENERATOR
5. PROVIDE PORTABLE 5KW GENERATOR FOR 120V EQUIPMENT LOCATED IN THIS ROOM DURING THE UTILITY OUTAGE. PROVIDE TURTLE FOR CORD AND PLUG CONNECTION OF EQUIPMENT. COORDINATE WITH OWNER.
6. PROVIDE LIFE SAFETY AND CONTROL CONDUITS FROM EXISTING "WP" WEATHERPROOF PULLBOXES TO NEW GENERATOR LOCATION.
7. APPROXIMATE LOCATION OF EXISITNG CONDUIT STUB-UP. FIELD VERIFY.



NO.	DATE	BY	DESCRIPTION OF REVISIONS

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CHECKED RLY

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STANDBY GENERATOR PROJECT

MAPLE GROVE MINNESOTA

CLIENT NO. MSA NO.

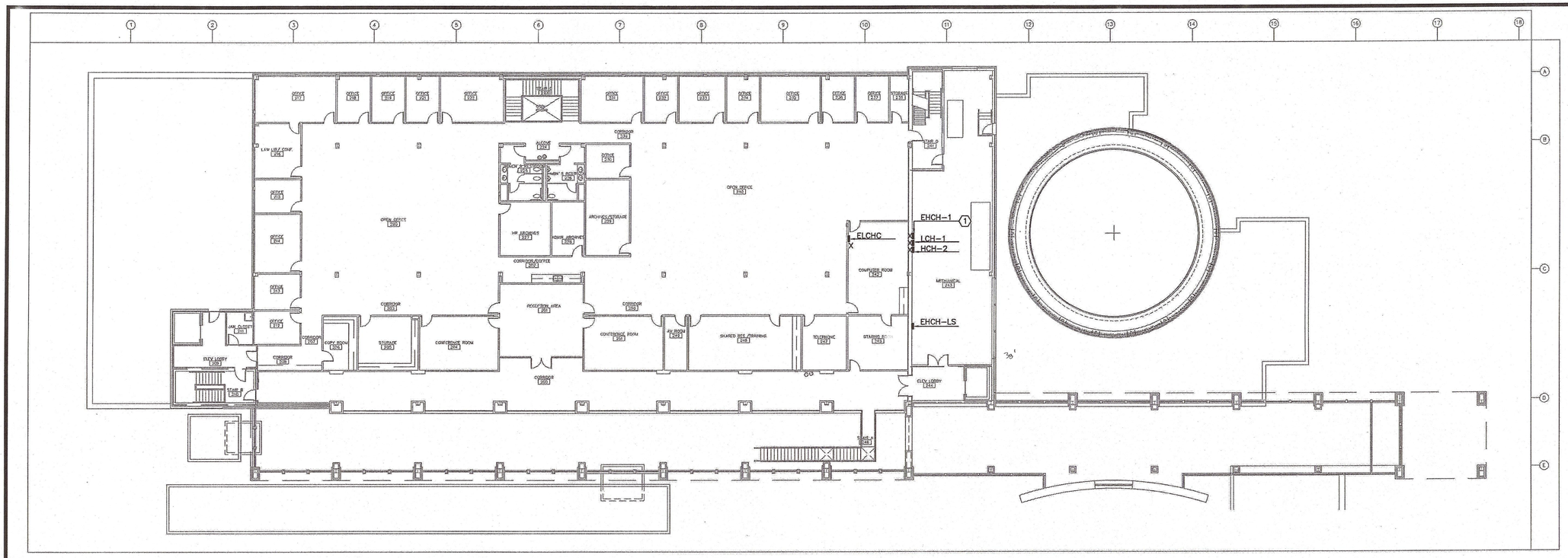
GOVERNMENT CENTER PUBLIC SAFETY ELECTRICAL PLANS

PROJECT NO. 15088.000

RECORD NO.

SHEET NO. E111

FIELD BOOK:
Plot Date: 11/13/2012
Drawing name: K:\g-m\Maple Grove\15088000\elec\dwgs\E111-POLICE-STATION.dwg
Xrefs: Wall #6-10_FLOORPLAN, CH_BA01, CH_BA02, CH_BA00



N

1 SECOND FLOOR ELECTRICAL PLAN

(1 / E112)

SCALE: 1/16" = 1'

0 8 16 32

SCALE IN FEET

PANELBOARD SCHEDULE

PANEL NAME EXISTING EHCH-LS

☑ SURFACE ☐ FLUSH

☐ FEED-THRU LUGS

☑ MAIN LUGS ☐ MAIN BREAKER

BUS 100 | VOLTAGE 480 L-L 277 L-N | PHASE 3 | WIRE 4

CKT NO.	CIRCUIT DESIGNATION	AMPS	POLES	LOAD-KVA			POLES	AMPS	CIRCUIT DESIGNATION	CKT NO.
1	2ND FLOOR EMERG & NIGHT LTG	20	1	0	A	0	1	20	PARKING GARAGE LIGHTING	2
3	2ND FLOOR EMERG & NIGHT LTG	20	1	0	B	0	1	20	PARKING GARAGE LIGHTING	4
5	MAIN FLOOR EMERG & NIGHT LTG	20	1	0	C	0	1	20	SPARE	6
7	MAIN FLOOR EMERG & NIGHT LTG	20	1	0	A	0	1	20	SPARE	8
9	MAIN FLOOR EMERG & NIGHT LTG	20	1	0	B	0	1	20	SPARE	10
11	MAIN FLOOR EMERG & NIGHT LTG CC	20	1	0	C	0	1	20	SPARE	12
13	SPARE	20	1	0	A	0	1	20	SPARE	14
15	STAIRWELL LIGHTING	20	1	0	B	0	1	20	SPARE	16
17	SPARE	20	1	0	C	0	1	20	SPARE	18
19	SPARE	20	1	0	A	0	1	20	SPARE	20
21	SPARE	20	1	0	B	0	1	20	SPARE	22
23	SPARE	20	1	0	C	0	1	20	SPARE	24
25	SPACE		1	0	A	0	1		SPACE	26
27	SPACE		1	0	B	0	1		SPACE	28
29	SPACE		1	0	C	0	1		SPACE	30
31	BLANK		1	0	A	0	1		BLANK	32
33	BLANK		1	0	B	0	1		BLANK	34
35	BLANK		1	0	C	0	1		BLANK	36
37	BLANK		1	0	A	0	1		BLANK	38
39	BLANK		1	0	B	0	1		BLANK	40
41	BLANK		1	0	C	0	1		BLANK	42

SUBTOTAL	A TBD	B TBD	C TBD
FEED THRU SUBTOTAL	A TBD	B TBD	C TBD
TOTAL KVA	A TBD	B TBD	C TBD
TOTAL AMPS	TBD		

THE ENTIRE SYSTEM SHALL BE ☑ FULLY ☐ SERIES RATED

AVAILABLE SHORT CIRCUIT CURRENT AT PANELBOARD LUGS IS 14 KAIC.

PANELBOARD SCHEDULE

PANEL NAME EXISTING EHPS-LS

☑ SURFACE ☐ FLUSH

☐ FEED-THRU LUGS

MAIN BREAKER RATING = 200

☐ MAIN LUGS ☑ MAIN BREAKER

BUS 225 | VOLTAGE 480 L-L 277 L-N | PHASE 3 | WIRE 4

CKT NO.	CIRCUIT DESIGNATION	AMPS	POLES	LOAD-KVA			POLES	AMPS	CIRCUIT DESIGNATION	CKT NO.
1	PANEL EHPS LOADS	20	1	0	A	0	1	20	POLICE STATION NIGHT LIGHTS	2
3	PANEL EHPS LOADS	20	1	0	B	0	1	20	POLICE STATION NIGHT LIGHTS	4
5	PANEL EHPS LOADS	20	1	0	C	0	1	20	POLICE STATION NIGHT LIGHTS	6
7	PANEL EHPS LOADS	20	1	0	A	0	1	20	SPARE	8
9	PANEL EHPS LOADS	20	1	0	B	0	1	20	SPARE	10
11	PANEL EHPS LOADS	20	1	0	C	0	1	20	SPARE	12
13	SPARE	20	1	0	A	0	1	20	SPARE	14
15	SPARE	20	1	0	B	0	1	20	SPARE	16
17	SPARE	20	1	0	C	0	1	20	SPARE	18
19	SPARE	20	1	0	A	0	1	20	SPARE	20
21	SPARE	20	1	0	B	0	1	20	SPARE	22
23	SPARE	20	1	0	C	0	1	20	SPARE	24
25	SPACE		1	0	A	0	3	60	PANEL EHCH-LS	26
27	SPACE		1	0	B	0	-	-	-	28
29	SPACE		1	0	C	0	-	-	-	30
31	SPACE		1	0	A	0	1		SPACE	32
33	SPACE		1	0	B	0	1		SPACE	34
35	SPACE		1	0	C	0	1		SPACE	36
37	SPACE		1	0	A	0	1		SPACE	38
39	SPACE		1	0	B	0	1		SPACE	40
41	SPACE		1	0	C	0	1		SPACE	42

SUBTOTAL	A TBD	B TBD	C TBD
FEED THRU SUBTOTAL	A TBD	B TBD	C TBD
TOTAL KVA	A TBD	B TBD	C TBD
TOTAL AMPS	TBD		

THE ENTIRE SYSTEM SHALL BE ☑ FULLY ☐ SERIES RATED

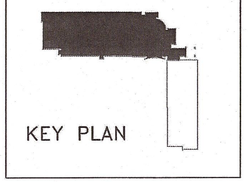
AVAILABLE SHORT CIRCUIT CURRENT AT PANELBOARD LUGS IS 14 KAIC.

GENERAL NOTES

A. SEE RISER DIAGRAM FOR FURTHER INFORMATION.

KEY NOTES

(1) INTERCEPT AND EXTEND EXISTING PANEL 'EHCH-1' CIRCUITS 1,2,4,6,8,10,16, TO EXISTING LIFE SAFETY PANEL 'EHCH-LS'. UTILIZE EXISTING CONDUIT(S).



FIELD BOOK:
Plotted: 11/14/2012
Drawing name: K:\gm\MapleGrove\15088000\elec\dwgs\E112-city hall.dwg
Xrefs: Wall #8-10_FLOORPLAN, CH_BA01, CH_BA02, CH_BA00

NO.	DATE	BY	DESCRIPTION OF REVISIONS

DESIGNED RCS DRAWN RCS CHECKED RLY

I HEREBY CERTIFY THAT THIS PLAN WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MINNESOTA

ROBERT L. YOUNG

DATE 11/14/12 LIC. NO. 26746

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STANDBY GENERATOR PROJECT

MAPLE GROVE MINNESOTA

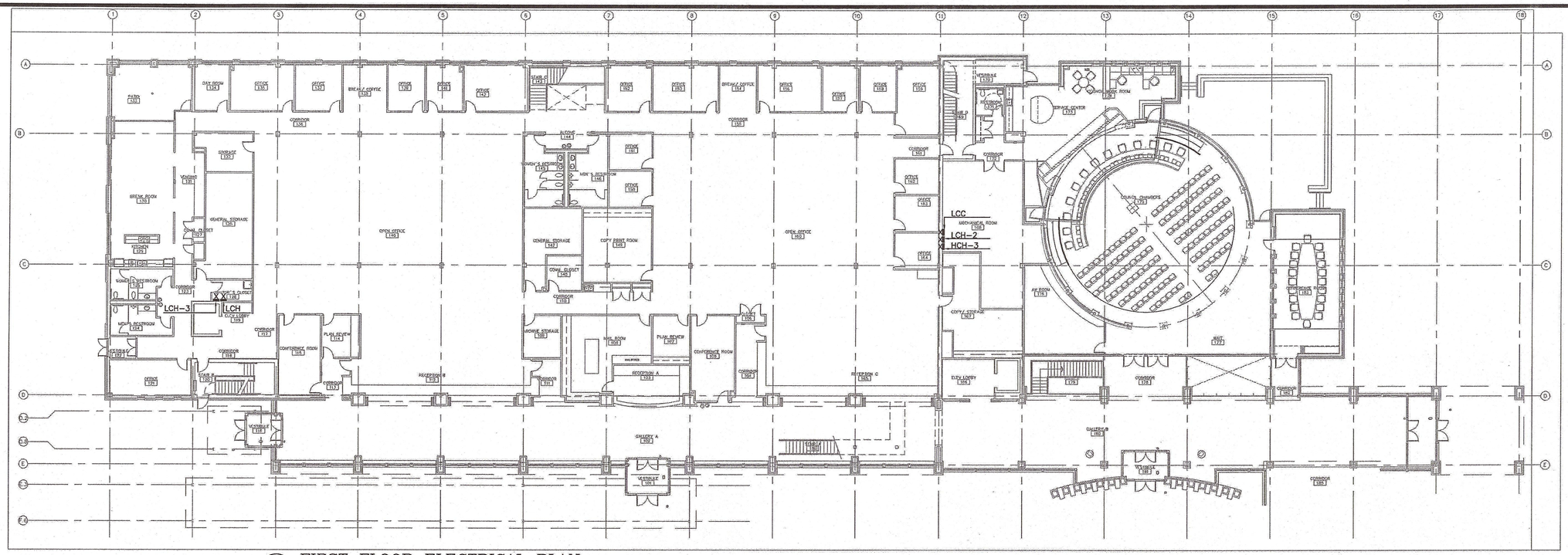
CLIENT NO. MSA NO.

GOVERNMENT CENTER
CITY HALL
ELECTRICAL PLAN

PROJECT NO. 15088.000

RECORD NO.

SHEET NO. E112

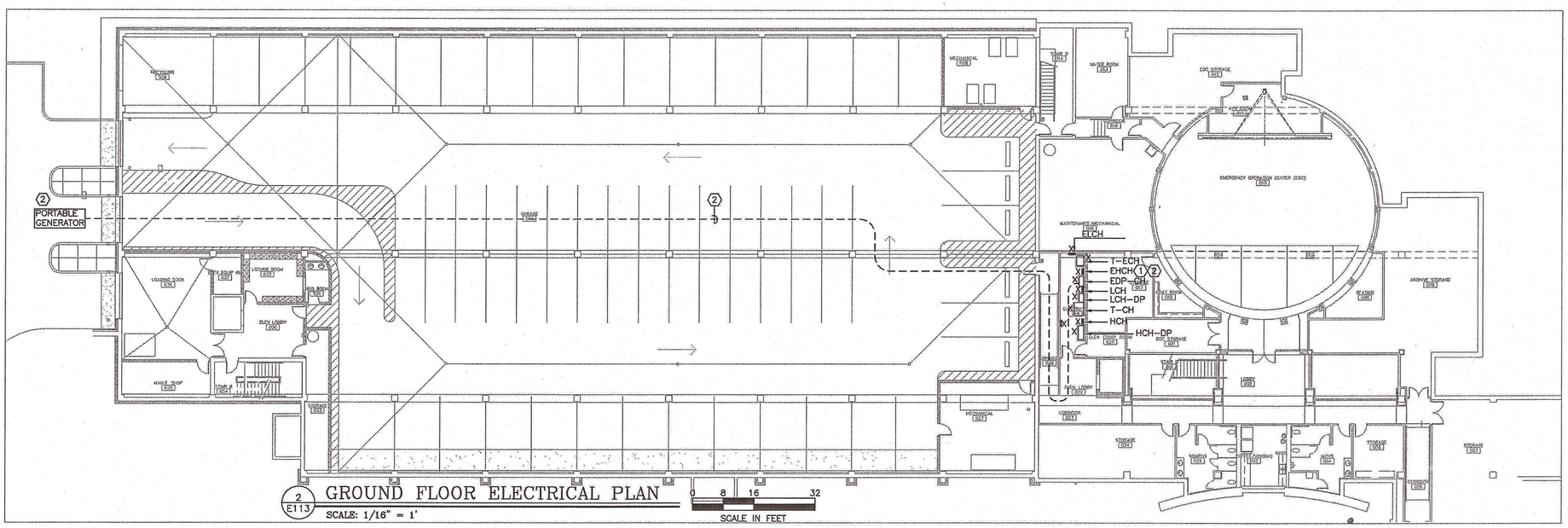


1 / E113 FIRST FLOOR ELECTRICAL PLAN

SCALE: 1/16" = 1'

0 8 16 32

SCALE IN FEET



2 / E113 GROUND FLOOR ELECTRICAL PLAN

SCALE: 1/16" = 1'

0 8 16 32

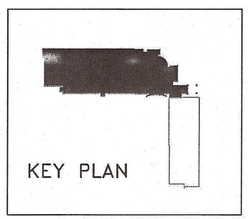
SCALE IN FEET

GENERAL NOTES

A. SEE RISER DIAGRAM FOR FURTHER INFORMATION.

KEY NOTES

1. INTERCEPT AND EXTEND EXISTING PANEL 'EHCH' CIRCUITS 25,26 TO EXISTING LIFE SAFETY PANEL 'EHCH-LS'. SEE DRAWING E112 FOR PANEL LOCATION. UTILIZE EXISTING CONDUIT(S).
2. PROPOSED TEMPORARY 60KW GENERATOR LOCATION AND TEMPORARY FEEDER ROUTING.



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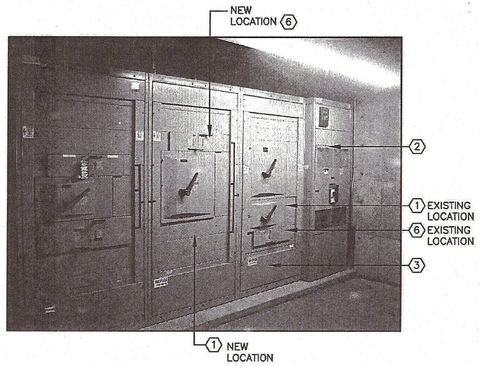
RECORD NO. SHEET NO. E113

FIELD BOOK:
PLOT DATE: 11/13/2012
Drawing name: K:\gov\MapleGrove\15088000\elec\dwg\E113-city hall.dwg
Xrefs: Wall #6-10_FLOORPLAN, CH_BA01, CH_BA02, CH_BA00



2 E114 EXISTING MAIN SWITCHBOARD

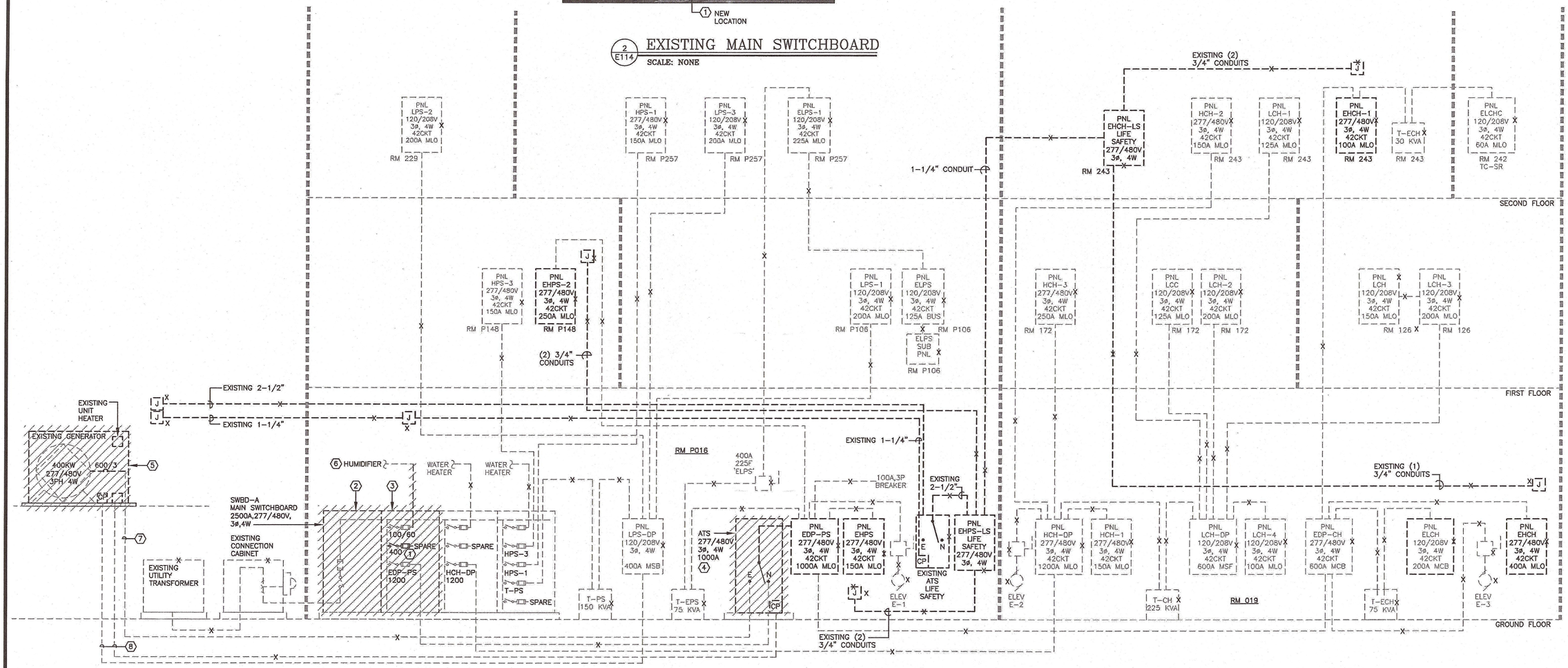
SCALE: NONE

GENERAL NOTES

1. EXISTING EQUIPMENT OR DEVICES TO REMAIN ARE DESIGNATED WITH DASHED LINES OR X. EQUIPMENT OR DEVICES TO BE REMOVED ARE DESIGNATED BY HATCHING.
2. REMOVE JUNCTION BOXES, CONDUIT AND WIRING NOT USED AS A RESULT OF THIS PROJECT. CONSULT OWNER REGARDING DISPOSAL OR SALVAGE OF REMOVED ELECTRICAL EQUIPMENT.

KEY NOTES

1. RELOCATE EXISTING 400A DISCONNECT TO ADJACENT SECTION FOR FUTURE USE. SEE 2/E114.
2. DISCONNECT AND REMOVE EXISTING SERVICE MAIN SECTION TO ACCOMMODATE THE INSTALLATION OF NEW EQUIPMENT. SEE 2/E114. RETURN REMOVED EQUIPMENT TO OWNER.
3. DISCONNECT AND REMOVE EXISTING DISTRIBUTION SECTION TO ACCOMMODATE THE INSTALLATION OF NEW EQUIPMENT. SEE 2/E114. RETURN REMOVED EQUIPMENT TO OWNER.
4. REMOVE EXISTING AUTOMATIC TRANSFER SWITCH. CONTRACTOR SHALL RETURN TO OWNER FOR FUTURE USE. MAINTAIN EXISTING WIRING FOR RECONNECTION TO NEW AUTOMATIC TRANSFER SWITCH.
5. REMOVE EXISTING 400KW GENERATOR. CONTRACTOR SHALL RELOCATE TO WELL 6 AND 10 SITE. DISCONNECT ALL ASSOCIATED CONDUIT AND WIRING TO GENERATOR. MAINTAIN FOR FUTURE USE UNLESS OTHERWISE NOTED.
6. DISCONNECT EXISTING HUMIDIFIER. MAINTAIN EXISTING WIRING FOR RECONNECTION TO SPARE DISCONNECT SWITCH IN ADJACENT SECTION. SEE 2/E114.
7. EXISTING (4) SETS OF 350KCMIL ALUMINUM. CONDUIT AND WIRING SHALL REMAIN.
8. EXISTING CONDUIT SHALL REMAIN. REMOVE EXISTING WIRING.



1 E114 EXISTING RISER DIAGRAM

SCALE: NONE

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STANDBY GENERATOR PROJECT

MAPLE GROVE MINNESOTA

CLIENT NO. MSA NO.

GOVERNMENT CENTER & EXISTING RISER DIAGRAM

PROJECT NO. 15088.000

RECORD NO. SHEET NO. E114

FIELD BOOK:
PLOT DATE: 11/14/2012
Drawing name: K:\g-m\Maple Grove\15088000\sheets\dwgs\E114- CITY HALL RISER-EXISTING.dwg
Xrefs:

GENERAL NOTES

1. ALL RACEWAYS SHALL CONTAIN A GROUNDING CONDUCTOR.
2. PROVIDE CONSPICUOUS, DURABLE, WRITTEN DOCUMENT INSTRUCTING OPERATOR ON PROCEDURE FOR MANUAL TRANSFER SEQUENCE TO AND FROM GENERATOR AND CORRESPONDING ISOLATION/CONNECTION FROM AND TO UTILITY TO RESTORE/MAINTAIN POWER
3. ANY REQUIRED SHUTDOWN OF EXISTING ELECTRICAL SERVICE WILL NEED TO BE COORDINATED BY THE CONTRACTOR WITH THE UTILITY SITE REPRESENTATIVE IN ADVANCE. THIS WORK SHALL BE CONDUCTED AFTER NORMAL BUSINESS HOURS OR ON WEEKENDS TO MINIMIZE DISRUPTION OF BUILDING OPERATIONS.

KEY NOTES

(15) PROVIDE NEW GENERATOR AT EXISTING GENERATOR LOCATION. NEW GENERATOR ENCLOSURE SHALL BE CUSTOM BUILT TO FIT OVER EXISTING CONDUIT LOCATIONS AND SHALL MAINTAIN THE EXISTING GENERATOR FOOT PRINT. SEE SPECIFICATIONS FOR FURTHER INFORMATION.

(16) EXISTING (4) SETS OF 350KCMIL ALUMINUM. CONDUIT AND WIRING SHALL REMAIN, RECONNECT TO NEW GENERATOR.

(17) EXISTING 3/4" CONDUIT SHALL REMAIN. PROVIDE 2#6 CU AND GROUND FOR CONNECTION TO GENERATOR BLOCK HEATER AND BATTERY CHARGER.

(18) EXTEND EXISTING ELEVATOR CONTROL CIRCUIT FROM EXISTING REMOVED ATS/PULLBOX LOCATION TO NEW AUTOMATIC TRANSFER SWITCH. (CONDUIT & WIRING NOT SHOWN).

(19) PROVIDE 3/4"PVC CONDUIT WITH (2) #14 FOR GENERATOR RUN SIGNAL, (1) #3106A COMMUNICATION CABLE FOR MODBUS COMMUNICATIONS.

(20) EXISTING 100/3 G.E. SPECTRA SERIES CIRCUIT BREAKER AT EXISTING PANEL 'EDP-CH FOR BACK FEEDING OF PANEL 'EHCH-1' AND 'ELCH'.

(21) ELECTRICAL CONTRACTOR IS RESPONSIBLE TO PLAN AND IMPLEMENT SAFE OPERATION OF KEY EQUIPMENT AND PERSONAL SAFETY DURING PROVISION OF TEMPORARY GENERATOR TO BACK UP REQUIRED SYSTEMS DURING CONSTRUCTION. PLANNING WILL INCLUDE: DISCONNECT FEEDER WIRING AND ISOLATE OR 'LOCK-OUT-TAG-OUT 'EDP-CH' FEEDER BREAKER TO PANEL 'EDP-PS' DURING EQUIPMENT MODIFICATIONS.

(22) GENERATOR START/STOP AND STATUS CONTROL.

(23) EXTEND CONDUIT AND WIRING FROM EXISTING JUNCTION BOX TO NEW GENERATOR. CUT EXISTING CONCRETE AND EMBED NEW CONDUIT IN CONCRETE SUCH IT WILL ELIMINATE A TRIPPING HAZARD.

(9) EXISTING JUNCTION BOX. INTERCEPT EXISTING LIFE SAFETY BRANCH CIRCUITS AND CONDUITS CURRENTLY FED FROM ADJACENT PANEL AND EXTEND TO JUNCTION BOX FOR REROUTING TO LIFE SAFETY PANEL AND PROVIDE BRANCH CIRCUIT CONDUCTORS. LIFE SAFETY CIRCUITS TO BE SEPARATED FROM NORMAL BRANCH CIRCUIT WIRING.

(10) GENERATOR REMOTE MONITOR PANEL.

(11) THE ELECTRICAL CONTRACTOR SHALL PROVIDE A TEMPORARY GENERATOR AND CABLING CONNECTION AS SHOWN FOR BACK-UP POWER FROM THE TIME THE EXISTING GENERATOR AND TRANSFER SWITCH ARE DISCONNECTED FROM THE EXISTING MSB BUS UNTIL THE NEW GENERATOR AND AUTOMATIC TRANSFER SWITCH AND WIRING ARE RECONNECTED TO THE EXISTING MSB BUS. THE SITE SHALL NOT BE LEFT WITHOUT BACK-UP GENERATOR POWER. CONTRACTOR SHALL COORDINATE WORK WITHIN THE MAIN ELECTRICAL ROOM WITH THE OWNER SUCH THAT THE WORK PERFORMED IS NOT DURING BUSINESS HOURS. SEE DRAWING E113 FOR PROPOSED LOCATION.

(12) SEE FLOOR PLAN DRAWINGS FOR CIRCUITS TO BE RELOCATED FROM THIS PANEL TO NEW LIFE SAFETY PANEL. REVISE EXISTING PANEL INDEX.

(13) REMOVE (2) EXISTING SPARE CIRCUIT BREAKERS. PROVIDE 40/2 CIRCUIT BREAKER AT EXISTING PANEL. PANEL IS G.E. 'AQF' SERIES.

(14) EXISTING 1" CONDUIT WITH (2) #14 FOR GENERATOR MONITOR PANEL POWER, (2) #14 GENERATOR RUN SIGNAL, (2) BELDEN #3106A COMMUNICATION CABLE FOR MODBUS COMMUNICATIONS BETWEEN THE GENERATOR AND THE TRANSFER SWITCH AND BETWEEN THE GENERATOR AND REMOTE MONITOR PANEL.

(1) NEW CUSTOM BUILT 2000A MAIN SECTION WITH OPTIONAL STANDBY CIRCUIT BREAKER, LIFE SAFETY CIRCUIT BREAKER, ELECTRONIC METERING AND GFI PROTECTION. MATCH EXISTING 65KAIC DISTRIBUTION SECTION CONTACT ELECTRO-MECHANICAL INDUSTRIES (EMI) FOR INFORMATION AND PRICING. INSTALLATION BY ELECTRICAL CONTRACTOR.

(2) NEW CUSTOM BUILT 1600A AUTOMATIC 4 POLE TRANSFER SWITCH (OPTIONAL STANDBY) AND DISTRIBUTION SECTION WITH DISTRIBUTION BREAKER AND TVSS. MATCH EXISTING 65 KAIC DISTRIBUTION SECTION. CONTACT ELECTRO-MECHANICAL INDUSTRIES (EMI) FOR INFORMATION AND PRICING.

(3) NEW CUSTOM BUILT BUS TIE TO BE INSTALLED BY ELECTRICAL CONTRACTOR. CONTACT ELECTRO-MECHANICAL INDUSTRIES (EMI) FOR INFORMATION AND PRICING.

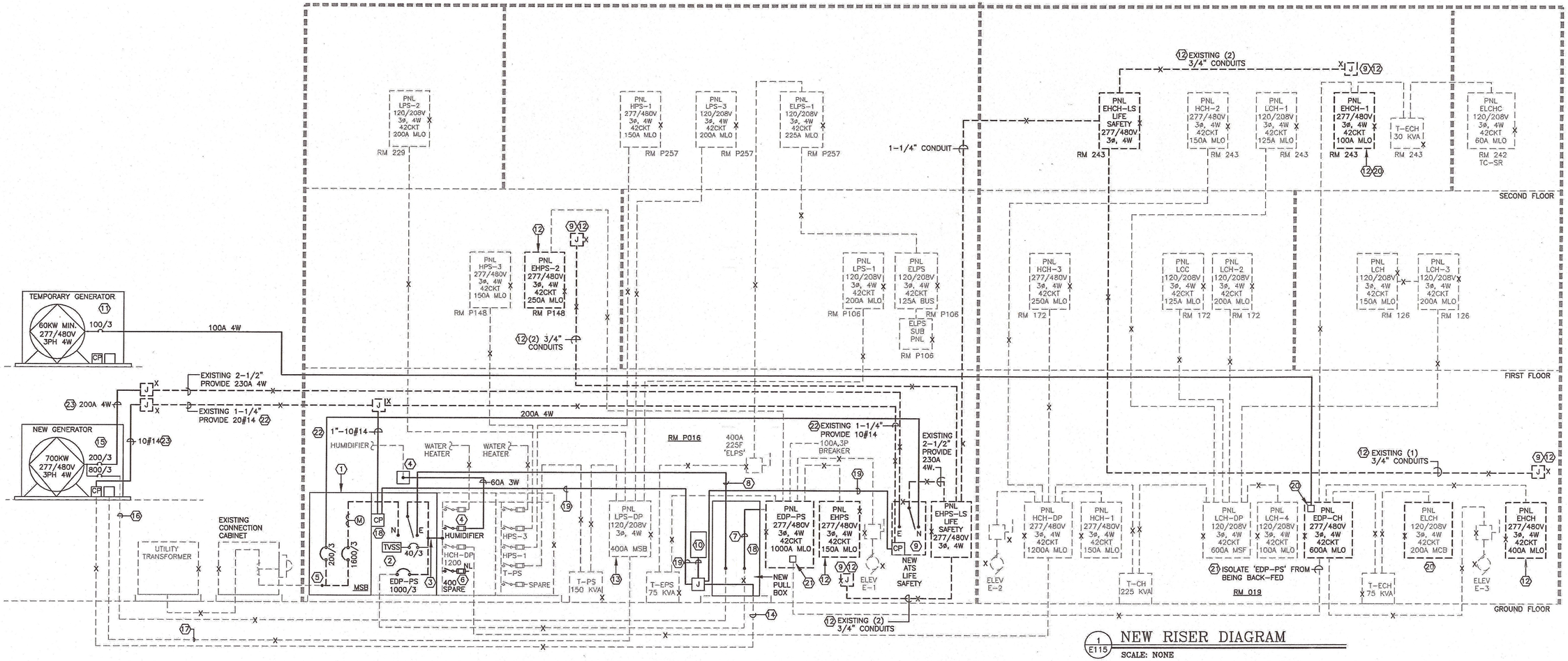
(4) INTERCEPT EXISTING HUMIDIFIER CONDUIT AND WIRING. EXTEND TO EXISTING SPARE DISCONNECT SWITCH AT ADJACENT DISTRIBUTION SECTION. RE-INSTALL 60A FUSES FROM REMOVED EXISTING DISCONNECT SWITCH

(5) RECONNECT EXISTING SERVICE CONDUCTORS TO NEW MAIN SECTION BUS.

(6) INSTALL THE REMOVED 400A DISCONNECT SWITCH IN THE ADJACENT SECTION.

(7) EXTEND EXISTING (4) SETS 4#350KCMIL ALUMINUM CONDUCTORS TO EXISTING PANEL 'EDP-PS'.

(8) EXTEND EXISTING (4) SETS 4#350KCMIL ALUMINUM CONDUCTORS FROM NEW PULLBOX LOCATION TO NEW AUTOMATIC TRANSFER SWITCH. PROVIDE REQUIRED RACEWAY.



1 E115 NEW RISER DIAGRAM

SCALE: NONE

NO.	DATE	BY	DESCRIPTION OF REVISIONS

DESIGNED RCS DRAWN RCS CHECKED RLY

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MAPLE GROVE MINNESOTA

CLIENT NO. MSA NO.

GOVERNMENT CENTER
NEW RISER DIAGRAM

PROJECT NO. 15088.000

RECORD NO. SHEET NO. E115

FIELD BOOK:
PLOT DATE: 11/14/2012
Drawing name: K:\gen\Maple Grove\15088000\elec\dwgs\E115- CITY HALL-RISER-NEW.dwg
Xrefs: